

Three layers of realisation:
hierarchy recovery, holonomy, information, contextuality
and categorical dynamics
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Abstract

This paper consolidates three bodies of work into one account. Each result carries its source and its status: proved, computed, measured, open, or closed by theorem.

1. *Hierarchy recovery* [H]. A gradient-free estimator (spectral initialisation, soft hand-off, inside-outside EM and split-merge) recovers a depth-4 probabilistic grammar to 0.001 nats per sequence of the true model in 131 s. Twelve transformer configurations never represent the level-3 latent symbol.
2. *Cohomological strand* [O]. Holonomy of known operations gives a genuine obstruction $[g] \in H^1(\Gamma, A)$. Marginal obstructions within one data source vanish (T0). The information site and the contextuality site are separated by a theorem.
3. *Results of the present work* [S]:
 - an exact identifiability criterion for holonomy estimated from data, with a calibrated test and its sample cost;
 - a direct construction from operations to outcomes, $[g] \neq 0 \iff \gamma \neq 0$, which is invisible to the probability layer;
 - for lossy operations: a closed form $CF = \max(0, 1 - |L|\varepsilon/2)$ on a cycle, and a holonomy-girth lower bound on general graphs;
 - the loop spectrum as a monoid-valued holonomy invariant;
 - a decorated Bregman-Čech nerve that carries orientation and phase, with a codimension criterion for when it can;
 - restriction and extension maps between the information and contextuality views;
 - a gated contextuality test for streams with *not-applicable* entries, with a pinned maximum-likelihood null.

The organising statement is that the three layers (operations, probability and outcomes) respond to the same data through different invariants, and that no invariant of one layer determines another.

Code: <https://github.com/vkawasthi/Trainingless-Transformer-NoGradientDescent/tree/main/AMB-Vigneaux/>

Provenance tags.

- [H]: *Recovering latent hierarchy without gradient descent* (hierarchy paper and handoff).
- [O]: *Holonomy obstructions for composed local operations* (holonomy paper).
- [S]: this work, implemented in the `amb.vigneaux` package. It has 219 tests, and every numerical claim tagged [S] is produced by that code.

Status tags: *proved*, *computed* (exact computation or verified identity), *measured* (empirical, no proof), *open*, *closed* (settled by theorem, not to be reopened).

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1 The standing constraint

Theorem 1.1 (T0: single-source global section). *[O, Thm 7] proved Let $\{P_C\}_{C \in \mathcal{M}}$ be context marginals estimated from one sample $\mathcal{D}$. The empirical joint $P_{\mathcal{D}}$ restricts to each P_C , so the family extends. Consequently the Čech obstruction $\gamma(s)$ vanishes for every section and the contextual fraction is 0, on any cover, acyclic or not.*

Proof. $P_{\mathcal{D}}$ is a global section by construction. □

Corollary 1.2. *[H/O] proved A single fitted model can never exhibit an obstruction to its own consistency. If an estimator and a detector share a data source, the detector measures model misfit, not structure.*

Remark 1.3 (T0 as a functor). [S] proved Restriction R from global laws to context families (Section 25) has image exactly the non-contextual models. So T0 is the statement that anything in the image of R has $\text{CF} = 0$ and $\gamma = 0$. The engine’s first version violated T0 by resampling a single-source world context by context: it produced a spurious CF up to 0.10, and the fitted model’s CF equalled its signalling defect exactly. After adding a single-source sampling mode, $\text{CF}(\hat{P}_N) \leq 2 \times 10^{-16}$.

2 The three layers

Layer	Invariant	Condition for non-triviality
1 Operations	$H^1(\Gamma, A) \cong A^{\beta_1}$, holonomy; the loop spectrum for lossy operations	The operations are known or identifiable (Prop. 17.1); $\beta_1 > 0$
2 Probability	$H^1(\mathcal{S}; \mathcal{F}_\alpha)$, one-dimensional, generated by entropy (Tsallis for the α -action)	The conditioning action is present; $I = \delta_t H$ measures its failure to be trivial
3 Outcomes	$\gamma \in \check{H}^1(\mathcal{U}, \mathcal{F}_{\tilde{C}})$; CF	The cover has no global context; data come from more than one source (T0)

Table 1: The stratification [O, §6], extended [S].

Remark 2.1 (Stratification, not bridge). [O, Rem. 18] proved Each invariant is non-trivial on its own layer and zero on the others, and the reasons are the hypotheses themselves. Layer 2’s site has a global joint, which gives the observable monoid an absorbing element; Theorem 15.3 then kills every trivially-acting coefficient system, which is where Layer 3’s coefficients would have to live. Layer 3’s site is built so that no global context exists. That is what permits $\gamma \neq 0$, and it simultaneously denies Layer 2 the joint it requires.

[S] adds that the layers are connected *around* Layer 2, not through it. There is an exact construction from Layer 1 to Layer 3 (Theorem 18.1), and Layer 2 is provably blind to it (Proposition 18.2).

3 Framework: sites, presheaves, categories and schemes

Reading guide. This section (about four pages, to page 12) collects the categorical, sheaf-theoretic and scheme-theoretic language that sits behind the paper. It is here so that readers who know this language can connect the results to it. *None of it is needed to follow the results.* Readers who want only the results can skip to Part I (hierarchy recovery) or directly to Section 36 (bias detection). Every statement below is either standard or proved elsewhere in the paper, and the only new computation, the gluing lattice (§3.3), is also reported with the bias results.

3.1 Sites, presheaves and gluing

Contexts and covers. Fix a finite set of measurements X . A *context* is a set of measurements that can be made jointly. Contexts, ordered by inclusion, form a finite poset $\mathcal{C}$, viewed as a category with a unique arrow $C' \rightarrow C$ when $C' \subseteq C$. A *measurement cover* $\mathcal{U} = \{C_1, \dots, C_n\}$ is a family of maximal contexts. The Grothendieck topology we use declares $\mathcal{U}$ and its restrictions to be the covering families. On a finite poset every Grothendieck topology is of this kind, and *sieves* (downward-closed families of contexts) are the natural bookkeeping device.

Presheaves. A presheaf is a functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ (or $\mathbf{Ab}$, $\mathbf{Mod}_R$): to each context a set of local data, to each inclusion a restriction map. Three presheaves carry the paper.

- The *event sheaf* $\mathcal{E}(C) = O^C$ (assignments of outcomes to the measurements in C). This one *is* a sheaf: compatible local assignments glue uniquely [1].
- The *distribution presheaf* $\mathcal{D}_R\mathcal{E}$, which sends C to the R -distributions on $\mathcal{E}(C)$, for a semiring R ($\mathbb{R}_{\geq 0}$ for probability, the Booleans for supports). An empirical model is a compatible family $e = (e_C)$.
- The *support subpresheaf* $\mathcal{S}_e \subseteq \mathcal{E}$, which sends C to the support of e_C .

Gluing (descent). A compatible family over a cover *glues* if it is the restriction of a single global element. In this language:

- e is *non-contextual* iff it glues in $\mathcal{D}_{\mathbb{R}_{\geq 0}}\mathcal{E}$;
- e is *strongly contextual* iff $\mathcal{S}_e$ has no global section at all [1].

Neither $\mathcal{D}_R\mathcal{E}$ nor $\mathcal{S}_e$ is a sheaf in general. Their failure to be a sheaf *is* contextuality, and a write-up must keep “presheaf” and “sheaf” apart.

The information structure as a site. On Vigneaux’s side [6] the category is a poset of observables (partitions or random variables, ordered by refinement), and the product XY is the join. The coefficients are a presheaf of modules F of functions on probability laws. The observable monoid acts on F by conditioning:

$$(X.f)(P) = \sum_x P(X=x)^\alpha f(P|_{X=x}),$$

and information cohomology is $\text{Ext}^\bullet(\mathbb{R}, F)$ in this ringed topos. The contextuality site has no global context; the information site has a global joint. That is the content of the stratification remark above: the two sites are different categories, not two presheaves on one.

The bias site. For bias, the objects are *regions* (source $\times$ topic $\times$ time window). The presheaf assigns to a region the bias measurements made there, and “gluing” means that one additive model, house lean plus topic effect, fits the whole region (Section 36).

3.2 Čech cohomology, connections and monodromy

Nerve and Čech complex. The nerve $N(\mathcal{U})$ has a k -simplex for every $(k + 1)$ covering sets with non-empty common intersection. For an abelian presheaf F , the Čech complex is $\check{C}^k = \prod_{\sigma \in N_k} F(\cap \sigma)$ with the alternating differential δ ; $\delta\delta = 0$. The AMB obstruction γ is a class in relative $\check{H}^1$ of the support subpresheaf with coefficients in a ring [2].

Connections on a graph, and holonomy. When the cover is a graph Γ (sources joined by comparisons), a 1-cochain b is a connection with values in an abelian group A . Its *holonomy* around a cycle is the sum of b along the cycle, and b is flat with trivial holonomy iff $b = \delta u$ is a coboundary. Then $H^1(\Gamma; A) \cong A^{\beta_1}$. In the bias analysis $A = (\mathbb{R}, +)$: the loop sum Φ is the holonomy, and house lean plus topic effect is the coboundary. Two consequences of $\delta\delta = 0$ are used repeatedly:

- a single source on the same units always glues (Proposition 35.2);
- loops that bound a filled cell (a unit measured by all the loop’s sources) are flat (Proposition 36.1).

The *arity grading* of Section 36 is therefore a filtration of the nerve by which simplices are filled.

Local systems and the monodromy functor. A local system on Γ is a functor $\rho : \Pi_1(\Gamma) \rightarrow \mathcal{V}$ from the fundamental groupoid to a target category $\mathcal{V}$. Its value on a loop is the *monodromy*. The paper uses three targets:

- finite permutation groups, for relabellings of labels in Corpus V and EWT (Section 31);
- $(\mathbb{R}, +)$ or $(\mathbb{R}_{>0}, \times)$, for bias and odds ratios (Section 36);
- **FinStoch**, for noisy transports (next subsection).

“Holonomy” and “monodromy” name the same construction; the target category decides what can be seen. A permutation of finite order is seen through its orbits. A real number is seen through its size and sign.

3.3 The gluing lattice

Definition 3.1 (Gluing lattice). For a presheaf F on a cover $\mathcal{U}$, the gluing lattice $\mathcal{G}(F, \mathcal{U})$ is the set of sub-covers $S \subseteq \mathcal{U}$ on which the local data of F admit a global section.

Restriction of a global section shows that $\mathcal{G}$ is *downward closed*. It is therefore determined by its maximal elements, the largest regions that glue. Its complement is determined by its minimal elements, the smallest obstructed regions. Statistically each membership is a test, and monotonicity can fail at the margin; we count such violations. We computed the lattice with `amb.vigneaux.gluing_lattice (examples/bias3/gluing_lattice_run.py)`:

- *NewsWCL50, cover by targets* (Trump, USA, Migrant caravan, Iran, Democrats). 12 of the 26 sub-covers glue, with 0 monotonicity violations. The maximal glueable sub-covers are {Democrats, Migrant caravan, Trump, USA} and {Democrats, Iran}. The minimal obstructed sub-covers are exactly {Iran, USA}, {Iran, Migrant caravan} and {Iran, Trump}.
- *NewsWCL50, cover by outlets*. 11 of 26 glue, with 0 violations. The only maximal glueable sub-cover is {NYT, USA Today, Fox, Breitbart}, and the minimal obstructed sub-covers are HuffPost paired with each other outlet.
- Together these place the global failure of the NewsWCL50 no-three-way model on one cell, HuffPost $\times$ Iran, which matches the loop analysis.

- *MBIC, cover by topics.* 63 of 91 topic pairs glue. The minimal obstructions are the other 28 pairs. Greedy search finds maximal glueable regions of 10 of the 14 topics, several of them different, so no single topic is the culprit and the failure is spread. Here monotonicity fails statistically: some obstructed pairs, for example abortion and white nationalism, lie inside glueable 10-topic regions. A local failure is diluted when it is pooled with regions that glue, which is why the minimal obstructions, and not a single global test, are the informative output.

The lattice refines the yes/no verdict of a global test into a map of *where* the local pictures stop fitting together.

3.4 Markov categories

FinStoch has finite sets as objects and stochastic matrices as morphisms. It is a Markov category in Fritz’s sense [16].

- A *statistic* is a deterministic morphism.
- Conditional products and sufficiency are categorical properties; the soft hand-off verification and O2 (Section 27) are statements about sufficient statistics in **FinStoch**.
- The transports T_{ab} of Section 31 are morphisms of **FinStoch**, and the loop operator $M = T_{AB}T_{BC}T_{CA}$ is the monodromy of the functor $\Pi_1(\Gamma) \rightarrow \mathbf{FinStoch}$.
- For deterministic, invertible transports this functor lands in permutations, and its class is the $[g] \in H^1(\Gamma, A)$ of the holonomy paper.
- Simpson’s atomic sheaf logic [13] gives the complementary semantic statement: a presheaf on standard Borel probability spaces is an atomic sheaf iff it sends independent squares to pullbacks.

3.5 The finite Giry monad: what the paper uses, and why it suffices

For readers who meet it here first: the *Giry monad* [14] sends a measurable space X to the space $\mathcal{G}(X)$ of probability measures on X .

- Its unit sends a point to its Dirac measure, and its multiplication averages a measure of measures.
- A map $X \rightarrow \mathcal{G}(Y)$ is a Markov kernel, and these form the *Kleisli category*, whose composition is the Chapman–Kolmogorov formula.
- On finite sets the Giry monad becomes the finite distribution monad $\mathcal{D}$. Its Kleisli category is **FinStoch**: finite sets as objects, stochastic matrices as morphisms, matrix product as composition [16].

All our data are finite, so this finite version is all the paper needs, and it uses it in four places:

- **FinStoch** as a *Markov category* in Fritz’s sense (previous subsection), which gives conditional independence and sufficiency their categorical meaning.
- *Sufficiency of the soft hand-off.* Identity V2 (Table 3) checks it to 6.7×10^{-16} , and Fritz’s Fisher–Neyman theorem is the statement being checked.
- *Merging as averaging in the monad.* A merged symbol’s rule row is the usage-weighted convex mixture of the two rows, the multiplication of $\mathcal{D}$ (Construction “merging as convex combination”, Sections 11, 40.3).

- *Transports as Kleisli morphisms.* The source-to-source transports T_{ab} are stochastic matrices, and the loop operator $M = T_{AB}T_{BC}T_{CA}$ is the monodromy of the functor $\Pi_1(\Gamma) \rightarrow \mathbf{FinStoch}$ (Section 31).

Which piece of the monad does what (a guide for re-implementing the engine). The Kleisli adjunction $\text{Hom}_{\mathbf{Meas}}(X, \mathcal{G}Y) \cong \text{Hom}_{\mathbf{Stoch}}(X, Y)$, with $F(f) = \eta \circ f$ and $U(Y) = \mathcal{G}Y$, is a dictionary: a Markov kernel $X \rightarrow Y$ is a measurable map $X \rightarrow \mathcal{G}Y$. The monad is $\mathcal{G} = UF$, and its multiplication is $\mu = U\varepsilon_F$. The adjunction is used silently wherever a kernel appears, but it computes nothing at an overlap. The operations of the engine use the monad's pieces as follows.

operation in the engine	categorical tool	uses μ ?
restrict a law to an overlap $U \cap V$ (marginalise)	the functor on maps: pushforward $\mathcal{G}(\pi)$ along the projection $X_U \rightarrow X_{U \cap V}$	no
check that two contexts agree on the overlap	equaliser of two pushforwards; the Čech sheaf condition is a <i>limit</i>	no
glue compatible marginals into a joint (extension)	conditional product $p_U \cdot p_V _{U \cap V}$, a Kleisli composite; one chosen section among many	yes
compose transports; loop operator $M = T_{AB}T_{BC}T_{CA}$	Kleisli composition $g \circ_{\text{Kl}} f = \mu \circ \mathcal{G}(g) \circ f$ (matrix product)	yes
merge two symbols (mixture of rule rows)	μ applied to a distribution over distributions	yes
deterministic maps (hard decode, coarse-graining)	the unit η : a function becomes a Dirac-valued kernel	η only
decode a block (posterior)	Bayesian inversion in FinStoch (needs a prior; Proposition 3.2)	no

Contextuality lives in the limit, not in μ . The obstruction (γ_{AMB} , the loop sums) is the failure of the limit of the presheaf $C \mapsto \mathcal{G}(X_C)$ over the cover to have an element: the local laws agree on every overlap but admit no joint. The multiplication enters only after a way of gluing has been *chosen* (the conditional product), and it then produces one candidate global section. A re-implementation should therefore keep three things apart:

- restriction and compatibility, which are functorial and need no μ ;
- composition and mixing, which are Kleisli operations and need μ ;
- inference, which is Bayesian inversion and needs a prior.

Two further statements make the decoding and the persistence route precise (`amb-vigneaux/giry.py`, `tests/test_giry.py`).

Proposition 3.2 (Decoding is Bayesian inversion). *proved; checked* The grammar is a Kleisli morphism, the composite of the kernels $P_\ell : V \rightarrow \mathcal{D}(V \times V)$ from the root to the leaves. For a block b with symbol z :

1. The inside vector $\iota(b)(z) = p(x_b | z)$ is the likelihood of the block's own leaves. Its normalisation is the soft hand-off, a sufficient statistic of x_b for z ($V2$).
2. The outside vector carries the rest of the sequence. The block posterior $\iota \cdot o / \langle \iota, o \rangle$ is the Bayesian inverse $k^\dagger(z | x) = p(z)k(x | z) / \sum_{z'} p(z')k(x | z')$ of the kernel $z \mapsto x_b$, with the outside vector as prior.

Proof. The tree factorises the joint law. Summing out everything except z , the leaves of b enter only through $\iota(b)$, and all other leaves only through $o(b)$. Both parts are checked against brute-force enumeration in `tests/test_giry.py`. $\square$

Remark 3.3 (What the soft decoder does and does not do). The inside vector of a block depends only on that block’s leaves, so it is unchanged, exactly, when any other leaves change (on $V = 8$, $L = 6$, the change is 0 at levels 1–3). A “soft decoder” that returns the normalised inside vector therefore sends every conditioning on other leaves to the identity. The functor laws then hold trivially: this is locality, not a functoriality theorem, and the map on morphisms is constant. What changes under conditioning is the block *posterior* of Proposition 3.2(2): its argmax differs between block-only and whole-sequence evidence on 8.5, 14.8 and 17.2% of sequences at levels 1–3. A comparison of a soft inside vector with a hard posterior mixes these two objects. The caption of Table 3 has been corrected accordingly.

Proposition 3.4 (Mutual-information persistence is functorial; HSIC persistence is not). *proved; checked* For a joint law of $X_1, \dots, X_k$, let $K(t)$ be the graph (or its flag complex) of pairs with $I(X_i; X_j) \geq t$. A coarse-graining f (a deterministic surjection on one variable) satisfies $I(f(X_i); X_j) \leq I(X_i; X_j)$ by the data-processing inequality. Hence $K_f(t) \subseteq K(t)$ for every t , and f induces a morphism of persistence modules $H_*(K_f(\cdot)) \rightarrow H_*(K(\cdot))$, contravariantly in f . By the stability theorem [15], the barcode moves by at most the sup-norm error of the estimated mutual informations. The same construction with HSIC weights is not functorial, because HSIC does not satisfy data processing: with Gaussian kernels, $\text{HSIC}(2 \text{sign} X, Y) > \text{HSIC}(X, Y)$ for $Y = X + \text{noise}$, and HSIC is not even invariant under the isomorphism $X \mapsto 10X$.

Proof. Data processing gives the entrywise inequality, which gives the inclusions of filtered complexes, and homology is functorial. The HSIC example is computed in `tests/test_giry.py`. The inclusion is checked on random joint laws over 20 draws and 25 thresholds each. $\square$

A thresholded single $H^1(K_\ell, \mathbb{F}_2)$ is not a coefficient morphism, because the step function is not additive. The persistence module of Proposition 3.4 is the functorial replacement, provided the dependence weight satisfies data processing.

3.6 Functors, natural transformations and higher structure

- *Restriction and extension* (Section 25) are functors between global laws and context families. Non-contextual models are the image of restriction.
- *Non-unification.* There is no morphism of coefficient systems (natural transformation) from $\mathcal{F}_{\mathbb{Z}}$ to $\mathcal{F}_{\alpha}$ that carries a non-zero AMB class to a non-zero Vigneaux class. The open problems ask whether an AMB scenario determines an information structure, and whether the correspondence is a morphism of sites.
- *Twisting.* The observable monoid is idempotent (a band), so every twisting cochain into a group is trivial (Proposition 33.1). Operations can twist only the coefficients, and for symmetric local solutions they cannot be seen (Proposition 33.2).
- *A_{∞} structure.* Every A_{∞} algebra is quasi-isomorphic to a strictly associative one (rectification). An algebra is *formal* when it is quasi-isomorphic to its own cohomology with all higher operations zero. In the multiplicative pointwise model the surprisal generates a formal structure: every Massey power $\langle s, \dots, s \rangle$ vanishes (Proposition 41.1).
- *Associahedra.* The only genuinely A_{∞} -combinatorial object in the paper is the space of parse trees. The vertices of the associahedron are the binary bracketings, its faces are planar trees, and a node of arity k spans a cell of dimension $k - 2$. The arity of Corpus V nodes is this dimension.

3.7 Schemes: where the language is literally correct

Toric statistical models are affine schemes. A log-linear model with design matrix $\mathcal{A}$ is the toric variety cut out by its toric ideal $I_{\mathcal{A}} \subset \mathbb{C}[p]$, i.e. the affine scheme $\text{Spec } \mathbb{C}[p]/I_{\mathcal{A}}$ [36]. By the fundamental theorem of Markov bases [35], a finite set of binomials generates $I_{\mathcal{A}}$ iff the corresponding moves connect every fiber of the sufficient statistics. For the no-three-way model on a $2 \times 2 \times 2$ table the ideal is principal, generated by the single quartic binomial; in log coordinates that binomial is the loop holonomy Φ . So the exact tests of Section 36 are random walks on fibers of a toric scheme. Over regions these models form a presheaf of affine schemes. Nothing in the paper yet needs more than the variety, for example non-reduced structure or base change of the model.

Families over $\text{Spec } \mathbb{Z}$: why the obstruction depends on the ring. The AMB obstruction depends on the coefficient ring (Section 16; the Hardy false negative). Here the scheme-theoretic view has content, and it is proved by the universal coefficient theorem. The Čech cochains of a finite cover with integer coefficients are free abelian, so for each prime p

$$0 \rightarrow H^1(\mathbb{Z}) \otimes \mathbb{F}_p \rightarrow H^1(\mathbb{F}_p) \rightarrow \text{Tor}(H^2(\mathbb{Z}), \mathbb{F}_p) \rightarrow 0.$$

The obstruction therefore behaves differently over $\mathbb{F}_p$ exactly at the primes dividing the torsion of the integral H^2 . Viewing γ as a family over $\text{Spec } \mathbb{Z}$, the special fibres differ from the generic fibre precisely at these torsion primes. For the CHSH scenario the marginal map is unimodular (all Smith invariants 1), so no prime is special (Proposition 35.1).

What is not claimed.

- There is no “structure sheaf of measures”: measures do not form a ring, and signed or p -adic measures always glue, so they cannot see contextuality (Proposition 35.1).
- Berkovich spaces enter only as the tree of p -adic discs, and the gluing radius p^{-m^*} of Proposition 41.2.
- The analogies with p -adic Hodge theory are recorded as analogies (Remark 33.6).

3.8 Dictionary

concept	where it appears
site of contexts, measurement cover	outcome layer (AMB); Sections 16, 21
event sheaf, distribution and support	contextuality, γ , CF; Sections 16, 34
presheaves	
information structure, module of	Layer 2; Sections 20, 33
functions on laws	
Čech H^1 , holonomy of an	Sections 31, 36
A -connection	
monodromy functor $\Pi_1(\Gamma) \rightarrow \mathcal{V}$	Sections 31 (permutations), 36 ($\mathbb{R}$)
filled simplices, arity filtration	Proposition 36.1
gluing lattice (sieves that satisfy descent)	§3.3; Section 36
Markov category FinStoch , sufficiency	Sections 27, 31
finite Giry monad, Kleisli category	§3.5
= FinStoch ; Bayesian inversion	
functors R, E ; natural transformations	Sections 25, 32
twisting cochains, A_∞ , formality	Sections 33, 41
toric affine schemes, Markov bases	Section 36
universal coefficients over $\text{Spec } \mathbb{Z}$	Sections 16, 35
natural transformations as phase changes	Section 37

Part I

Hierarchy recovery [H]

4 Setup and exact floors

All experiments use the Random Hierarchy Model of Cagnetta et al. [9]. It is a probabilistic context-free grammar on a fixed complete binary tree of depth L , with V symbols per internal level, m productions per symbol, and a leaf alphabet of size A . Because the tree shape is fixed and positional, the latent structure is a Markov chain from leaves to level 1 and on up to the root.

- *Corpus A*: $L = 4$, $V = 64$, $A = 48$, $m = 12$, 2×10^5 sequences of length 16.
- *Corpus B*: $L = 4$, $V = 8$, $A = 12$, $m = 3$, 2×10^5 sequences.

The exact conditional entropy $H(x_t \mid x_{<t})$ is computed by inside–outside (Theorem 12.1). For Corpus A: $H_{\text{local}} = 1.6034$, $H_{\text{higher}} = 3.5527$, $H(x_0) = 3.8161$ nats. The latent-symbol entropies given the prefix, for levels 1–4, are 2.7835, 2.2923, 2.2900, 2.7910 against chance $\ln V = 4.1589$. About 1.87 nats of level-3 information are recoverable from the prefix.

5 Transformers do not recover levels 3–4

Statement [H] *measured*. Across twelve configurations the level-3 probe never leaves the shuffled-label floor, although 1.87 nats of level-3 information are present in the prefix. A linear probe on the embedding was *withdrawn*: a random matrix supports the same probe accuracy.

configuration	local	higher	level-3 probe
2 layers, $d = 64$ (baseline)	2.173	3.831	0.028
4 layers	2.078	3.797	—
8 layers, frozen embedding	2.037	3.755	—
8 layers, frozen emb., fresh sampling	2.022	3.772	0.027
$d = 128$, 4 layers	2.039	3.770	—
frozen random embedding (2 layers)	2.186	3.824	—
constructed order coordinate	2.162	3.815	—
auxiliary supervision, levels 3–4	2.088	3.777	0.056
latent marginalising head (stopped at 34k)	3.227	3.835	—
dilated causal CNN	2.104	3.827	0.032
curriculum over tree depth	2.085	3.803	0.034
curriculum control (no easy cases)	2.043	3.789	0.033
exact floor	1.603	3.553	—

Table 2: Corpus A, 48,000 steps each. Shuffled-label controls read 0.021–0.034; chance is $1/V = 0.0156$. [H]

Hypothesis (not established). Recovering the level- ℓ symbol requires marginalising over the parses consistent with the prefix. A feedforward stack has only an upward pass, while inside–outside couples an upward and a downward pass.

6 The estimator and its verification

Construction 6.1 (Estimator). [H]

1. *Spectral initialisation*: build the pair–context co-occurrence matrix, take its leading singular directions, and k -means the pairs into V groups.
2. *Soft hand-off*: pass the posterior to the next level, $q_\ell \propto \sum_{x,y} P_\ell[\cdot, x, y] q_{\ell-1}^{\text{lo}}(x) q_{\ell-1}^{\text{hi}}(y)$.
3. *EM*: inside–outside E-step and normalised-counts M-step, keeping the sweep with the best held-out likelihood.

check	statement	max error
V0	EM never decreases the training log-likelihood	$+1.8 \times 10^{-2}$ (min step)
V1a	a free subtree has inside value exactly 1	1.8×10^{-15}
V1b	restriction = marginalisation	8.9×10^{-16}
V1c	gluing = independence join, $P(x_{0:16}) = \sum_{u,v} \mu(u,v) \beta_L(u) \beta_R(v)$	1.0×10^{-14}
V2	the soft hand-off is a sufficient statistic	6.7×10^{-16}
V4a	merge cost = $N I(\text{children}; B \mid q(B))$	1.2×10^{-14}

Table 3: Identities of the tree model class [H] computed. They verify the implementation and that the fitted object is a sheaf section. They do not test whether the data are tree-generated. The normalised inside vector of a block depends only on that block’s leaves, so it is unchanged, exactly, when any other leaves change; this is locality, not a separate functoriality result. The block’s *posterior* (inside $\times$ outside) does change as the rest of the sequence is revealed: on $V = 8$, $L = 6$ its argmax differs between block-only and whole-sequence evidence on 8.5, 14.8 and 17.2% of sequences at levels 1–3 (mean total variation 0.10–0.18).

variant (Corpus A, 2×10^4 sequences)	local	higher	held-out $\log p/\text{seq}$
joint EM from random init	2.566	3.831	−51.18
bottom-up, frequency-rank clusters	2.598	3.833	−51.45
spectral + soft hand-off + EM (10 sweeps)	2.227	3.802	−48.23
spectral + soft hand-off + EM (50 sweeps)	2.162	3.807	−47.75
stagewise + vertical re-clustering	2.240	3.802	−48.34
transformer, 48k steps	2.022	3.772	—
true grammar	1.599	3.553	−41.21

Table 4: Estimator variants on Corpus A [H].

7 Instruments for detecting hierarchy

- **V6, leaf pairs** [H] *measured*. The test is a parametric bootstrap of empirical mutual information against datasets sampled from the fitted model.
 - *Planted positive*: copying leaf 7 into leaf 8 with probability 0.3 gives $z = +220$, against +6.5 elsewhere; even the true grammar is refuted there ($z = +390$).
 - *Limitation*: on Corpus A, pairwise leaf MI across level-2 and root boundaries is about 10^{-4} even under the true grammar, while the plug-in bias floor is about 0.055. So V6 has no power above level 1.
- **V7, blocks** [H] *measured*. Each block is decoded by its own inside vector, and adjacent blocks are compared under the disjointness invariant.
 - Decoded with the true grammar, blocks sharing a parent carry about 1 nat, and others 0.002–0.014.
 - The fitted model captures 79%, 1.4% and 3.9% of this at levels 1–3.
- **β_1 of the block complex** [H] *measured*. A tree predicts $\beta_1 = 0$ at every threshold.
 - Corpus B, unplanted: 0 at all 25 thresholds.
 - Two planted block couplings: $\beta_1 = 1$ at 5 thresholds, with the null at 0. A single coupling never produces a cycle.
 - β_1 is computed on a Rips complex, so it is a calibrated statistic and not an approximation to Čech persistence (Theorem 12.20).

8 The regime determines learnability

	level 1	level 2	level 3	level 4
Corpus A: observations per parameter	1.4	0.31	0.15	0.08
Corpus B: observations per parameter	1389	1563	781	391

Table 5: Identifiability regime [H]. Levels fail at 0.15 observations per parameter and succeed at 781.

On Corpus B at 200 sweeps, the level-3 block dependence captured rises from 0.0854 to 0.1827, and its discrepancy falls from $z = +20.2$ to +5.4. The earlier failure was therefore an unestimable

regime, then undertraining. Merges at levels 1–3 cost real conditional MI; the root alphabet is redundant, with $8 \rightarrow 2$ improving held-out likelihood by 0.004.

9 Closing the gap: basin selection and split–merge

start	held-out $\log p/\text{seq}$	local / higher
exact truth ($\varepsilon = 0$)	−15.324	0.3322 / 1.5833
truth + 50% / 80% / 90% uniform	−15.325 / −15.325 / −15.330	—
truth + 95% / 99% uniform	−15.775 / −18.113	—
uniform	−20.706	0.5091 / 2.0791
spectral init (ours)	−17.511	0.3952 / 1.7937

Table 6: Diagnosis: EM started at the truth [H] measured. The truth is a fixed point with a very wide basin.

A hard partition of observed child pairs cannot represent a grammar in which pairs are shared across parents (4 of 16 at level 1).

- Sharpening the clustering hurts: sibling context gives −18.06.
- Blurring it helps: blending with uniform at weight 0.9 gives −15.87.

Construction 9.1 (Split–merge). [H] For each level:

1. duplicate the symbols with a perturbation;
2. run EM;
3. greedily merge back by Bregman information over *all* $2V$ symbols, so merges can recombine across the split;
4. accept only if held-out likelihood improves.

stage	held-out $\log p/\text{seq}$	merge cost (nats/seq), levels 1–4
warmup, 40 EM sweeps	−17.187	—
round 1	−15.491	0.160, 0.113, 0.008, 0.003
round 2	−15.325	0.042, 0.011, 0.007, 0.005
round 3	−15.3253	0.052, 0.020, 0.007, 0.004
true grammar	−15.3243	—

Table 7: Split–merge on Corpus B [H] measured: 131s, local/higher 0.3322/1.5835 against a true 0.3322/1.5833.

The recovered model passes all four instruments:

- V3 matches the true column at every boundary.
- V6 flags 0 of 18 pairs, with largest $|z| = 1.7$.
- V7 at level 3 reads 0.5863/0.5882 against a true 0.6140.
- β_1 against the fitted null is 0 of 25, where the local-optimum fit gave 6 of 25.

10 Why clustering is necessary at scale

A dense inside pass costs about $2NV^3$. At $N = 10^6$ and $V = 1024$ a sweep costs about 4.3×10^{15} . Compression to $K = 16$ gives about 8×10^{11} over 50 sweeps, a speedup of $(V/K)^3 = 262,144$. For m -ary grammars the order rises to $m + 1$, so compression buys $\log_K V$ orders of headroom.

Compression is not free. Merge costs are level-dependent and must be measured, not assumed; V4a supplies the measurement. At $L = 16$ the root is observed only about 15 times per 10^6 tokens, so compression removes the compute barrier but not the identifiability barrier.

11 O6 baseline: where split–merge breaks

We built O6 as a benchmark (`benchmarks/o6`). It consists of:

- a seeded RHM generator that exports the true latent symbol of every test node;
- a scorer that reports the held-out likelihood gap and, at each level, the NMI of the model’s MAP symbol against the truth, beside the true grammar’s own (Bayes-ceiling) recovery;
- tiers O6-S ($V=64$, $L=8$), O6-M (256, 12) and O6-L (1024, 16).

We ran our gradient-free baseline up a ladder on 2 CPUs with 7 GB of memory. The baseline is dense inside–outside EM with split–merge.

rung	protocol	test gap (nats/seq)	top-level recovery (model / ceiling)	s/sweep
$V=16, L=4$	full	0.020	0.55/0.88	1.4
$V=32, L=4$	short	0.150 (0.237 with $4\times$ data)	0.69/0.98	8.3
$V=64, L=4$	short	1.488	0.72/0.99	40
$V=128, L=4$	—	out of memory once split	—	180
$V=8, L=6$	full	0.005	0.39/0.74	2.0
$V=8, L=8$	full	2.11	0.41/0.77	0.8
$V=8, L=10$	short	60.7	0.03/0.60	0.8

Findings:

- *Likelihood does not certify structure.* Near-truth likelihood, with gaps of 0.02 and 0.005, coexists with top-level recovery far below the ceiling, because the top of the hierarchy carries little likelihood. The benchmark therefore scores structure first. *Correction:* the root level is not identifiable at all (Proposition 40.2), so the root parts of these shortfalls are not recoverable by any method, and structure is scored on levels $1 \dots L - 1$ (Section 40.3).
- *Where it breaks.* Optimisation breaks at $V \approx 64$: four times more data did not help at $V=32$. Memory breaks at $V=128$. Top-level structure fails from $L \approx 6-8$, and the full protocol repairs likelihood but not the top two levels.
- *Scaling.* Time per sweep grows as $V^{2.17}$. O6-S is beyond this hardware, at an estimated week or more.
- *O6-L needs a sparse or structured parameterisation, whatever the optimiser.* Dense tables have $1.8 \cdot 10^{10}$ parameters and about 10^{-4} observations per top-level parameter at $N = 10^5$. The true grammar has 3072 rules per level.

The baseline, the generator and the scorer are released so that other methods can be run on the same footing.

12 Theorems and constructions used [H]

Theorem 12.1 (Inside–outside; Baker, Lari–Young). *used* For a PCFG on a fixed tree, the inside values β and outside values α give exact marginals: the posterior at a node is $\propto \alpha\beta$.

Theorem 12.2 (EM ascent; Dempster–Laird–Rubin). *used* Each EM iteration does not decrease the observed-data log-likelihood.

Theorem 12.3 (Convergence; Wu). *used* Under regularity conditions, EM converges to a stationary point, and the limit depends on the initialisation.

Theorem 12.4 (Extension; Vorob’ev). *used* Every consistent family of marginals on a hypergraph extends to a global distribution if and only if the hypergraph is acyclic. A fixed tree is acyclic.

Theorem 12.5 (Independent pullback; Simpson Thm 6.4, Prop. 8.5). *used* A presheaf on standard Borel probability spaces is an atomic sheaf iff it maps independent squares to pullbacks. The independent pullback is the conditional product.

Definition 12.6 (Supports; Simpson Def. 7.1). A support of a random variable is its terminal factorisation through a sample space.

Theorem 12.7 (Conditional products; Fritz §12). *used* In a Markov category with conditionals, conditional products exist and satisfy the Dawid–Studený axioms. FinStoch is such a category.

Theorem 12.8 (Sufficient statistics; Fritz Def. 14.3, Thm 14.5). *used* A statistic is a deterministic morphism, and the abstract Fisher–Neyman factorisation holds. The soft hand-off is a statistic, and V2 verifies its sufficiency.

Theorem 12.9 (Minimal sufficiency; Fritz Def. 16.2, Bahadur). *used* A minimal sufficient statistic is least informative among the sufficient ones; a complete sufficient statistic is minimal.

Remark 12.10 (Fritz and cohomology). Fritz’s framework supplies semantics for conditional independence and sufficiency, but no cohomology. It does not bear on a map from information cohomology into a persistence module.

Construction 12.11 (Independence join; Wright §7.1). For tables agreeing on a shared marginal, $(p \bowtie q)_{ijk} = p_{ij}q_{jk}/r_j$ is the maximum-entropy weak pullback.

Construction 12.12 (Merging as convex combination; Wright §7.2). Merged rows are mixtures, averaged in the Giry monad.

Theorem 12.13 (Bregman information; Bregman, Banerjee et al.). *used* For KL, the optimal cluster representative is the mean, and the clustering loss equals the mutual information between the data and the cluster assignment.

Construction 12.14 (Merge criterion). Merging symbols costs

$$\sum_s n_s D_{\text{KL}}(R_s \| R_{\text{class}}) = N I(\text{children}; B \mid q(B)).$$

The cost is zero iff the child rows are equal, which is strong (downward) lumpability (Kemeny–Snell). Observational losslessness is strictly broader: upward-equal merges lose nothing at positive cost (Proposition 27.5).

Construction 12.15 (Agglomerative merging; Slonim–Tishby). Greedy least-loss merging is the agglomerative information bottleneck; split–merge (Petrov et al.) is its grammar analogue.

Theorem 12.16 (Ostrowski). *used Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to the real absolute value or a p -adic one, and the resulting completions are pairwise inequivalent.*

Construction 12.17 (Bias correction; Miller–Madow). Plug-in MI on a $K \times L$ table is biased upward by about $(K - 1)(L - 1)/2N$.

Construction 12.18 (β_1). $\beta_1 = \dim \ker \partial_1 - \text{rank } \partial_2$, computed over $\text{GF}(2)$ on the 2-skeleton.

Proposition 12.19 (The lattice quasi-morphism does not exist). *[H, Prop. 1] proved, closed Let $\rho : \mathbb{Q} \rightarrow L$ be order-preserving for the real order and constant on each p -adic ball. Then ρ is constant.*

Proof. A p -adic ball is dense in $\mathbb{R}$. Choose $x < y < z$ with x, z in one ball B_1 and y in another ball B_2 . Monotonicity gives $c_1 \leq c_2 \leq c_1$. $\square$

Theorem 12.20 (Bregman TDA; Edelsbrunner–Wagner). *used KL restricted to the simplex is of Legendre type. Čech and Delaunay complexes of Bregman balls have the homotopy type of the union (Contractibility Lemma). The Rips filtration can be arbitrarily far from Čech (Result 2). Measured [H]: maximum β_1 is 20–21 for Čech and 0 for Rips.*

Refuted, do not retry [H]:

- sibling-only clustering context (-18.06);
- deterministic annealing ($+0.22$ nats, independent of T_0);
- stagewise plus vertical re-clustering (-48.34 against -47.75);
- posterior entropy as a confidence criterion: posterior entropies are 0.198 – 1.003 at levels capturing under 4% , whereas $I(\text{parent}; \text{children})$ reads $10.1, 8.1, 4.5, 5.3$ against a true 8.9 – 9.0 ;
- curriculum transfer: the control gives $2.043/3.789$ against $2.085/3.803$.

Part II

Holonomy and the separation of sites [0]

13 The holonomy obstruction

Definition 13.1 (Operation system). [O, Def. 1] (Γ, A, g) consists of a finite connected graph $\Gamma = (V, E)$, an abelian group A , and $g : E \rightarrow A$ with $g_{ba} = -g_{ab}$.

Definition 13.2 (Global field). [O, Def. 2] A map $c : V \rightarrow A$ with $c(b) - c(a) = g_{ab}$ for every edge.

Theorem 13.3 (Holonomy obstruction). *[O, Thm 3] proved, computed A global field exists iff the holonomy $h(L) = \sum_{e \in L} \pm g_e$ vanishes on every cycle. The obstruction is $[g] \in H^1(\Gamma, A) \cong A^{\beta_1}$, with $\beta_1 = |E| - |V| + 1$.*

Proof. Integrate g along a spanning tree. A non-tree edge is consistent iff its fundamental cycle has zero holonomy, and the fundamental cycles form a basis of the cycle space. $\square$

Verified against brute force on a room with $|V| = 4$, $|E| = 5$, $\beta_1 = 2$, $A = \mathbb{Z}/4096$. The holonomy is $(0, 0)$ when a field exists, and $(0, 1)$ and $(2, 1)$ when none does [O]. The same computation in [S] gives the same classes, up to a unimodular change of cycle basis.

Theorem 13.4 (Holonomy as a connecting class). *[O, Thm 4] proved Consider the exponential sequence $0 \rightarrow \mathbb{Z}_M \rightarrow \mathbb{R}_M \rightarrow \mathcal{S}_M^1 \rightarrow 0$ on a manifold M with boundary. Since $\mathbb{R}_M$ is soft, the connecting map $\delta_M : H^0(M, \mathcal{S}_M^1) \rightarrow H^1(M, \mathbb{Z})$ is an isomorphism modulo global real lifts. Naturality under $r : \partial M \hookrightarrow M$ gives $r^*(\delta_M[g]) = \delta_{\partial M}(r^*[g]) = \text{Hol}(\alpha, \partial M)$.*

Remark 13.5 (What Theorem 13.4 does not say). [O, Rem. 5] $\delta_M[g]$ lies in $H^1(M, \mathbb{Z})$, not in $\text{Ext}^1(\mathcal{S}_M^1, \mathbb{Z}_M)$. It is not the AMB obstruction: the “support reduction” $\mathcal{F}_{\text{prob}} \rightarrow \mathbb{Z}$ is not additive, so it is not a morphism of sheaves.

Proposition 13.6 (T0 does not apply to operation systems). *[O, Prop. 8] An operation system has no sample; a witness would be a global field, which is the object in question.*

Proposition 13.7 (Vorob’ev does not apply). *[O, Prop. 9] Here the relevant structure is the cycle space of Γ : $\beta_1 = 0$ makes H^1 trivial, while $\beta_1 > 0$ leaves it free.*

Proposition 13.8 (Possibilistic fragility does not apply). *[O, Prop. 10] computed The AMB obstruction is a functional of the support, so smoothing kills it. Measured: γ falls from 9/9 to 0/27 at $\varepsilon = 10^{-9}$, while $\text{CF} = 1 - 3\varepsilon$ survives to $\varepsilon = 1/3$. The holonomy class is a sum of the operations themselves and has no support to blur.*

attempted coefficients	why it failed	verified
p -adic lattice $\rho : \mathbb{Q} \rightarrow L$	order-preserving and constant on balls implies constant	proof
probability simplex Δ^{V-1}	a convex set with no inverses, so no δ	—
support indicator $\omega = \int_{\text{supp}} \mathbf{1} dP$	identically 1 on every face of every family	computed
rounding $\mathbb{R} \rightarrow \mathbb{Z}$	not additive, so not a module map	—
$\mathbb{Z}/n$ on edges	abelian, δ exists, the class varies	computed

Table 8: The coefficient group is what mattered [O, §3].

Remark 13.9 (β_1 changes role). [O, Rem. 11] In the marginal setting, β_1 of the incidence complex is a function of the cover alone. It equals 4 for the obstructed 3×3 parity family and for both controls, so it cannot map to γ . In an operation system, β_1 is the rank of the obstruction group.

14 The two coboundary operators

Definition 14.1 (Conditioning action). [O, Def. 12] $(S_0 \cdot F)(P) = \sum_j P(S_0 = v_j) F(P \mid S_0 = v_j)$.

Definition 14.2 (The two coboundaries). [O, Def. 13]

$$(\delta F)(S_0; \dots; S_k) = S_0 \cdot F(S_1; \dots; S_k) + \sum_{i=1}^k (-1)^i F(\dots; S_{i-1} S_i; \dots) + (-1)^{k+1} F(S_0; \dots; S_{k-1}).$$

δ_t is the same expression with the first term replaced by $F(S_1; \dots; S_k)$. Both operators square to zero.

Theorem 14.3 (Baudot–Bennequin). *[O, Thm 14]; [5] used; verified numerically [S]*

1. $\delta c = 0$, so $H^0 \cong \mathbb{R}$.

2. $\delta H = 0$ (the chain rule), and under locality hypotheses $H^1(\mathcal{F}, \delta)$ is one-dimensional, generated by H .

3. $\delta_t H = I$, and $(\delta - \delta_t)H = (S_0 - 1) \cdot H(S_1) = -I$.

[S] verifies numerically: $\delta H = 0$ and $\delta_t H = I$ to 10^{-10} ; $(XY) \cdot F = X \cdot (Y \cdot F)$; $\delta^2 = 0$; and Rényi-2 entropy is not a cocycle.

Remark 14.4 (Easy to get backwards). [O, Rem. 15] $I_2 = \delta_t H$, not δH . Higher I_k are coboundaries for δ or for δ_t according as k is odd or even. No regularity or symmetry condition separates H from I .

15 Non-unification and vanishing

Theorem 15.1 (Non-unification). [O, Thm 16] *proved* No morphism of coefficient systems carries a nonzero AMB class in $\mathcal{F}_{\mathbb{Z}}$ to a nonzero class in $\mathcal{F}_{\alpha}$.

Remark 15.2 (Holonomy is not the bridge). [O, Rem. 17] The holonomy class shares its coefficient type with AMB and its locality with Vigneaux, but it is a cocycle of transformations, not of sections or information functions. So it is a third object.

Theorem 15.3 (Vanishing on the trivially-acting complex). [O, Thm 27] *proved, computed* Let $\mathcal{S}$ be an information structure whose observables form a monoid under joint, with an absorbing element, and let A be an abelian group with trivial action. Then $H^n(\mathcal{S}; A) = 0$ for $n > 0$ and $H^0 = A$.

Proof. With trivial action the cochains are constants, and the complex is the bar complex of a monoid with an absorbing element, whose classifying space is contractible. $\square$

Verified for the join-semilattices of two and three binary variables: $H^0 = \mathbb{Z}$ and $H^1 = H^2 = 0$ over $\mathbb{Q}$, and over $\mathbb{Z}$ by Smith normal form, with no torsion.

Corollary 15.4 (Extensions do not help). [O, Cor. 28] *proved, closed* The connecting map of any extension of $\mathcal{F}_{\alpha}$ by a trivially-acting $\mathcal{F}_{\mathbb{Z}}$ lands in $H^{k+1}(\mathcal{F}_{\mathbb{Z}}) = 0$.

Remark 15.5 (The dichotomy). [O, Rem. 29] Either the coefficients act trivially, and Theorem 15.3 applies; or they do not, and the AMB class does not live there. An information structure presupposes that every set of variables has a joint. A measurement scenario presupposes that some do not.

Remark 15.6 (Scope of Theorem 15.3). [S] The theorem concerns the *trivially-acting* complex. Under the non-trivial conditioning action, H^1 is one-dimensional (Theorem 14.3); under the α -deformed action it is generated by Tsallis entropy (Section 20). $H^{\geq 2}$ is open for both actions (Baudot–Bennequin, Problem 1). The finite linear algebra of `h2_rational.py` does *not* transfer to the deformed action: cochains are then functions of P , and the spaces are infinite-dimensional without a restricted function class.

Problem 15.7 (Site correspondence). [O, Prob. 21] *open* Does an AMB scenario determine an information structure in Vigneaux’s sense?

Problem 15.8 (Cover preservation). [O, Prob. 22] *open* Is the correspondence a morphism of sites? This depends on Problem 21.

Problem 15.9 (Coefficient comparison). [O, Prob. 23] blocked A morphism $\Psi : \mathcal{F}_{\mathbb{Z}} \rightarrow \mathcal{F}_{\alpha}$ intertwining the module structures has an undefined left-hand side.

Problem 15.10 (Extension formulation). [O, Prob. 24] settled negatively by Cor. 28

Problem 15.11 (Parity criterion). [O, Prob. 25] A mod- p family is contradictory iff every variable lies in an even number of contexts and $\sum_C s_C \not\equiv 0 \pmod{p}$. Brute force caught two failed examples. [S] reproduces the mod-3 grid exactly (Section 16).

Problem 15.12 (Holonomy from estimated operations). [O, Prob. 26] answered in part by §17

Closed by theorem, not to be reopened [O]:

- the p -adic bridge (Prop. 12.19, Theorem 12.16);
- branch cohomology with simplex coefficients: a rooted tree poset has an extremal object, so $H^{k>0} = 0$ for every coefficient system;
- a map from β_1 of the incidence complex to γ ;
- a class for $I(X; Y \mid Z)$ obtained by regularity restrictions;
- the support-reduction identification of boundary holonomy with γ .

Part III

Results of the present work [S]

16 Engine and verification of the outcome layer

The `amb.vigneaux` package implements three things:

- the outcome layer: support presheaves, logical and strong contextuality, and γ decided exactly over $\mathbb{Z}$ (by integer diagonalisation), $\mathbb{Q}$ and $\mathbb{Z}/p$;
- the contextual fraction by LP, with its dual Bell inequality;
- the probability layer: conditioning action, δ , δ_t and co-information.

It reproduces the standard values computed:

- PR box and GHZ: strongly contextual, $\text{CF} = 1$, $\gamma \neq 0$ at every section.
- Tsirelson box: $\text{CF} = \sqrt{2} - 1$; noisy Tsirelson: $\text{CF} = \max(0, \sqrt{2}v - 1)$; noisy PR box: $\text{CF} = \max(0, 2v - 1)$.
- Hardy model: logically contextual with $\text{CF} = 1/6$, and a *false negative* of $\mathbb{Z}$ -cohomology. The returned witness family uses a -1 coefficient and is verified compatible, which agrees with the note in `glue.py`.
- Mod-3 grid (Problem 15.11): $\gamma \neq 0$ at 9/9 over $\mathbb{Z}$ and $\mathbb{Z}/3$; 0/9 over $\mathbb{Q}$, $\mathbb{Z}/2$ and $\mathbb{Z}/5$. Both controls give 0; $\text{CF} = 1, 0, 0$; the fragility of Proposition 13.8 is reproduced exactly.

17 Problem 26: holonomy from estimated operations

Operations are observed through a *design*: integer functionals $a_k \in \mathbb{Z}^E$, with observations $y_k = a_k \cdot g + \epsilon_k$ and a noise law independent of g .

Proposition 17.1 (Ring-theoretic identifiability). *proved, computed* Let $A \in \{\mathbb{R}, \mathbb{R}/\mathbb{Z}, \mathbb{Z}/n\}$ with $R = \mathbb{Q}, \mathbb{Z}, \mathbb{Z}/n$ respectively. The holonomy $h(L) = \ell_L \cdot g$ is identifiable iff $\ell_L \in \text{span}_R\{a_k\}$.

Proof. If $\ell_L = \sum \lambda_k a_k$, then $h = \sum \lambda_k (a_k \cdot g)$. Conversely, A is an injective cogenerator over R . If $\ell_L \notin \text{span}$, there is g' with $a_k \cdot g' = 0$ for all k and $\ell_L \cdot g' \neq 0$, so g and $g + g'$ are observationally equivalent. $\square$

Observing only each loop traversed twice identifies h over $\mathbb{Q}$ and over $\mathbb{Z}/\text{odd}$, but not over $\mathbb{R}/\mathbb{Z}$ or $\mathbb{Z}/\text{even}$. Witness pairs are constructed in the tests.

Corollary 17.2 (T0 for operations). *proved*

- A design that reads each vertex through one route (a spanning-tree section) spans no cycle, so no loop is identifiable.
- An estimator that fits a field and sets $\hat{g} = \delta \hat{c}$ returns $\hat{h} \equiv 0$. Measured: 10^{-17} on a world with $h = 0.2$, while edge-local estimates give 0.205.

Proposition 17.3 (Calibrated test and sample cost). *measured; large-sample* The Wald statistic $T = \hat{h}^\top \Sigma^{-1} \hat{h}$, with $\Sigma = C \text{diag}(\text{se}^2) C^\top$, is $\chi^2_{\beta_1}$ under $[g] = 0$ when the circular-mean standard error is the delta-method one, $\text{var} = (1 - \rho_2)/(2m\rho_1^2)$.

- Calibration: 4.2–6.4% false positives at nominal 5% for $\sigma \in [0.05, 0.35]$ turn ($m = 30, 500$ replicates). The naive standard error reaches 74%.
- Power follows the non-central χ^2 .
- Readings per edge for 80% power:

$$N_{80} = \frac{|L|}{h^2} v(\sigma) c(\alpha, \text{power}, \beta_1), \quad v(\sigma) = \frac{(1 - e^{-8\pi^2\sigma^2})e^{4\pi^2\sigma^2}}{8\pi^2}.$$

This assumes equal per-edge noise and edge-disjoint loops. Measured power at the predicted N_{80} is 0.77–0.86.

- The Cramér–Rao bound equals $e^{4\pi^2\sigma^2}/(8\pi^2)$ asymptotically, and the circular mean is 89–100% efficient, so the exponent is intrinsic.

Proposition 17.4 (Discrete groups). *proved; measured looseness* For $\mathbb{Z}/n$ with a symmetric channel, the mode estimator is exact with probability at least $1 - |E|(|A| - 1)(1 - (\sqrt{p_t} - \sqrt{p_w})^2)^m$. The bound holds but is about 10× loose (0.018 against a measured 0.001 at $m = 80$).

18 From operations to outcomes, around the probability layer

Theorem 18.1 (The operation-system model e_g). *proved, computed* For $(\Gamma, \mathbb{Z}/n, g)$, let the measurements be the vertex colours, the contexts the edges, and let e_{ab} be uniform on $\{(x, x + g_{ab})\}$. Then:

1. e_g is strongly contextual iff $[g] \neq 0$;

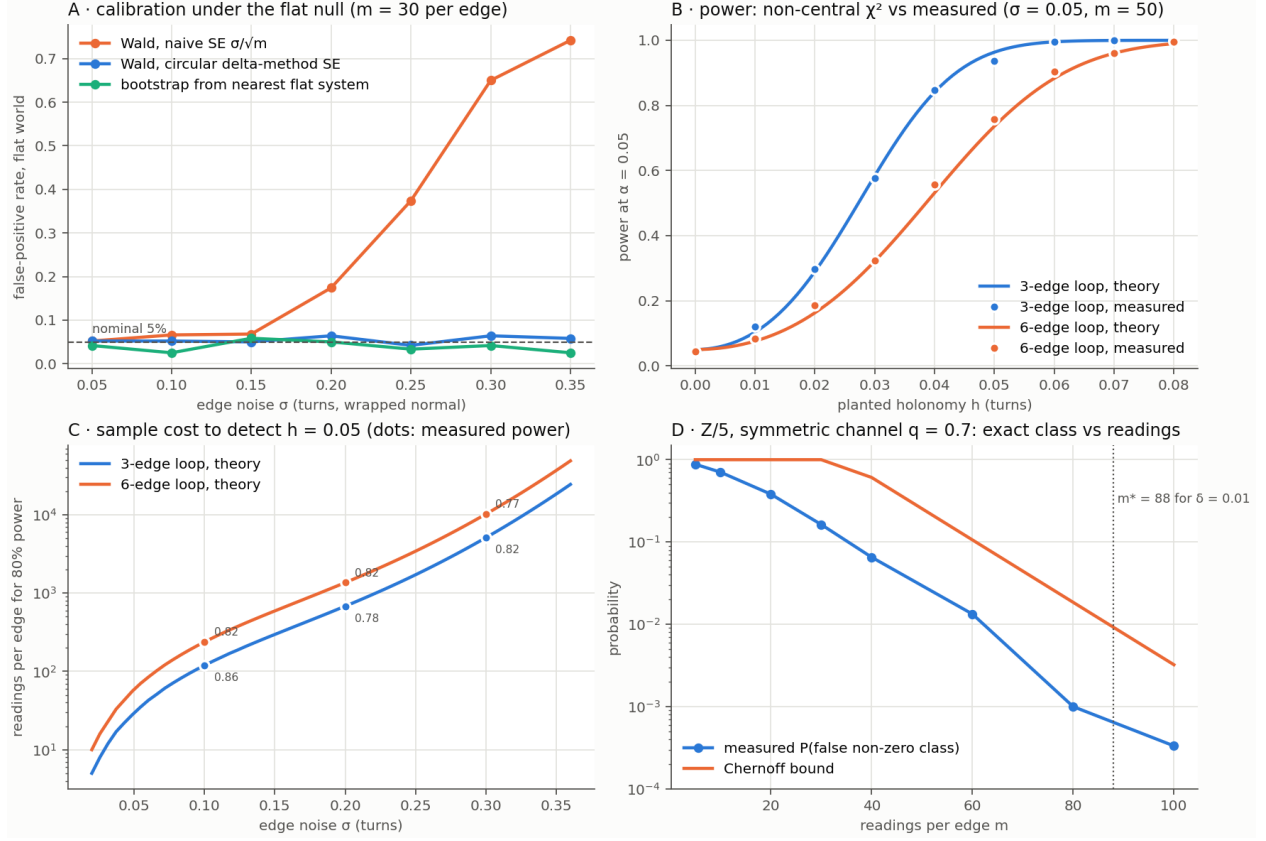


Figure 1: Problem 26 [S]. (A) calibration; (B) power against theory; (C) sample cost; (D) discrete class error against the Chernoff bound.

2. for $[g] \neq 0$, $\gamma(s) \neq 0$ for every section over every nonzero coefficient ring;
3. e_g is an instance of an all-versus-nothing model over $\mathbb{Z}/n$ [3].

Proof. (1) A global section is exactly a global field.

(2) Take a compatible R -family with $r_{C_0} = s$. Its restrictions to vertices are functions $f_v : \mathbb{Z}/n \rightarrow R$. Each support is the graph of a bijection, so compatibility forces $f_b(y) = f_a(y - g_{ab})$. Propagating the delta function $f = \delta_{x_0}$ from C_0 around a loop with $h \neq 0$ gives $\delta_{x_0} = \delta_{x_0+h}$, which is impossible when $R \neq 0$. $\square$

Proposition 18.2 (Layer 2 is blind to $[g]$). *proved In e_g , every edge has $I(c_a; c_b) = \log n$ whatever g is. More generally, any per-context functional invariant under relabelling outcomes (every Shannon or Tsallis functional) is unchanged by post-composing an edge with a permutation.*

The ring-invariance in Theorem 18.1(2) contrasts with the mod-3 grid, where γ vanishes over $\mathbb{Q}$. Ring dependence of γ is a property of the family, not of γ .

19 Lossy operations

Each edge carries a doubly stochastic channel K . Colours are uniform, so the Bayesian inverse is $K^\top$. The loop operator M_L is the product around the loop, and the edge-context model e_K has law $\frac{1}{n}K_{ab}$ on edge (a, b) .

Proposition 19.1 (Character eigenvalues). *proved* For covariant channels $C_{\mu_e} S_{g_e}$ on a finite abelian A , every character χ is an eigenvector of M_L with $\lambda_\chi = (\prod_e \hat{\mu}_e(\chi)) \chi(h)$. For the noisy shift $K = (1 - \varepsilon)S_g + \varepsilon U$, $M_L = (1 - \varepsilon)^{|L|} S_h + [1 - (1 - \varepsilon)^{|L|}] U$ and $\lambda_\chi = (1 - \varepsilon)^{|L|} \chi(h)$.

Corollary 19.2. *The labelled eigenvalue is base-point free, and its phase is $\chi(h)$ when every $\hat{\mu}_e(\chi)$ is real and positive. The unlabelled multiset is base-point free for any channels, since AB and BA share their nonzero spectrum, but on $\mathbb{Z}/p$ it sees only the order of h . Asymmetric noise adds its own phase: the readout is 2.285 instead of 2.*

Proposition 19.3 (Information closed forms). *proved* $I(c_a; c_b) = \log_2 n + (1 - \varepsilon + \frac{\varepsilon}{n}) \log_2(1 - \varepsilon + \frac{\varepsilon}{n}) + (n - 1) \frac{\varepsilon}{n} \log_2 \frac{\varepsilon}{n}$. The loop-composite information is the same with $\varepsilon_L = 1 - (1 - \varepsilon)^{|L|}$. Both are independent of h .

Theorem 19.4 (Contextual fraction on a cycle). *proved* On a cycle of length L with noisy shifts and holonomy $h \neq 0$ in a finite abelian A ,

$$\text{CF}(e_K) = \max\left(0, 1 - \frac{L\varepsilon}{2}\right),$$

independent of $|A|$ and of h .

Proof. Upper bound on NCF. Weight each cell by its discrepancy $d = y - x - g$: $w(0) = 0$, $w(-h) = 1$, and $w = \frac{1}{2}$ otherwise. Discrepancies sum to $-h \neq 0$, so every colouring has weight at least 1. The dual value is $L \frac{\varepsilon}{n} (1 + \frac{n-2}{2}) = \frac{L\varepsilon}{2}$.

Lower bound (packing). Take colourings with one violation of discrepancy $-h$ (mass ε/n per edge), and colourings with two violations $(a, -h - a)$ (mass $\varepsilon/(2n(L-1))$ per ordered edge pair), spread over translations. These fill every off-graph class exactly, and the on-graph load $\frac{L\varepsilon}{2} - \varepsilon(1 - \frac{1}{n})$ fits within the capacity $1 - \varepsilon(1 - \frac{1}{n})$ iff $L\varepsilon \leq 2$.

Beyond the threshold. The model is affine in ε and non-contextual at $\varepsilon = 2/L$ and at $\varepsilon = 1$. $\square$

Theorem 19.5 (Any graph, lower bound). *proved* $\text{CF}(e_K) \geq \max(0, 1 - \ell_{\min} \varepsilon / 2)$, where $\ell_{\min}$ is the shortest simple cycle with nonzero holonomy.

Observation 19.6. *measured* Equality holds in 44 configurations: θ -graphs, K_4 , shared edges, shared vertices, disjoint non-flat loops, and $\varepsilon \leq 0.8$. Only $\ell_{\min}$ matters, not the cycle basis. The general packing is open.

Counterexample 19.7 (What does not hold). *computed*

- *Information does not determine CF.* A 3-cycle and a 5-cycle over $\mathbb{Z}/3$ at $\varepsilon = 0.2$ have the same per-edge information, but $\text{CF} = 0.70$ and 0.50 .
- *Non-covariant noise is contextual on its own.* Flat systems reach CF up to 0.24 , and the phase readout of $h = 2$ spreads over $[1.1, 3.1]$.
- *Entropy is not a function of Fourier moduli.* A homometric pair has identical $|\hat{p}|$ but $H = 2.7714$ against 2.7689 bits.

Proposition 19.8 (Orientation for $\mathbb{Z}/3$ and $\mathbb{Z}/5$). *proved, computed* For odd primes, $h \neq -h$ whenever $h \neq 0$, so the labelled phase always detects orientation. Reversing the loop inverts the readout ($1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ on $\mathbb{Z}/5$). The unlabelled multiset and CF do not detect orientation.

- On $\mathbb{Z}/3$ a non-flat holonomy is exactly a chirality, read from the sign of $\text{Im } \lambda_1$.

layer	invariant	response to ε	detects h until
1	labelled eigenvalue of M_L	modulus $(1 - \varepsilon)^{ L }$, phase fixed	$\varepsilon \rightarrow 1$
2	per-edge and loop I	decreases, independent of h	never
3	CF	$1 - \ell_{\min}\varepsilon/2$	$\varepsilon = 2/\ell_{\min}$
3	γ	support becomes full	$\varepsilon = 0$ only

Table 9: The same noise acting on four invariants [S].

- On $\mathbb{Z}/5$ the phase splits into a magnitude class, $\text{Re } \lambda_1/|\lambda_1| = 0.309$ for ± 1 and -0.809 for ± 2 , and an orientation.
- The magnitude class is meaningful only with fixed state labels, since $x \mapsto 2x$ carries $h = 1$ to $h = 2$.

20 The support dial: Tsallis entropies

Proposition 20.1. *computed; the cocycle property is Vigneaux's* Let $S_\alpha = (1 - \sum p^\alpha)/(\alpha - 1)$ with the deformed action $(X \cdot_\alpha F)(P) = \sum_x P(X=x)^\alpha F(P \mid X=x)$.

- S_α is a δ_α -cocycle (defect below 3×10^{-15}), but not a cocycle for Shannon's action (defect up to 1.8).
- $S_0 = |\text{supp}| - 1$, and $S_\alpha \rightarrow H$ as $\alpha \rightarrow 1$.
- Mass ϵ added to an empty cell changes S_0 by 1, and S_α by $\sim \epsilon^\alpha$.
- $S(X) + S(Y) - S(XY)$ reaches -1.71 at $\alpha = 0.5$.

α therefore interpolates between AMB's possibilistic lens ($\alpha = 0$) and Shannon ($\alpha = 1$). Every S_α is symmetric, so Proposition 18.2 still holds.

21 Support strata: where each lens can see anything

An empirical model $e = (e_C)_C$ lies in a product of simplices $\prod_C \Delta(O^C)$, intersected with the no-signalling polytope. Its *support* is $\sigma(e) = (\text{supp } e_C)_C$. The *stratum* of e is the set of models with the same support, that is, the relative interior of a face.

- The *interior* is full support in every context.
- The *vertices* are one outcome per context.

Proposition 21.1 (Support stratification). [S]proved

1. Face functions. The support presheaf $\mathcal{S}_e$, its global sections, logical and strong contextuality, and γ over every ring depend on e only through $\sigma(e)$. Strong contextuality is inherited by smaller supports: if $\sigma(e) \subseteq \sigma(e')$ componentwise and e' is strongly contextual, so is e .
2. Interior. With full support, every local section extends to a global section of $\mathcal{S}_e$. So e is not logically contextual and $\gamma(s) = 0$ for every s , over every ring. The contextual fraction can still be positive.
3. Vertices. If each $e_C = \delta_{s_C}$ and e is no-signalling on a cover of the measurements, the s_C glue to a global assignment g with $e = \delta_g|_C$. Then:

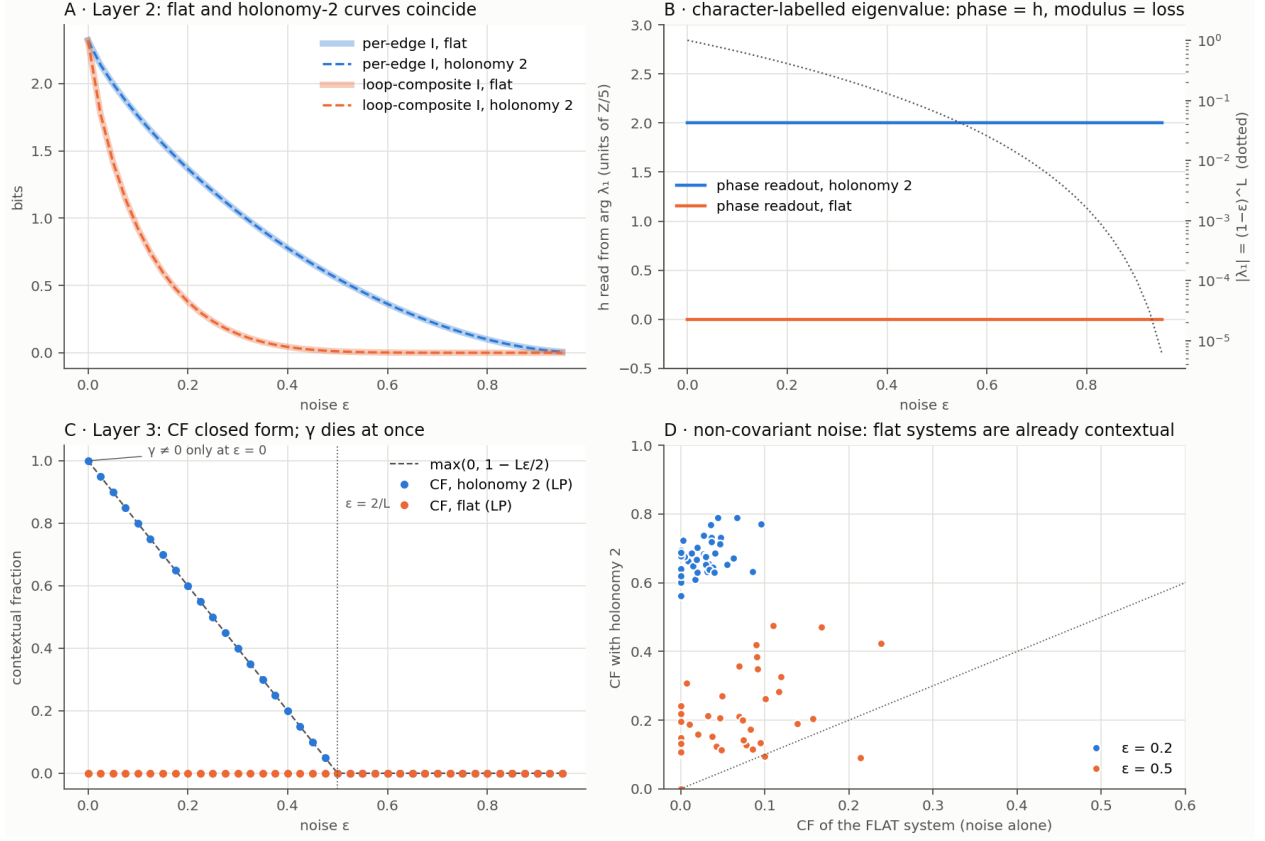


Figure 2: Noisy operations on a 4-cycle over $\mathbb{Z}/5$, with $h = 2$ against a flat system [S].

- e is non-contextual, $CF = 0$ and $\gamma = 0$;
- every Shannon and Tsallis functional of every context vanishes, together with every mutual and conditional information, since each is a combination of entropies of point masses.

4. The probability lens varies within strata. CF is not a face function. It is convex (Abramsky–Barbosa–Mansfield), hence continuous on the relative interior of the model polytope. S_α for $\alpha > 0$ and H are continuous on closed simplices. $S_0 = |\text{supp}| - 1$ is constant on strata and jumps exactly across them.

Proof. 1. $\mathcal{S}_e(U)$ is the set of sections over U whose restrictions to each context lie in $\text{supp } e_C$. Every listed invariant is defined from $\mathcal{S}_e$. If $\sigma(e) \subseteq \sigma(e')$ then $\mathcal{S}_e \subseteq \mathcal{S}_{e'}$, so a global section of $\mathcal{S}_e$ is one of $\mathcal{S}_{e'}$.

2. Extend $s \in O^C$ arbitrarily to $g \in O^X$. Every restriction of g lies in a full support, so $g \in \mathcal{S}_e(X)$. A section that extends to a global section has trivial obstruction and is no logical witness.
3. No-signalling gives $\delta_{s_C}|_{CNC'} = \delta_{s_{C'}}|_{CNC'}$, so $s_C|_{CNC'} = s_{C'}|_{CNC'}$. Because the contexts cover X , the s_C glue to a unique g , and δ_g is a global law. $H(\delta) = S_\alpha(\delta) = 0$ for all α .
4. The Tsirelson box and white noise lie in the same (interior) stratum, with $CF = 0.414$ and 0. Continuity of S_α for $\alpha > 0$ and the jump of S_0 are immediate from the formulas.

□

Corollary 21.2. *proved* A nonzero γ and logical contextuality occur only on intermediate strata: neither full support nor single outcomes.

- The PR box has no global section and 8 cohomological witnesses.
- The Hardy model has one logical witness, missed by γ over $\mathbb{Z}$.

Contextuality in the interior is visible only to the probability lens (CF); the possibilistic lens is blind to it. S_0 reads the stratum and $S_{\alpha>0}$ reads the position within it, which completes the support dial of Section 20.

Proposition 21.3 (Two-outcome supports are all-or-nothing). *[S]proved* Consider a graph scenario with binary measurements and pairwise contexts. Let e be no-signalling with $|\text{supp } e_C| = 2$ for every context.

1. Every measurement is either pinned (deterministic) or free (both values occur). An edge between two free measurements has a bijection support $\{y = x + c_C\}$ with $c_C \in \mathbb{Z}/2$. An edge between a pinned and a free measurement has a product support. No edge joins two pinned measurements.
2. e is contextual if and only if the $\mathbb{Z}/2$ connection c on the free subgraph has a non-flat cycle. It is then strongly contextual, with $\text{CF} = 1$ and $\gamma \neq 0$. Otherwise every local section extends and $\gamma = 0$. Logical-but-not-strong contextuality, as in Hardy, needs a support of size at least 3.
3. On a component with a non-flat cycle, no-signalling forces every free marginal to be $\frac{1}{2}$, so the stratum is a single point there. On a flat component the stratum is a one-parameter family $q \in (0, 1)$ with $\text{CF} = 0$.
4. Per context, $H = I = h(q)$ and $S_0 = 1$. The non-flat point and the flat point $q = \frac{1}{2}$ have identical per-context Shannon and Tsallis profiles, yet $\text{CF} = 1$ and 0.

Proof. 1. A size-2 subset of $\{0, 1\}^2$ either fixes one coordinate or is the graph of a bijection. No-signalling makes pinned/free a property of the measurement.

2. Global sections are the pinned values together with a flat section of the $\mathbb{Z}/2$ torsor on the free subgraph, which exists if and only if every cycle is flat. If no flat section exists, then $\mathcal{S}_e(X) = \emptyset$, and $\text{CF} = 1$ by Abramsky–Barbosa–Mansfield. If one exists, a torsor section is fixed by any one value on each component, so every local section extends.
3. A bijection edge transports the marginal $p \mapsto p$ or $p \mapsto 1 - p$, and an odd cycle gives $p = 1 - p$.
4. The joint on a bijection edge is determined by its marginal. Relabelling invariance gives the last claim, which is Proposition 18.2 at $n = 2$.

□

So at two outcomes, Layer 3 is exactly $\mathbb{Z}/2$ holonomy. The information lens sees only the free parameter q , and q is pinned to $\frac{1}{2}$ precisely when the holonomy is non-trivial.

Checked: the CHSH parity family (`test_two_outcome_supports_all_or_nothing`):

- flat models are non-contextual for $q = 0.5, 0.7, 0.9$;
- the odd model at $q = 0.9$ signals (defect 0.8);
- at $q = \frac{1}{2}$ the odd model is the PR box, with $\text{CF} = 1$ and a profile identical to the flat model's.

Remark 21.4 (What this replaces). ~~withdrawn~~ A proposed “sheaf–geometric bridge” identified γ_{AMB} with the boundary holonomy of a smooth circle bundle through a support-reduction functor χ . It placed contextuality at single-outcome supports and tied it to the divergence of $\partial H/\partial p_i$ at $\partial\Delta$. It is rejected here, for four reasons:

- Its central equation equates an integer class taken mod $\mathbb{Z}$, which is 0, with a holonomy that is never defined.
- The AMB cover is a finite cover of a measurement set, not an open cover of a manifold, and χ is never constructed.
- Information cohomology uses measurable cochains and functional equations, so the divergence of a derivative cannot create a class. Its $H^{\geq 2}$ is open, and “ $I = dH$ ” conflates de Rham d with δ and δ_t (we have $I = \delta_t H$).
- By Proposition 21.1(3), single-outcome models are exactly where both lenses vanish.

The correct discrete statement relating holonomy and γ is the connecting-class result for operation systems (Theorem 18.1). The correct role of the simplex boundary is the stratification above.

Checked in `tests/test_strata.py`: interior (Tsirelson, noisy PR, noisy GHZ); vertices (random deterministic assignments on the CHSH, GHZ and $\mathbb{Z}/3$ grid scenarios); a CF-versus-face counterexample; and the intermediate faces.

22 Magnitude and phase: what Layer 2 reads

Proposition 18.2 cannot be evaded by choosing a different symmetric functional. This section says what Layer 2 *does* read, and how little must be added to recover CF.

Proposition 22.1 (Logical Bell bound). *[S]proved Let $\varphi_1, \dots, \varphi_n$ be formulas on contexts $C_1, \dots, C_n$ that are jointly unsatisfiable, as in Abramsky [27], and let $p_i = e_{C_i}(\varphi_i)$. Then $\text{CF}(e) \geq \sum_i p_i - (n - 1)$.*

Proof. Write $e = (1 - \lambda)e^{NC} + \lambda e'$ with $\lambda = \text{CF}(e)$. A global assignment satisfies at most $n - 1$ of the φ_i , so the non-contextual part contributes at most $n - 1$ to $\sum_i p_i$, and the rest contributes at most n . Hence $\sum_i p_i \leq (1 - \lambda)(n - 1) + \lambda n$. $\square$

The bound costs linear time. A Liar cycle ($x_1 = x_2, \dots, x_n = \neg x_1$) is the case of odd $\mathbb{Z}/2$ holonomy. For $n = 4$ it is the PR box [27], which is Proposition 21.3.

Proposition 22.2 (CF from moduli and one bit). *[S]lower bound proved; equality computed for $n = 3, \dots, 6$ Take a binary n -cycle with uniform marginals and edge correlations $s_i = \mathbb{E}[(-1)^{a+b}]$.*

1. *Layer 2 determines each modulus $|s_i|$ and nothing else about the s_i . The Shannon form is $I_i = \log 2 - h\left(\frac{1+|s_i|}{2}\right)$. The Tsallis-2 form, with Vigneaux’s action, is $I_{2,i} = S_2(b) - a \cdot 2 S_2(b) = (1 + s_i^2)/4$. The signs are invisible by Proposition 18.2, and their product is the holonomy bit.*
2. *The contextual fraction is*

$$\text{CF} = \begin{cases} \max\left(0, \frac{1}{2}(\sum_i |s_i| - (n - 2))\right) & \text{odd holonomy,} \\ \max\left(0, \frac{1}{2}(\sum_i |s_i| - 2 \min_i |s_i| - (n - 2))\right) & \text{flat.} \end{cases}$$

The right-hand side is a lower bound by Proposition 22.1, applied to the odd sign pattern that is best aligned with the s_i . Equality is computed for $n = 4$: the LP agrees to 10^{-15} on 60

random models in `tests/test_magnitude_phase.py`, and on 300 in an exploratory run. For $n = 3, 4, 5, 6$ it agrees to $4 \cdot 10^{-16}$ on 160 random cycles (`tests/test_weak_edge.py`).

Consequences.

- The holonomy bit moves CF by at most $\min_i |s_i|$. On 300 random CHSH models it moved CF by 0.078 on average and by 0.82 at most. With a noisy edge, Layer 2 alone nearly determines CF.
- A flat cycle with one uniform edge ($|s_j| = 0$) and the other edges perfect has $\text{CF} = \frac{1}{2}$.
- The Tsallis-2 form needs no inverse entropy: $|s_i| = \sqrt{4I_{2,i} - 1}$. Independent bits give $I_2 = \frac{1}{4}$, not 0, which is non-additivity. S_2 also has an exactly unbiased estimator, $\sum_x n_x(n_x - 1)/N(N - 1)$, so it avoids the Miller–Madow bias of plug-in Shannon estimates.

Remark 22.3 (Grassmannian charts: magnitude is chart-invariant, phase is the chart coordinate).
proved (identities); interpretation

- *Exponential charts.* $\text{Gr}(k, n) = GL_n/P$ has big cell $\exp(\mathfrak{n}^-)P$ with $\mathfrak{n}^- \cong \text{Hom}(\mathbb{F}^k, \mathbb{F}^{n-k})$. For Grassmannians $\mathfrak{n}^-$ is abelian with $X^2 = 0$, so $\exp X = I + X$ and the chart $X \mapsto \text{row}[I \mid X]$ is linear.
- *Point counts.* Over $\mathbb{F}_q$, $|\text{Gr}(k, n)| = \sum_{\lambda} q^{|\lambda|} = q^{k(n-k)}(1 + O(q^{-1}))$.
- S_2 is a dimension. For flags of type $k = (k_1, \dots, k_s)$,

$$\dim \text{Flag}(k) = \sum_{i < j} k_i k_j = \frac{1}{2} \left(n^2 - \sum_i k_i^2 \right), \quad \text{so} \quad S_2(k/n) = \frac{2}{n^2} \dim \text{Flag}(k)$$

exactly. This underlies Vigneaux’s limit $\frac{2}{n^2} \log_q \binom{n}{pn}_q \rightarrow S_2(p)$ [6]. Shannon entropy is the log of a set count, the multinomial, and $q \rightarrow 1$ recovers it.

- *Edges as chart points.* For the operation systems over $\mathbb{F}_p^k$, the support of edge e is the graph of G_e , the point $\text{row}[I \mid G_e] \in \text{Gr}(k, 2k)$. Its exponential coordinate is the operation G_e . Affine maps are homogenised to $\mathbb{F}_p^{k+1}$.
- *Two lenses.*
 - Holonomy is the product of coordinates around a loop. The space of global sections has dimension reduced by $\text{rank}(G_{\text{loop}} - I)$; for a translation by $h \neq 0$ the loop matrix $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$ fixes 1 of 2 dimensions.
 - Dimension, S_2 and entropy are GL -invariant, hence the same in every chart. That is Layer 2’s blindness in geometric form.
 - Chart invariants read magnitude (Layer 2). Coordinates composed around loops read phase (Layer 3).
- *Global structure.* No single chart covers $\text{Gr}(k, n)$, and the transitions are non-linear (Plücker). The global invariants, Schubert classes and tautological Chern classes, live in the gluing, just as holonomy does.

Not claimed. Vigneaux proves that differential entropy and dimension are the only degree-1 classes for Gaussian modules without coordinates. Whether dimension is a cocycle in a finite-field information structure built from our operation systems is settled in Proposition 22.4 below.

Proposition 22.4 (Dimension over $\mathbb{F}_p$). *[S]proved; computed Laws on $\mathbb{F}_p^n$ whose support is an affine subspace form a sub-functor: it is closed under linear pushforward and under conditioning on a linear observable. On it, $D[X](P) = \log_p |\text{supp } X_* P|$ is a 1-cocycle for the Shannon action,*

$$D[XY] = D[X] + \sum_x P(x) D[Y](P \mid X = x).$$

The reason is that every fibre $A \cap X^{-1}(x)$ is a coset of $A \cap \ker X$, so all fibres have the same dimension. On general laws the identity fails (defect up to 0.42). The count $S_0 = |\text{supp}| - 1$ is a cocycle for the $\alpha = 0$ action on all laws.

Computed (magphase.py, tests/test_magphase_module.py): 300 random affine-support laws on $\mathbb{F}_3^3$ with random linear observables give defect $\leq 7 \times 10^{-16}$; general laws give up to 0.42.

Consequence. Dimension is a Layer-2 cocycle in our finite-field setting, on affine supports. Its non-triviality would follow from the Baudot–Bennequin locality convention (local 0-cochains are constants), which we have not checked separately. Per edge it is constant, $D = 1$ for every graph of a bijection of $\mathbb{F}_p$, so it is blind to holonomy, as Proposition 18.2 requires. Holonomy appears only in the dimension of the *glued* section space.

Dimension deficit from estimated supports. [S]measured Setup (examples/dimension-deficit.py):

- a 4-cycle over $\mathbb{F}_3$ and $\mathbb{F}_5$, with $N \in \{30, 90, 300\}$ samples per edge and symmetric noise $\varepsilon \in \{0.1, 0.3\}$;
- edge supports thresholded at $\tau \in [0.2, 0.6]$, and an edge accepted only if its support is the graph of an affine bijection;
- the loop map in homogeneous coordinates, and the statistic $\text{rank}_{\mathbb{F}_p}(M_{\text{loop}} - I)$;
- a null band from a parametric bootstrap under the nearest flat system.

Three systems are planted:

- *flat*: deficit 0, p global sections;
- *translation holonomy*: deficit 1, 0 global sections, strongly contextual;
- *multiplicative holonomy* ($a_{\text{loop}} = 2$): deficit 1, exactly 1 global section (the fixed point), logically but not strongly contextual.

Findings:

- Whenever all supports were recovered, the deficit was correct in every replicate. The null band was 0 at every (ε, N, τ) .
- The statistic fails by *abstaining*: at $\mathbb{F}_5$, $\varepsilon = 0.3$, $N = 30$, recovery is 2–9%.
- γ over $\mathbb{Z}$ and over $\mathbb{Z}/p$ agrees with the deficit in all six cases. The holonomy test covers only translations.
- The global-section count separates strong from logical contextuality, which the deficit alone does not.

Scope. Under symmetric noise a failed recovery is a non-bijection, never a wrong bijection, so the null band is trivially 0. Non-covariant noise could produce wrong bijections and is untested.

Estimation: S_2 versus Shannon. [S]measured Moduli $|s_i|$ and CF of a binary 4-cycle were estimated from N samples per edge, with 400 replicates for each of three configurations (`examples/s2-vs-shannon.py`). Four estimators were compared: unbiased I_2 , Shannon plug-in, Miller–Madow, and the labelled sample correlation.

- At $N \geq 100$ all four agree to within 0.003 in RMSE of CF.
- At $N = 30$ the S_2 route has the smallest bias in s^2 on weak edges (0.017 against 0.035). It has the *largest* CF RMSE on strong and near-threshold edges (0.087 against 0.067, and 0.153 against 0.125), because CF needs $|s| = \sqrt{s^2}$ clipped to $[0, 1]$.
- S_2 *does not pay off here*. With uniform marginals, each 2×2 table has a one-dimensional sufficient statistic, the agreement count. Every lens is then a function of it, and unbiasedness of s^2 does not carry through the square root.

23 A contradiction meter with provenance

Input. A claim (u, i, j, r) means that provenance unit u (a source, document or turn) asserts $x_i \oplus x_j = r$ about binary propositions. Claims may be single assertions (deterministic mode) or repeated samples.

Three separate failures [S]:

1. *Self-contradiction*: an odd cycle among one unit’s own claims.
2. *Overlap disagreement* ($d^0 \neq 0$, the deficit): two units give the same edge different parities, significant at $z > 3$ in sampled mode. It is recorded and the edge is *not* glued.
3. *Contextuality across provenance* ($d^0 = 0$, $H^1 \neq 0$): the overlaps agree, but the glued graph has an odd cycle contained in no single unit. This is a Liar cycle (Proposition 21.3).

Each cycle is reported with:

- the logical-Bell lower bound on CF (Proposition 22.1);
- a union-bound confidence that all of its parities are right;
- a blamed edge, only when every other edge comes from a declared trusted unit.

Coherent fabrications are invisible by construction.

v1 changes [S]:

1. *Pooled noise*. The flip rate $\hat{\varepsilon}$ is estimated by EM over all claims, with each edge’s true parity latent. An edge’s error probability is then the posterior $\lambda^d / (1 + \lambda^d)$, where $\lambda = \hat{\varepsilon} / (1 - \hat{\varepsilon})$ and d is the majority margin. This replaces the worst-case binomial.
2. *Shortest odd cycles*. For every edge, the meter finds the shortest odd cycle through it by BFS in the parity double cover. It replaces fundamental cycles.
3. *Multiple testing*. Cycles are certified at level α/k over the k candidates (Bonferroni).

Benchmark measured (`examples/meter_benchmark.py`).

- *Data*: 12 propositions and 4 honest units asserting 6 true parities each, with one added unit per scenario; 200 instances per cell. Results are in Table 10.
- *Pooling*: it fixes the $m = 5$ row. At $\varepsilon = 0.1$, self-contradictions go from 0.00 to 0.78 and cross-unit cycles from 0.33 to 0.84.

- *Clean cases*: honest-only firing stays at or below 0.04, and coherent fabrications are not flagged.
- *Limits*: at $\varepsilon = 0.2$ with $m \leq 5$, power is low.
- *Validation*: `tests/test_meter.py` checks the CF lower bound against the LP on 3- and 5-cycles.

m, ε , noise model	honest	self	overlap	cross (correct edge blamed)	fabrication
1, 0 (deterministic)	0.00	1.00	1.00	1.00 (1.00)	0.00
5, 0.1, worst case	0.00	0.00	0.74	0.33 (0.32)	0.01
5, 0.1, pooled	0.01	0.78	0.86	0.84 (0.84)	0.01
3, 0.1, pooled	0.04	0.37	0.61	0.55 (0.55)	0.04
15, 0.2, pooled	0.01	0.95	0.98	0.94 (0.94)	0.00
5, 0.2, pooled	0.04	0.21	0.56	0.19 (0.19)	0.04

Table 10: Detection rates by planted scenario. The “honest” column is the false-positive rate of the full meter.

Relational claims: “A said X” and “C caused D”. [S]_{measured} (`relclaims.py`, `examples/basil_relational.py`) **Extraction.** Claims are extracted from spaCy dependency parses with transparent rules:

- *attribution*: a say-verb with a subject and a clausal complement, or “according to”; deny, reject and dispute, or a negated verb, flip the polarity;
- *causation*: cause, lead to, result in, trigger, spark, fuel, passive by-agents, and because of or due to; a negated verb flips the polarity.

Stated axioms:

- A1: one statement has one speaker;
- A2: causation is not mutual;
- A3: there are no causal 3-cycles.

Conflicts detected:

- *polarity*: the same proposition asserted and denied;
- *speaker*: A1 is violated;
- *reversal*: A2 is violated;
- *cycle*: A3 is violated by claims from two or more sources.

A speaker is used for A1 only when resolved. Role nouns, documents and channels (“the attorney”, “the statement”, “according to the Times”) are unresolved without coreference. Say-verbs nested in another say-verb’s clause, and speakers inside quotations, belong to the outer speaker.

Validation. Ten planted mini-articles: one per conflict type, plus controls for a paraphrase, a causal chain, passive-active agreement, a role noun and a nested quote. All ten are classified correctly (`tests/test_relclaims.py`).

BASIL (100 events, 3 outlets each).

- *Claims*: 1,667 attribution claims and 149 causal claims were extracted.

- *Overlap*: only 43/1605 attribution contents and 3/146 causal pairs are made by two or more outlets in the same event. The opportunity to contradict is small, which is consistent with BASIL’s finding that bias works through selection and framing.
- *First pass*: 17 speaker flags, all reviewed by hand and none a contradiction. Thirteen were one person named two ways, two came from nested quotes, and two were different people saying similar lines.
- *After resolution*: one flag, Lofgren against Pelosi on “power grab”. These are two different quotes, and both can be true.
- *Verdict*: no factual contradiction between outlets was found. The detector’s value here is negative evidence, together with the overlap statistic. The rules are shallow, and recall on real text is unmeasured.

General binary claims. [S]proved (classification); computed Each unit’s formulas define a local support $S_u \subseteq \{0, 1\}^{C_u}$ (`meter_general.py`; formula syntax: $\sim, \&, |, \hat{\cdot}, \rightarrow, \leftrightarrow$). A deletion-based minimal unsatisfiable set of units is classified as one of:

- *self*: $S_u = \emptyset$;
- *overlap*: two units with disjoint projections onto their shared propositions;
- *propagation*: arc consistency (semijoin reduction) empties a unit;
- *cyclic*: arc consistency leaves every S_u non-empty and consistent on overlaps ($d^0 = 0$), yet no global assignment exists.

By Beer–Fagin–Maier–Yannakakis [8], the cyclic type requires a cyclic hypergraph; this is the strongly contextual case. With assertion frequencies p_u , the core carries the logical-Bell bound $CF \geq \sum p_u - (k - 1)$.

Gaussian fast path over $\mathbb{F}_2$. [S]computed Weights of Gaussian type are closed under gluing, and gluing them is matrix algebra rather than a sum over assignments. The Grassmann–Gaussian weights of Korepanov and Sadykov [31], whose 4-simplex weights are Grassmann–Gaussian exponents and whose Pachner-move relations are therefore algebraic identities between Gaussians, are an example of this principle; we use only this structural point, not their parameterisations. Over $\mathbb{F}_2$ the analogue is a unit whose support is an affine subspace, such as a parity, XOR or literal claim or a conjunction of these.

- *Method*: for such units, contradiction cores come from Gaussian elimination with row tracking. The combination y with $yA = 0$ and $yb = 1$ is the certificate. A core is already minimal when $\text{rank} = |\text{rows}| - 1$ (a circuit); otherwise it is minimised by deletion.
- *Timing*: brute force takes 0.35 s at 16 propositions and grows as 2^n . The fast path handles a 5,000-claim Liar ring among 15,000 consistent distractor claims (20,000 propositions) in 0.66 s.
- *Validation*: cores match brute force and are minimal on 40 random parity systems (`tests/test_meter_general.py`).
- *Scope*: non-affine claims such as $A \rightarrow B$ keep the exact but exponential path. General satisfiability is NP-hard, and the CF LP is not reduced by this.

First real run: BASIL, overlap channel. [S]_{measured} **Setup** (examples/basil.overlap.py):

- BASIL [28]: 97 events with all three outlets (Fox, NYT, HuffPost) annotated; gold annotations only.
- Sources are the outlets. A proposition is “the article is negative toward target T in event t ”.
- Literals are encoded as parity edges to an anchor proposition, so cycles cannot arise and only the overlap channel is exercised.
- Null: outlet labels permuted within events, 1000 permutations.

reading	Fox–NYT	Fox–HPO	NYT–HPO	separation (p)
A: article-level author feeling (deterministic)	12/26	6/17	1/27	0.370 (0.002)
B: phrase-level spans (sampled, $\hat{\varepsilon} = 0.114$)	11/112	9/115	7/113	0.026 (0.21)

Table 11: Overlap disagreements over shared propositions. Separation is the mean Fox-versus-others rate minus the NYT–HPO rate.

- At the article level Fox separates from the other two ($p = 0.002$). In span-level polarity it does not ($p = 0.21$).
- Disagreeing article pairs have different annotated relative stance in 19/19 (A) and 26/27 (B) cases, against a base rate of 240/291. The binomial tails are 0.026 and 0.037.
- *Caveats.*
 - Stance and feelings come from the same annotators, so the stance check is not independent.
 - The counts are small: 17–27 shared propositions in reading A.
 - These disagreements are differences in slant, recorded as overlap. They are neither factual contradictions nor hallucinations.
 - Cycles need relational claims, which BASIL’s annotations do not provide.

One example in `tests/test_meter_general.py` matters for the attention proposals reviewed earlier: “ $A \rightarrow B, B \rightarrow C, C \rightarrow \neg A$ ” is *satisfiable* (take A false), so it is not a contradiction. Adding the claim “ A ” produces a propagation-type core. The parity version, $A \leftrightarrow B, B \leftrightarrow C, C \leftrightarrow \neg A$, is cyclic.

24 The decorated Bregman–Čech nerve

Tokens are points $p_i \in \Delta(\text{states})$ with a time τ_i . The radius of a simplex is $\min_c \max_i D_{\text{KL}}(p_i \| c)$.

Lemma 24.1. *proved, computed* The balls $\{p : D_{\text{KL}}(p \| c) \leq r\}$ are convex, since KL is convex in its first argument, so the nerve lemma applies. For a pair, the optimum lies on the segment between the two points; it is found by bisection and certified by the minimax dual $\max_{\lambda} [\lambda D_{\text{KL}}(p \| m) + (1 - \lambda) D_{\text{KL}}(q \| m)]$ to 10^{-9} .

Construction 24.2 (Decoration). • Edges are oriented by time (“A before B”) and labelled by the best-aligning group element, with its margin over the runner-up.

- Triangles carry their boundary holonomy; a nonzero value is a non-flat event.

- Each H_1 bar is labelled by the holonomy of its birth cycle.

Proposition 24.3 (Invariants for $\text{Aff}(\mathbb{Z}/p)$). *proved* Loop labels depend on the base point only by conjugation. The conjugacy classes are: the identity; all nontrivial translations, as one class; and one class per multiplier $u \neq 1$. For odd p , $\text{Aff}(\mathbb{Z}/p)^{ab} = (\mathbb{Z}/p)^\times$, so only u is a function of the H_1 class. Relabelling tokens preserves the class and not the element (tested).

Proposition 24.4 (Walls). *proved; consequence measured* A permutation σ of n states fixes a face of codimension $n - \#\text{cycles}(\sigma)$. In $\text{Aff}(\mathbb{Z}/3) \cong S_3$ the reflections are transpositions, fixing codimension-1 walls. The quotient is a Coxeter chamber, and every planted holonomy is absorbed into a flat labelling (0/100 recovered). Every non-identity element of $\text{Aff}(\mathbb{Z}/5)$ has codimension ≥ 2 .

Observation 24.5. *measured* On $\text{Aff}(\mathbb{Z}/5)$, rings with every edge margin $\geq 10^{-3}$ are recovered in 254/255 cases; rings with an ambiguous edge are wrong in 57/145. On $\mathbb{Z}/3$ translations, both orientations are recovered and reversing time inverts the label.

Limits *measured*:

- Dirichlet noise on tokens (concentration $2000 \rightarrow 200$) raises ring-label error from 15% to 48–63%.
- The $\mathbb{Z}/2$ barcode readout is confounded by shortcut loops of the quotient and by coefficients: 18/20 correct on $\mathbb{Z}/5$ translations and 8/12 on $\mathbb{Z}/3$.
- In the non-abelian case, bars with non-identity labels can die at flat triangles.

Semantics adopted. Negation is *partial*: it is undefined on U and sends U to an explicit *not-applicable* state. Operations therefore form an inverse monoid, not a group, and are lossy.

25 Two facets: restriction and extension

Let $i : \Sigma \rightarrow \mathcal{S}$ include the cover into the information site.

Proposition 25.1. *proved, computed*

1. Restriction $R: P \mapsto (\pi_C P)_C$ has image exactly the non-contextual models.
2. Extension $E: e \mapsto \text{Ext}(e) = \{P : R(P) = e\}$ is a convex polytope, nonempty iff $\text{CF}(e) = 0$. If $\text{CF} = 0$, there is $b \geq 0$ with $\sum b = 1$ and $R(b) \leq e$, and equal context totals force $R(b) = e$.
3. $P \in E(R(P))$, and $R(E(e)) = \{e\}$ whenever $E(e) \neq \emptyset$.
4. The canonical member of $E(e)$ is the maximum-entropy extension, found by iterative proportional fitting (Csiszár). IPF fails to converge on contextual models.

Observation 25.2 (Graded extension). *measured* The optimal non-contextual parts, of mass $1 - \text{CF}$, form a face. In CHSH it is a single point for every contextual model tested (39/39); in the three-party GHZ scenario it never is (0/10). The maximum-entropy point of the face gives a unique representative.

These maps act on models, not on cohomology, so they are consistent with Theorem 15.1. The attached drafts' forward map, which sent γ/CF to the spectral modulus, is not a functor.

26 Not-applicable entries: the gated test

Records are windows of labels (T, F, could-T, could-F, U, n/a), with a *mask* marking variables that are not in the context. The mask is distinct from the *n/a value*.

- Construction 26.1** (Pipeline). 1. GYO reduction of the maximal observed contexts. If the cover is acyclic, the pipeline stops (Theorem 12.4); any $CF > 0$ is then inconsistency, not contextuality.
2. Possibilistic test across support thresholds, with its null band.
3. CF by LP, with the signalling defect reported beside it.
4. Null: each record keeps its mask, and its values are redrawn from the maximum-likelihood consistent law. That law is found by missing-data EM (Theorem 12.2) and pinned to maximum entropy.

stream	gate	CF	verdict
random N/A, complete records present	acyclic	—	stop: the AMB half is empty
honest, one N/A per record	cyclic	0.029	null mean 0.027, $p = 0.39$: stop
structured, shift twist	cyclic	0.858	above all 30 null replicates (max 0.044)
structured, negation with $U \rightarrow n/a$	cyclic	0.859	above the null, but signalling 0.18

Table 12: The gated test on 9,000-record streams [S].

Proposition 26.2 (The deficit). *proved (zero set); measured (the rest)* The deficit is the per-record log-likelihood lost by the best consistent law, $\min_P \sum_C w_C D_{KL}(e_C \| P_C)$.

- Zero set. For no-signalling tables it vanishes iff $CF = 0$: the minimum is attained, and KL is zero iff the tables coincide.
- Not a function of CF. At $CF = 0.5$ on noisy-shift cycles it is 0.101, 0.068, 0.051 for $L = 3, 4, 5$.
- Closed form at $\varepsilon = 0$: it equals $\log(L/(L-1))$ (measured to 10^{-4}).
- Sampling floor. On honest data it falls as about $1/N$ (0.0079, 0.0010, 0.00015 for $N = 10^3, 9 \cdot 10^3, 8.1 \cdot 10^4$); on contextual data it stays flat at about 0.282.
- Bootstrap intervals: $[0.270, 0.287]$ for the shift twist and $[0.291, 0.309]$ for the negation twist.

Observation 26.3 (The ridge). *measured* When contexts are pairs, the three-way interaction is unidentified. EM from different starts reaches the same likelihood with laws 0.005–0.023 apart in L_1 . This is the upper-level identifiability failure of Part I in three variables. The maximum-entropy pin removes the start dependence (tested); without it, a null on larger contexts would measure ridge drift.

Observation 26.4 (Thresholded γ). *measured* Near a cell’s expected mass, sampling flips cells in and out of the support and manufactures logical witnesses. The null fires in 20/20 replicates at threshold 0.005 on honest data, and at 0.1 on the contextual streams’ null. γ is admissible only where the null band is near zero: at 0.02 and 0.15, both structured streams give $\gamma \neq 0$ at 5/5 against a null of 0/20. Report the full sweep.

27 O2: when is a merged model minimal sufficient?

Fix a level l of a binary tree grammar and one node n at that level. Write $B \in [V]$ for its symbol, X for the leaves of its subtree and O for all other leaves. The grammar gives $X \perp O \mid B$, so

$$P(x, o) = \sum_b P(b) D[b, x] U[b, o], \quad D[b, x] = P(x \mid b), \quad U[b, o] = P(o \mid b).$$

A *partition merge* $f : [V] \rightarrow [V']$ replaces B by $f(B)$ and pools rows with weights $P(b \mid f(B))$. It is *exact* if $X \perp O \mid f(B)$, so the merged model reproduces the law of (X, O) .

Proposition 27.1 (Lossless merges). *[S]proved A block S of f is exact if either*

- (i) *the downward rows $D[b]$, $b \in S$, are equal (in particular, if the child rows $R_l[b]$ are equal), or*
- (ii) *the upward rows $U[b]$, $b \in S$, are equal.*

Proof. The block contributes $M_S = \sum_{b \in S} P(b) D[b] U[b]^\top$ to $P(X, O)$. Under (i), $M_S = d_S (\sum_{b \in S} P(b) U[b])^\top$. Under (ii), $M_S = (\sum_{b \in S} P(b) D[b]) u_S^\top$. Either way $M_S = P(S) P(X \mid S) P(O \mid S)^\top$. $\square$

Case (ii) is exact at a position. In a grammar with one table per level it needs $P(b \mid f(B))$ to be the same at every node of the level. This holds automatically at the root, which is a single node.

Proposition 27.2 (The minimal sufficient statistic is a rank object). *[S]proved*

1. *The minimal sufficient statistic of X for O is $x \mapsto P(O \mid x) = q(x)^\top U$, where $q(x) = P(B \mid x)$. Equivalently it is $q(x)$ modulo $\ker U^\top$.*
2. *Its values span a space of dimension $r^* = \text{rank } P(X, O) = \text{rank}(G^{1/2} K^{1/2})$, where $G = D D^\top$ is computed by the recursion $G_l = R_l(G_{l-1} \otimes G_{l-1}) R_l^\top$ and K is the outside Gram.*
3. *Any exact merge has $V' \geq r^*$.*

Proof. 1. $P(o \mid x) = \sum_b P(b \mid x) U[b, o]$ by the Markov property. Minimality is the standard likelihood-ratio characterisation.

2. $P(X, O) = \text{diag}(p_X) [P(O \mid x)]_x$, so the span of the $P(O \mid x)$ has dimension $\text{rank } P(X, O)$. Write $D = G^{1/2} W$ and $U' = K^{1/2} W'$ with W, W' row-orthonormal on the relevant ranges. Then $P(X, O) = W^\top G^{1/2} K^{1/2} W'$.

3. An exact merge factors $P(X, O)$ through V' states. $\square$

Theorem 27.3 (When a partition reaches the minimum). *[S]proved Suppose the rows $D[b]$ are linearly independent.*

1. *A block S is exact if and only if $U[b]$ is constant on S .*
2. *A partition merge attains $V' = r^*$ if and only if the number of distinct upward rows equals $\text{rank } U$. That is, the whole rank deficiency of U must come from equal rows.*

Proof. With D_S the matrix of rows $D[b]$, $b \in S$, we have $M_S = D_S^\top W_S$ with W_S having rows $P(b) U[b]$. Since D_S has independent rows, $\text{rank } M_S = \text{rank } W_S$. This is 1 if and only if the $P(b) U[b]$ are proportional, that is, the probability vectors $U[b]$ are equal. The coarsest exact partition is therefore “equal upward rows”. Its size is the number of distinct rows, while $r^* = \text{rank } D^\top \text{diag}(P(b)) U = \text{rank } U$. $\square$

Counterexample 27.4. `computed` Take $V = 3$ with independent D , and $U[c] = \frac{1}{2}(U[a] + U[b])$ with $U[a] \neq U[b]$. Then $r^* = 2$, but none of the three 2-block partitions is exact. The minimal sufficient statistic $q_a + \frac{1}{2}q_c$ mixes symbols fractionally, so no merge can produce it. This is checked in `tests/test_o2.py`.

Proposition 27.5 (Bregman cost is a complete-data quantity). *[S]proved* The split-merge cost of merging a block S is $\sum_{b \in S} n_b \text{KL}(R_L[b] \parallel \bar{R}_S) = n_S I(B; \text{children} \mid B \in S)$. It vanishes if and only if case (i) holds. It is blind to case (ii): an upward-exact merge has zero observed-data loss but positive cost whenever the child rows in S differ.

Proof. The identity is the Bregman-information form of the mutual information, as in Banerjee et al. [18]. Zero-loss under (ii) is Proposition 27.1. $\square$

Corollary 27.6 (The root collapses). *proved* The root has an empty outside, so U has rank 1 and every root partition is exact. The root can always be replaced by one symbol with coupling $\mu(u, v) = \sum_b \pi_b R_L[b, u, v]$, since $P(x) = \sum_{u, v} \mu(u, v) \beta_u \beta_v$. The complete-data cost of this collapse is $I(B_{\text{root}}; \text{children}) > 0$.

Corpus B, true grammar. `computed` Corpus B was regenerated with the builder’s flags. The rules are identical, because they are drawn before sampling.

level	reachable V	distinct child rows	rank D	distinct upward rows	r^*
1 (left, right)	7	7	7	7	7
2 (left, right)	8	8	8	8	8
3 (left, right)	8	8	8	8	8
4 (root)	8	8	8	1	1

Table 13: O2 ranks on the true grammar of Corpus B (`examples/o2_ranks.py`).

Level-1 symbol 2 is never produced by a level-2 rule. A first pass that kept it showed $r^* = 7 < 8$ with 8 distinct rows. That is unreachability, not a non-partition deficiency.

By Theorem 27.3:

- At levels 1–3 no nontrivial exact merge exists, and the reachable true grammar is already minimal sufficient.
- At the root the minimum is 1.

On the regenerated validation split (2000 sequences):

- the 8-symbol root and the single μ -root give identical held-out $\log p$ per sequence, -15.282238 (difference 0 to machine precision);
- the complete-data cost of the collapse is 1.799 nats per root node;
- across all 127 two-block root partitions the cost ranges over $[1.106, 1.563]$ nats, while the observed loss is exactly 0 for every one.

This explains the level-4 anomaly in [H]: merging $8 \rightarrow 2$ at the root cost 0.72 nats/node yet improved held-out likelihood. Proposition 27.5 predicts exactly this, and it is why split-merge must accept merges on held-out likelihood rather than on cost.

Status of O2.

- Settled for the true grammar: the characterisation (Theorem 27.3), the bound $V' \geq r^*$, and the Corpus B verdict.
- Open: the same test on fitted tables, where exact equalities become statistical ones. A test for equal upward rows needs an outside-Gram estimator and a null, in the same pattern as V6.
- Open: whether real corpora exhibit Counterexample 27.4. That would require a soft, non-partition compression.

28 Experiment: grammar-guided transformers on Corpus B

Question. Does the split–merge grammar help a transformer beyond what the transformer learns alone?

Setup [S] measured.

- Corpus B: 19,000 training sequences, 1,000 for early stopping, and the 2,000-sequence validation split.
- The grammar is fitted by split–merge (3 rounds) and scores -15.421 against the truth’s -15.282 .
- The model is a causal transformer with $d = 64$ and 4 heads, trained for 4,000 AdamW steps at batch size 256.
- The guidance is the fitted grammar’s prefix posteriors $P(B_\ell(t) \mid x_{<t})$ of the four ancestors of the next leaf, computed by inside–outside with unobserved leaves set to 1.
- Variants:
 - *base*: no guidance;
 - *aux*: auxiliary heads distil the posteriors, and nothing is used at test time;
 - *feed*: the log-posteriors are added to the input embedding, so the grammar is used at test time.
- *Oracle* rows use posteriors from the true grammar.
- The metric is held-out excess NLL over the true grammar, in nats per sequence. It is split by the level of the lowest common ancestor of leaves $t - 1$ and t .

Findings.

- *The baseline is data-limited, not undertrained.* Doubling the steps changes nothing ($0.454 \rightarrow 0.464$). Its largest deficits are at the level-2 and level-3 boundaries, where the fitted grammar’s level-3 excess is 11 times smaller.
- *Distillation helps without using the grammar at test time.* The paired difference is -0.187 ± 0.021 at 2 layers and -0.096 ± 0.021 at 4 layers, and the level-3 deficit falls by about 40%. Oracle targets give the same result (0.269). So the gain is not limited by the quality of the fit.
- *Feeding the posteriors reaches the grammar, not beyond.* Against the grammar alone the paired differences are -0.046 ± 0.013 , $+0.006 \pm 0.014$ and -0.040 ± 0.013 . Even oracle posteriors leave 0.13 nats, which is the cost of the transformer’s read-out.

model	total excess	lvl 1	lvl 2	lvl 3	lvl 4
fitted grammar alone	0.139	0.028	0.096	0.011	0.004
base, 2 layers (seeds 0, 1; 8k steps)	0.454, 0.490; 0.464	0.14–0.17	0.18–0.19	0.12	0.01–0.02
base, 4 layers	0.369	0.129	0.125	0.090	0.023
aux, 2 layers (seeds 0, 1)	0.267, 0.299	0.07–0.08	0.11–0.13	0.06–0.08	0.01–0.02
aux, 4 layers	0.274	0.084	0.113	0.057	0.014
feed, 2 layers (seeds 0, 1)	0.145, 0.093	0.02–0.04	0.03–0.06	0.03	0.01–0.02
feed, 4 layers	0.099	0.024	0.034	0.031	0.008
oracle aux / oracle feed, 2 layers	0.269 / 0.132	—	—	—	—

Table 14: Excess NLL over the true grammar on 2,000 validation sequences. The standard error of a total is about 0.01–0.02.

- The transformer improves on the grammar’s level-2 error and loses at levels 3–4.

Scope.

- One corpus in the identifiable regime (781 observations per parameter at level 3).
- One fit; two seeds at 2 layers and one at 4.
- Corpus A is untested. There the fit itself fails at level 3, and guidance cannot supply what the fit lacks.
- Not tested: the specific proposal of Bregman-merge E-steps on hidden states with a multiplicative attention mask. Section 27 argues against its merge criterion.

29 Experiment: recursive training (model collapse) on Corpus B

Question. Can the support dial serve as a label-free collapse metric? Does the small- α lens see collapse before Shannon entropy or likelihood?

Setup [S] measured.

- Generation 0 is the split–merge fit on real data.
- Each generation samples N sequences from the current model and refits by EM (30 sweeps, warm-started).
- Three arms:
 - *replace*: train on the newest synthetic sample only;
 - *accumulate*: train on real data plus all synthetic samples so far;
 - *fresh*: each generation trains on fresh real data. This is the null, and it has the same retraining noise with no feedback loop.
- Metrics, computed on 20,000 model samples of 8-leaf blocks:
 - S_α for $\alpha \in \{0, 0.5, 1, 2\}$;
 - recall of the true block support and of its rarest half, the tail;
 - held-out log-likelihood of *real* data.

Which lens first? A generation is flagged when its change from generation 0 falls below -3 times the RMS drift of the fresh arm (`collapse_analysis2.py`).

$N = 1000$, generation 20	$\log p$ (real)	S_0 of 8-blocks	Shannon H	tail recall
generation 0	-15.421	4206	8.072	0.977
replace	-17.266	3716 (-11.7%)	7.801 (-3.4%)	0.842
accumulate	-15.453	4141	8.041	0.970
fresh (null, generation 19)	-15.451	4185	8.058	0.974

Table 15: Collapse under replacement, with none under accumulation or fresh data. At $N = 4000$ replacement collapses more slowly: $\log p$ goes from -15.421 to -15.629 and tail recall from 0.977 to 0.972 after 20 generations, while accumulation stays at -15.434 . At $N = 20,000$, six generations move $\log p$ by only 0.011.

- Under replacement, S_0 , $S_{0.5}$, H , S_2 and $\log p$ on real data are *all* flagged at generation 3. The accumulate arm is flagged 0 of 20 times by every metric.
- At generation 20 the z values are: $S_0 -15.0$, $S_{0.5} -14.6$, $H -15.0$, $S_2 -18.6$, $\log p -48.5$, tail recall -18.5 .
- The largest *relative* change is in S_0 (-11.7% , the fraction of distinct block types lost). This matches the tail-recall loss (-13.8%) and the likelihood loss.
- *The hypothesis that the support lens detects collapse earlier is not supported.* Every order of α detects at the same generation. S_0 is the most interpretable reading and S_2 the most precise.
- A naive null that uses sampling noise alone, with no refitting, makes H and S_2 flag the harmless drift of the accumulate arm. The retraining null is necessary.

Scope.

- One synthetic grammar and one generation-0 fit. The generation-0 model carries spurious mass (precision 0.963), and early generations shed some of it.
- This is a corpus-level signal, not per-document detection of machine text.

30 Experiment: unknown branching (Corpus V)

Grammar [S]. Every production is $b \rightarrow c_0 c_1 \cdots c_k$: one left child, and $k \in \{0, \dots, 5\}$ right children. The depth is fixed ($L = 3$), with $V = 8$ symbols, $A = 12$ leaves and $m = 3$ productions per symbol. Each production draws its own k , so sentence length varies (mean 77, range 3–149) and the tree is hidden.

Exact inference. Store each symbol as a span matrix, $B[i, j] = P(\text{symbol yields } x_{i:j})$.

- Concatenating children is a matrix product: $B_l[b] = \sum_r p_r B_{l-1}[c_0] \cdots B_{l-1}[c_k]$.
- A sparse chart version gives exact inside–outside in 1.7 ms per sentence.
- It agrees with brute-force enumeration to 10^{-16} , and expected counts agree with finite differences.
- Every validation sentence has exactly one derivation under the true grammar. The true log-likelihood is -30.87 nats per sentence (0.40 per token).

Learners (head plus right children; banded span tables; EM with autodiff expected counts; checked against brute force):

- *Markov*: $k \sim K[b]$, $c_0 \sim H[b]$, $c_t \sim R[b, c_{t-1}]$.

- *Positional*: $k \sim K[b]$, $c_0 \sim H[b, k]$, $c_t \sim R[b, k, t]$. It contains the true grammar except where a symbol has two productions with the same k ; there are 13 such collisions over the three levels.

model (1000 training sentences)	held-out $\log p$ / sentence
true grammar	−30.87
positional, from gold trees (an EM fixed point)	−42.71
Markov, from gold trees (3 EM sweeps: −82.19)	−82.56
positional, EM from random, 20 sweeps (seeds 0, 1)	−82.80, −90.43
bigram / unigram	−151.9 / −189.2

Table 16: Corpus V measured.

Chunks and split–merge measured (`vchunks.py`, `vsplitmerge.py`).

- *Chunk induction*, level by level:
 1. candidate substrings of length at most 6;
 2. EM for a unigram chunk model over segmentations;
 3. pruning of chunk types that are concatenations of other types, which counters the bias toward under-segmentation;
 4. spectral clustering by neighbouring chunks.
- *Level 1*: all 24 true productions are recovered, with token precision 1 and cluster purity 1.
- *Level 2*: all 24 productions are recovered with precision 1, but one true symbol is split across two clusters and two true symbols share a cluster.
- *The grammar*: productions are read off the chunks, and the root is collapsed to one symbol, which is lossless by the root corollary of Section 27. It scores −31.77, parses every held-out sentence, and EM leaves it unchanged.
- *Split–merge*: splits and merges of symbol assignments, accepted only when exact held-out likelihood improves. Three moves bring it to −30.91, against the truth’s −30.87.
- *Residual error*: every final cluster is pure, but one true symbol stays split in two. At 500 held-out sentences, likelihood does not separate this nearly lossless split, as Section 27 would predict. A parsimony tie-break would settle it.

learner (1000 training sentences)	held-out $\log p$ / sentence
true grammar	−30.87
chunks + split–merge	−30.91
chunks only	−31.77
positional, from gold trees	−42.71
positional, EM from random	−82.80

Table 17: Corpus V: bottom-up chunk induction removes the basin problem.

Findings.

- *Model class*. A production’s children form a fixed tuple, so a first-order chain loses 51.7 nats per sentence even with gold trees. Conditioning on arity and position loses 11.8, which comes from the arity collisions.

- *Search.* EM from random starts climbs well past the bigram, but after 20 sweeps it sits about 40 nats below the best model in its own class. That is the basin-selection problem of Corpus B again, and it motivates split-merge or chunk-based initialisation.

31 Monodromy: representations of π_1 and twisted cohomology

Construction [S] (`monodromy.py`). Transport along an edge $a \rightarrow b$ is the conditional expectation $T_{ab}[x, y] = P(b = y \mid a = x)$. Composing around the fundamental cycles gives

$$\rho : \pi_1(\Gamma) = F_{\beta_1} \rightarrow \text{End}(\mathbb{C}^n).$$

It is unitary exactly when every edge is a bijection; lossy edges give contractions.

Proposition 31.1 (What the representation adds). *proved; computed*

1. Abelian holonomy (shifts on $\mathbb{Z}/n$): $\rho(\ell_i) = T_{h_i}$. The representation is determined by the class $[g]$, and in the character basis it is $\text{diag}(\omega^{kh_i})$, a pure phase with no permutation part. Reversing a loop inverts the phases.
2. Non-abelian holonomy ($\text{Aff}(\mathbb{Z}/p)$): ρ sees more than the abelianised class. Two systems with the same multiplier class on each loop can have different traces and different commutators; see the test with loop traces $(0, 1)$ against $(5, 1)$. A multiplier -1 acts on characters as the reflection $\chi_k \mapsto \chi_{-k}$. Translations do not.
3. Twisted cohomology. Twist the coboundary by the character k : $(\delta f)_e = f(b) - \omega^{kg_e} f(a)$. On a connected graph, $h_k^0 = 1$ iff $k[g] = 0$, and $h_k^1 = E - V + h_k^0$.
 - Non-trivial holonomy kills H^0 and lowers H^1 by one.
 - The twisted H^0 is exactly the global-section test. The torsion becomes visible over $\mathbb{C}$, which needs complex (or rotation) coefficients unless $n = 2$.

Proof. 1. Shifts commute, and $F^*T_hF = \text{diag}(\omega^{kh})$.

2. The abelianisation of $\text{Aff}(\mathbb{Z}/p)$ is $(\mathbb{Z}/p)^\times$; the example is computed.

3. $\ker \delta$ consists of the flat sections of a rank-one local system, which is non-trivial iff some loop has $\omega^{kh} \neq 1$; then use rank-nullity. □

Observation 31.2 (The mod-3 grid is not a loop-holonomy system). *computed* The grid's contexts overlap as $K_{3,3}$, which has $\beta_1 = 4$. With overlap-conditional transports, every loop operator is the rank-one projection to uniform, with spectrum $\{1, 0, \dots\}$. The spectra are identical for the consistent grid (81 global sections) and the obstructed one (0). The obstruction $\sum \text{rows} \neq \sum \text{cols}$ is one global parity certified by all six contexts together. No proper cycle carries it, so a representation with per-plaquette phases does not exist for this model. The right tools are γ and $\mathbb{F}_p$ elimination.

Proposition 31.3 (The local shadow: certificate phase). *[S]proved; computed* Let every context C carry only local linear data over $\mathbb{F}_p$: a row A_C and a right-hand side b_C . Examples are an operation edge $x_v - a x_u = g$, a parity row $\sum x = r$, and a literal. No global law, section or trivialisation is used.

1. Fredholm alternative: the family glues iff $y \cdot b = 0$ for every certificate y with $yA = 0$. The obstruction is therefore the vector $\Phi(b) = Yb \bmod p$ over a basis Y of certificates. It is a sum of local contributions $y_C b_C$.

2. Local perturbation: $b_C \mapsto b_C + \varepsilon$ changes Φ by $y_C \varepsilon$. Contexts outside $\text{supp}(y)$ do not enter. With a real lift of the local data, the shadow $e^{2\pi i y \cdot b/p}$ is continuous in every b_C .
3. Shifts: the certificates span the cycle space, and $\Phi \neq 0$ iff the holonomy is non-trivial. The loop shadow of Proposition 31.1 is the special case.
4. Grid: there is a single certificate, $y = (-1, -1, -1, 1, 1, 1)$ over rows and columns, so $\Phi = \sum \text{cols} - \sum \text{rows}$. Its support is all six contexts, and every proper sub-family is consistent for every local data.

Proof. Part 1 is linear algebra over a field: $b \in \text{col}(A)$ iff $b \perp \ker A^\top$. Part 2 is linearity. For part 3, the rows $x_v - x_u$ form the incidence matrix, whose left kernel is the cycle space. Part 4 is computed; removing any context gives full row rank. $\square$

Consequence. The perturbation and every contribution are local. The *detection* of an obstruction, however, needs the whole support of a minimal certificate. That support is a cycle for loop systems and the entire cover for the grid. No statistic computed on a proper sub-family can see the grid obstruction. This is the precise sense in which a local shadow can be complete. Tests are in `tests/test_monodromy.py`.

Theorem 31.4 (Covariant perturbation split). *[S]proved (1–4); 5 as stated Let Γ be connected, with covariant edge channels $K_e = (1 - \eta_e)S_{g_e} + \eta_e U$ on a finite abelian group A . Here g_e is the phase, $\eta_e \in [0, 1]$ is the noise (the magnitude), and U is uniform.*

1. Factorisation. The loop channel is $K_L = (1 - \eta_L)S_{h_L} + \eta_L U$, with $1 - \eta_L = \prod_{e \in L} (1 - \eta_e)$ and $h_L = \sum_{e \in L} \pm g_e$. On every non-trivial character, $\lambda_\chi(L) = (1 - \eta_L) \chi(h_L)$; on the trivial character it is 1.
2. Layer 2 reads magnitude only. Every per-context Shannon or Tsallis quantity, per edge or of the loop channel, is a function of the η_e alone.
3. The class reads phase only. The operation class $[g]$, and with it the certificate phase Φ , depends on the g_e alone. The empirical invariants do not: the support of K_e is full whenever $\eta_e > 0$ and is the graph of S_{g_e} when $\eta_e = 0$. So the stratum depends on the magnitudes. $\gamma \neq 0$ requires a non-flat cycle all of whose edges are noiseless; with any noise on every non-flat cycle, $\gamma = 0$ and the model is in the interior.
4. Split. Changing the phase of an edge e_0 changes $[g]$ iff e_0 is not a bridge, and changes no Layer-2 quantity. Changing its noise changes Layer 2 and never changes $[g]$.
5. Coupling through CF (single cycle):
 - (a) Equal noise, non-flat: $\text{CF} = \max(0, 1 - L\eta/2)$ (Theorem 19.4).
 - (b) $A = \mathbb{Z}/2$, per-edge noise: $\text{CF} = \max(0, 1 - \sum_e \eta_e/2)$ if non-flat, and $\text{CF} = \max(0, \frac{1}{2}(\eta_{\max} - \sum_{e \neq e_{\max}} \eta_e))$ if flat. This is Proposition 22.2 in noise coordinates, so a flat cycle is contextual when one edge is noisier than all the others together.
 - (c) $A = \mathbb{Z}/p$, $p \geq 3$, per-edge noise: neither formula holds. With one fully uniform edge and the rest noiseless, $\text{CF} = 1 - 1/p$ whether the cycle is flat or not.

Proof. 1. Use $S_g U = U S_g = U^2 = U$; the product of two covariant channels is covariant with $(1 - \eta_1)(1 - \eta_2)$ and $g_1 + g_2$. Reversed edges are Bayesian inverses, giving S_{-g} . Characters diagonalise shifts, and U kills every non-trivial character.

2. Under uniform input the joint law of an edge is $|A|^{-1}K_e$, whose multiset of entries depends only on η_e . Relabelling invariance (Proposition 18.2) finishes it, and the loop channel has the same form by part 1.
3. $[g]$ is a function of g . The support statements follow from $\eta_e > 0 \Rightarrow K_e > 0$ entrywise, and an edge with full support imposes no constraint. The strata proposition then gives the rest.
4. Changing g_{e_0} by δ changes $[g]$ by δ times the class of e_0 , which is non-zero iff e_0 lies on a cycle.
5. (a) is Theorem 19.4. (b) is Proposition 22.2 with $|s_e| = 1 - \eta_e$; it is computed exact on cycles of length 3–5, with 10 random draws each. (c) is computed. When $\eta_e = 1$ the edge’s law is a product, so its non-contextual share is at most the $1/p$ of mass on the consistent diagonal.

□

Correction. The per-edge formula $1 - \sum \eta_e/2$ that we had conjectured matched one $\mathbb{Z}/3$ configuration, but it is false for $p \geq 3$. The claim “CF = 0 when $\Phi = 0$ ” is false even for $\mathbb{Z}/2$. The split is clean for Layer 2 against the class. CF couples magnitude and phase *non-linearly*, and depends on both even for flat systems. Tests are in `tests/test_perturbation_split.py`.

Corrections to the proposal we were given:

- $H^1(\Lambda; \mathbb{Z})$ of a space is torsion-free ($\text{Hom}(H_1, \mathbb{Z})$). The torsion in the grid lives in the AMB relative class with coefficients in the $\mathbb{Z}$ -linearised support presheaf.
- A twist by a real 1-cochain squares to zero only when it is closed, and phases $e^{2\pi i/3}$ need $\mathbb{C}$, not $\mathbb{R}$.
- Twisting lowers H^1 rather than making it nonzero.
- For shifts, the monodromy has no permutation part.
- Transports by conditional expectation and their loop spectra are already in Section 17 and the lossy-operations section.

The new content is part 2 of Proposition 31.1, the twisted-Betti identity, and Observation 31.2.

Experiment: **arity-graded monodromy with three sources.** [S]_{measured}
(`examples/vbranch/vmono.py`) **Setup.**

- Level-1 chunks of Corpus V carry a true symbol and an arity k .
- Three sources A, B, C label chunks in their own alphabets: $\pi_s(b)$ with probability $1 - \varepsilon$, uniform otherwise.
- Each edge of the cycle $A - B - C - A$ is compared on a *disjoint* third of 3000 documents. No chunk is labelled by all three, so there is no shared global section.
- *Positive*: on documents shared with A , source C applies a cyclic shift τ of its alphabet to odd-arity chunks only. *Null*: no twist.
- *Estimator*: each edge alignment is a one-to-one assignment on co-label counts (Hungarian), and the holonomy is their composition. It is read on labels whose grade count exceeds twice the estimated noise floor. The arity grades are pooled, k even or odd, and each single k .

Findings.

- In the odd grade the estimated holonomy *equals* the planted twist, conjugated into A ’s alphabet, on every active label (20 of 20 checks).

grade	null, $\varepsilon = .1$	twist, $\varepsilon = .1$	null, $\varepsilon = .3$	twist, $\varepsilon = .3$
pooled	0.00	0.85 (3.7/8 moved)	0.00	1.00 (4.4/8 moved)
k even	0.00	0.00	0.00	0.00
k odd	0.00	1.00 (7/7 moved)	0.00	1.00 (7/7)
$k = 1, 3, 5$	0.00	1.00 (all moved)	0.00	1.00 (all moved)
$k = 0, 2, 4$	0.00	0.00	0.00	0.00

Table 18: Rate at which the estimated holonomy differs from the identity, over 40 replicates.

- *Pooling across arity destroys the phase.* The pooled holonomy is a garbled permutation that moves only half the labels, and the twist appears instead as lost magnitude: $|\lambda_2|$ of the loop operator falls from 0.573 to 0.425 at $\varepsilon = 0.1$, and from 0.148 to 0.101 at $\varepsilon = 0.3$. This is Theorem 31.4 run in reverse: mixing phases turns into magnitude.
- The arity grading is therefore not decoration. It is the index that keeps the phase readable. A graded twist is invisible as a class to the pooled estimator, and exact to the graded one.

Development note. Two readouts produced false positives and were corrected before the table above. Reading the argmax of the composed loop matrix is biased toward labels that are common in the grade. Symbols with no production of arity k pick up noise-only rows. The table uses the corrected estimator.

Real data: Universal Dependencies English-EWT [30]. [S]_{measured} (examples/ewt_monodromy.py) **Setup.**

- *Data:* 16,622 sentences in five genres. Each word carries two annotation layers, X (the Penn XPOS tag) and B (UPOS plus morphological features), and a grade k , its number of right dependents.
- *Loop:* for an ordered pair of genres, $M = T_{X \rightarrow B}^{g_1} T_{B \rightarrow X}^{g_2}$. This is a 2-cycle in the provenance cover, estimated on disjoint sentences. A tag is *moved* if the round trip sends most of its mass elsewhere, with a margin of 0.1. Only the 36 tags whose pooled round trip is the identity are read.
- *Nulls:* (a) genre labels shuffled across sentences; (b) a *composition-preserving* null that keeps each word’s B , form and arity and redraws its XPOS from the pooled $P(X \mid B, \text{form})$, so every genre has the same conventions.
- *Positive control:* in email, NN words with odd k are relabelled NNP.

Results.

- *Planted swap:* found in all 4 email-first genre pairs, and *only* in the odd-arity grades (k odd, $k = 1, k \geq 3$). It is invisible pooled and in the even grades, as in the synthetic three-source run.
- *Real flags:* null (a) never fires, yet 4–7 real moves appear per grade. The largest are WP→WDT in newsgroup and weblog, and SYM→NFP between reviews and answers. Inspection shows these are *composition*, not convention. B conflates relative *who* (WP) with *that/which* (WDT) under PRON PronType=Rel, and genres differ in how often they use relative *who*. The SYM class mixes “/”, emoticons and “\$” in genre-specific proportions.
- *After null (b):* one borderline real flag remains (weblog→reviews WP→WDT at $k = 0$, margin 0.14) out of about 700 comparisons.

- *Verdict*: EWT's genres follow consistent XPOS conventions.

Lessons.

- The null must preserve composition down to the finest level at which the layer map is non-injective, here word forms. Otherwise the conflation in the layer map look like holonomy.
- Arity grading is what makes an arity-specific convention change visible on real distributions.

Arity grading. In Corpus V, a single parse's cover (each production instance is a context) is *acyclic*: GYO reduces all 50 sampled parses. So π_1 of the cover is trivial, and an arity-graded monodromy of one tree-generated source is trivial (T0 and Vorob'ev). A non-trivial graded representation needs a cyclic cover, and the three-source experiment above supplies one.

32 Bridges between $\mathcal{F}_{\mathbb{Z}}$ and $\mathcal{F}_{\alpha}$: the options

Why Vigneaux is blind, structurally. Information cohomology studies *functionals* on the probability functor of the reduced structure $S \setminus \{1\}$. An empirical model is a *section* of that functor (a no-signalling family), and contextuality is the failure to *extend* the section to the joint observable. Cohomology of functionals does not see extension failures of sections. This is why every per-context S_{α} is blind (Proposition 18.2) and why non-unification holds. A bridge must act on the extension problem itself.

Probability complex $\rightarrow \mathcal{F}_{\mathbb{Z}}$.

- The probability complex is the no-signalling polytope over the cover. Its face lattice maps onto support presheaves, and $\mathcal{F}_{\mathbb{Z}}$ is their $\mathbb{Z}$ -linearisation.
- A perturbation is felt in $\mathcal{F}_{\mathbb{Z}}$ exactly when it crosses a face (Proposition 21.1). The response is piecewise constant, and maximally fragile for logical contextuality.
- Two ways to feel it continuously: the certificate phase with a real lift (Proposition 31.3), and the deficit dial below.

Proposition 32.1 (The Rényi deficit dial). *[S]proved (i-iv); computed Let $\Delta_{\alpha}(e) = \min_q \sum_C w_C D_{\alpha}(e_C \| q_C)$, the minimum taken over global laws q , with Rényi divergence D_{α} . Here D_1 is KL, $D_0(p \| q) = -\log q(\text{supp } p)$, and $w_C = 1/n$ over n contexts. For $\alpha \in [0, 1]$ the objective is convex in q .*

- (i) Δ_{α} is non-decreasing in α .
- (ii) For $\alpha \in (0, 1]$: $\Delta_{\alpha}(e) = 0$ iff e is non-contextual.
- (iii) $\Delta_0(e) = 0$ iff e is not strongly contextual, that is, iff the support presheaf has a global section. So the $\alpha = 0$ end is the $\mathcal{F}_{\mathbb{Z}}$ -side strong obstruction, stated in Vigneaux's language.
- (iv) If e is strongly contextual, then $\Delta_0(e) \geq \log \frac{n}{n-1}$.

Proof. (i) Rényi divergences are non-decreasing in the order, and a minimum of non-decreasing functions is non-decreasing.

- (ii) For $\alpha > 0$, $D_{\alpha}(p \| q) = 0$ iff $p = q$, and a global law with all the given marginals exists iff e is non-contextual.
- (iii) $\Delta_0 = 0$ iff some q has $q_C(\text{supp } e_C) = 1$ for every C , iff q is carried by global assignments whose restrictions all lie in the supports.

- (iv) By Jensen, $\frac{1}{n} \sum_C -\log q_C(\text{supp } e_C) \geq -\log \frac{1}{n} \sum_C q_C(\text{supp } e_C)$. The sum is the q -expected number of satisfied supports, which is at most $n-1$ when no global assignment satisfies them all. This is the logical Bell inequality. $\square$

model	strong	CF	Δ_0	$\Delta_{0.5}$	Δ_1 (KL)
PR box	yes	1.000	$0.2877 = \log \frac{4}{3}$	0.2877	0.2877
mod-3 grid, obstructed	yes	1.000	$0.1823 = \log \frac{6}{5}$	0.1823	0.1823
Hardy	no	0.167	0	0.0133	0.0183
Tsirelson	no	0.414	0	0.0172	0.0321
noisy PR, $v = 0.8 / 0.6 / 0.4$	no	0.6 / 0.2 / 0	0	0.041 / 0.004 / 0	0.073 / 0.007 / 0

Table 19: The dial computed (`deficits.py`; tests in `tests/test_deficits.py`). The bound (iv) holds with equality for the PR box and the grid. For these uniform strong models the profile is flat in α ; for contextuality that is not strong it rises from 0.

Why the dial works where per-context S_α fails: the objective couples all contexts through one q , so Proposition 18.2 does not apply. Its KL end is the relative-entropy measure of van Dam, Gill and Grünwald [29]. The perturbation is local, but the minimiser q is a hypothetical global law. Its convex dual is a family of local functions, a Bell-type functional, so the certificate stays local.

Both directions from one object: repair hints. `[S]computed (deficits.repair_hints)` The minimiser q^* of the KL deficit, weighted by trust w_C , gives both directions at once.

- *Probability $\rightarrow$ outcome*: the straight path $e_t = (1-t)e + tq_C^*$ is the smallest (KL) change that flips the verdict. Along it, $\text{CF}(e_t) = (1-t)\text{CF}(e)$, measured as 0.6, 0.45, 0.3, 0.15, 0.06, 0 at $t = 0, \dots, 1$.
- *Outcome $\rightarrow$ probability*: the per-context cost $w_C D_{\text{KL}}(e_C \| q_C^*)$ says *where* the inconsistency sits. The per-cell change $q_C^* - e_C$ says *what* to change.

Planted tests:

- A $\mathbb{Z}/3$ four-cycle with one edge tampered (shift +1, noise 0.2). With equal trust the cost splits equally over the four edges, as it must (Proposition 42.1). With the other edges trusted ($w = 20$), the cost moves to the tampered edge (0.0125 against 0.0002), and the suggested change moves mass from the cells $y = x + 1$ to $y = x$. That is, it undoes the shift. Declaring a different edge untrusted moves the blame there instead: the hint is relative to the declared trust, not an oracle.
- The mod-3 grid with column 2's residue changed. With the other contexts trusted, the hint moves 95% of column 2's mass from residue 1 to residue 0, restoring the planted change.

$\mathcal{F}_{\mathbb{Z}} \rightarrow$ **shadows in $\mathcal{F}_\alpha$** . There is no module morphism (non-unification). The shadows, and what each one sees:

- S_0 reads the face, not the class.
- The dimension cocycle over $\mathbb{F}_p$ (Proposition 22.4) sees the class only on the glued object.
- The certificate phase (Proposition 31.3) is a representation, not a cocycle.
- The magnitude–phase formula (Proposition 22.2) gives CF from Layer-2 moduli plus one bit.

- The Rényi deficit dial is a Vigneaux-language quantity whose $\alpha = 0$ zero set is the strong $\mathcal{F}_{\mathbb{Z}}$ obstruction.

Not pursued, and why:

- A Čech×Vigneaux double complex again studies functionals, so the blindness argument above applies.
- Persistence of γ over support thresholds is available (`threshold_sweep`), but γ is not monotone in the threshold, so no stability theorem is available.

33 The operation-twisted Vigneaux complex: a finite test

Setting. Vigneaux’s topological addendum puts an information structure on open sets with union as product, and conditioning $(V.\varphi)(f) = \varphi(f|_{V^c})$. The cocycle rule is $\varphi[V \cup W] = \varphi[V] + V.\varphi[W]$. The cup-product addendum takes the presheaf F with action $(X.f)(P) = \sum_x P(X=x)f(P|_{X=x})$ and defines

$$(\varphi \smile \psi)[X_1 | \dots | X_{n+m}] = \varphi[X_1 | \dots | X_n] \otimes X_1 \cdots X_n.\psi[X_{n+1} | \dots | X_{n+m}],$$

with $d(\varphi \smile \psi) = d\varphi \smile \psi + (-1)^n \varphi \smile d\psi$. We asked whether an operation system (the edge relabellings $x \mapsto x + g_i$ of Section 31) can twist this complex so that its cohomology sees holonomy.

Proposition 33.1 (Band). *In both addenda the observable monoid is idempotent ($XX = X$, $V \cup V = V$). A twisting cochain in the classical sense, a monoid homomorphism ρ into a group, is therefore trivial: $\rho(X) = \rho(XX) = \rho(X)^2$ gives $\rho(X) = 1$. Operations can twist only the coefficients, by relabelling values, not the observable algebra.*

Finite computation (`twisted_info`). Take the reduced structure on an L -cycle with values in $\mathbb{Z}/n$: variables c_i , edges $E_i = c_i c_{i+1}$. Inside context E_i the law of c_{i+1} is read through the shift g_i . The coefficient module has labelled terms

$$\varphi[O](Q) = c_O + \sum_s w_O(s) Q(s)^\alpha \left(+ \sum_s v_O(s) Q(s) \log Q(s) \text{ at } \alpha = 1 \right).$$

A labelled term with non-constant w_O is exactly what could feel a relabelling. We impose $\varphi[XY] = \varphi[X] + X.\alpha\varphi[Y]$ for every ordered pair in every context, at random laws, and compute Z^1 modulo the parametrisation gauge.

module	flat ($g = 0$)	twisted (holonomy 1 or 2 in $\mathbb{Z}/3$)	label-dependent part
$\alpha \in \{0.5, 2, 3\}$	1	1	0
$\alpha = 1$, linear terms only	0	0	0
$\alpha = 1$, with $Q \log Q$	1	1	0

The one cocycle is $K(1 - \sum Q^\alpha)$, the Tsallis entropy, with the same K on every observable; at $\alpha = 1$ it is Shannon. The same holds on a $\mathbb{Z}/2$ odd three-cycle. Linear labelled terms (expectations) die at $\alpha = 1$ because $XX = X$ forces a cocycle to vanish on atoms, and a linear functional that vanishes on atoms is zero. This corrects our earlier prediction that they would carry a twisted Čech part.

Proposition 33.2 (Coefficient twists are invisible). *Suppose the local solution space of the cocycle equations on each context is fixed by every relabelling of values. This holds whenever the solutions are the symmetric Tsallis family, as in Vigneaux’s classification and the computation above. Then the twisted and flat complexes have the same Z^1 . By the Leibniz rule, cup products of these classes are symmetric too, so the cup algebra they generate is also twist-invariant.*

Proof. Twisting changes only how the shared variable c_{i+1} is read inside E_i , by a bijection σ of values. The cocycle equations on E_i are pulled back by σ . A σ -invariant solution set is unchanged, so the local systems of solutions agree and so do the glued spaces. Symmetry is preserved by $\otimes$ and by the conditioning action. $\square$

Reading. This is Layer-2 blindness again, now at the level of cohomology rather than functionals. The holonomy of a relabelling system lives in the Čech H^0 of the local system of value relabellings: the $\mathbb{Z}/3$ cycle has 3 global sections flat and 1 twisted (`twisted_betti`, Section 31). It does not live in Vigneaux H^1 . To make the probability layer feel a twist, the coefficients must be *non-symmetric*. We test values in $\mathbb{R}^\Omega$ next.

Non-symmetric coefficients: values in $\mathbb{R}^\Omega$

A cochain now returns a function of the outcome: $\varphi[O](Q) \in \mathbb{R}^{\Omega_O}$, pulled back to the context. Relabellings permute coordinates. The action is pointwise conditioning,

$$(X.f)(P)(\omega) = f(P|_{X=X(\omega)})(\omega),$$

with no averaging; its expectation is Vigneaux's action. Proposition 33.1 applies to the action too: $X.X.f = X.f$ leaves no room for a permutation or a character inside the action. The twist can therefore enter only through *gluing*, that is, through how the shared variable c_{i+1} is read inside E_i . The ansatz is local, with label-dependent coefficients:

$$\varphi[O](Q)(v) = a_0(v) + a_1(v) \log Q(v) + a_2(v) Q(v) \log Q(v) + \sum_s b_1(v, s) Q(s) + \sum_s b_2(v, s) Q(s)^2.$$

We impose the pointwise cocycle rule at every point of the support, on the $L = 4$ cycle, on three families of laws:

- full support;
- the *graph stratum*: laws supported on the graph of the edge relation, the deterministic, single-outcome stratum of Section 21;
- graph laws mixed with ε of full-support noise.

Dimensions are taken modulo functionals that vanish on every law used: we count generalised singular values of the constraint matrix relative to the evaluation matrix (`pointwise_cocycles`).

$\mathbb{Z}/n$, holonomy h	orbits of h	full support	graph stratum	$O(\varepsilon)$ near-cocycles, $\varepsilon = 3 \cdot 10^{-4}$
$\mathbb{Z}/2$, $h = 0 / 1$	2 / 1	1 / 1	8 / 4	—
$\mathbb{Z}/3$, $h = 0 / 1$	3 / 1	1 / 1	21 / 7	3 / 1
$\mathbb{Z}/4$, $h = 0 / 2 / 1$	4 / 2 / 1	1 / 1 / 1	36 / 18 / 9	4 / 2 / 1

Only the holonomy matters, not the individual edge shifts: on $\mathbb{Z}/3$, shifts $(1, 1, 1, 0)$ have holonomy 0 and give 21.

Proposition 33.3 (The probability layer reads the holonomy on the graph stratum).

1. On full-support laws the only pointwise cocycle in the ansatz is the surprisal $-K \log Q(v)$, for flat and twisted cycles alike.
2. On the graph stratum, $Z^1 \cong V_0^{(h)}$, where V_0 is the space of admissible local functionals on c_0 (those vanishing on atoms) and h acts jointly on laws and points. If V_0 is a free module under relabelling, as for our ansatz, then

$$\dim Z^1 = \frac{\dim V_0}{n} \cdot \#\{\text{orbits of } h\}.$$

Proof. (2) On the graph stratum every context law is a law p on the latent values pushed along the edge bijection σ_i . Conditioning on any non-trivial observable gives an atom, and a cocycle vanishes on atoms because $XX = X$. The rules for (c_a, c_b) and (c_b, c_a) then read

$$\varphi[E_i](P)(\omega) = \varphi[c_i](p)(a) = \varphi[c_{i+1}](\sigma_{i*}p)(\sigma_i a),$$

so the family $(\varphi[c_i])$ is transported along the edges. Going round the loop gives invariance of $\varphi[c_0]$ under h . Conversely, an h -invariant admissible functional defines $\varphi[c_i]$ by transport and $\varphi[E_i]$ by the display, and the remaining pair rules reduce to vanishing on atoms. The invariants of a free $\mathbb{R}[\langle h \rangle]$ -module have dimension equal to its rank.

(1) With full support, conditioning on c_a is not an atom. Writing the log-coefficients as $\lambda(v)$, the two orderings of the pair rule force $\lambda_a(a) = \lambda_b(b)$ for all a, b , so λ is constant. The remaining terms are excluded by the computation. $\square$

Near the stratum. With ε of noise the exact label-dependent cocycles disappear, and only the surprisal is left. They leave a trace, however: the number of generalised singular values that scale linearly in ε equals the number of orbits of h . For example, on flat $\mathbb{Z}/3$ they are 0.0007, 0.0008, 0.0012 at $\varepsilon = 10^{-4}$ and 0.007, 0.007, 0.011 at 10^{-3} , against a bulk near 0.03. With this sample design the gap closes by $\varepsilon \approx 10^{-2}$.

Remark 33.4 (Exactness depends on locality; corrected). Vigneaux's cochains are local: $\varphi_X[X_1 | \dots | X_k]$ depends only on the law of $X_1 \cdots X_k$. For $k = 0$ the empty product is the trivial observable, so a 0-cochain is a constant, and pointwise $\delta c = c - c = 0$. Hence $B^1 = 0$ and $H^1 = Z^1$. In Vigneaux's convention the classes above are non-exact H^1 classes, and on the graph stratum $\dim H^1 \propto \#\{\text{orbits of } h\}$. A non-exact H^1 class with $\mathbb{R}^\Omega$ coefficients therefore does see the twist. (This is the same convention under which Shannon entropy spans H^1 .)

If one drops locality in degree 0 and allows contextwise 0-cochains f_C that depend on the context law, a cone lemma makes every cocycle locally exact. Let E be the maximal observable of C and put $f_C = -\varphi[E]$. Then

$$\delta f[X] = \varphi[E] - X.\varphi[E] = \varphi[XE] - X.\varphi[E] = \varphi[X].$$

In that convention the twist information moves to degree 0, to the Čech H^0 of the relabelling local system (`twisted_betti`, Section 31). An earlier version of this remark stated only the second convention. The numbers are the same in both conventions; only the degree differs.

Twist $\rightarrow$ outcome. A relabelling twist perturbs the outcome layer exactly through its holonomy. On the graph stratum the global assignments are the fixed points of $x \mapsto x + h$: there are n of them if $h = 0$ and none otherwise, so any $h \neq 0$ makes the support strongly contextual, with AMB class $[h] \in H^1(\text{cycle}; \mathbb{Z}/n)$. Shifts whose holonomy is 0, such as $(1, 1, 1, 0)$ on $\mathbb{Z}/3$, are coboundaries, i.e. pure relabellings, and change nothing. With ε of uniform noise on every edge, $\text{CF} = \max(0, 1 - L\varepsilon/2)$ (Theorem 19.4). Measured on the $L = 4$ cycle for $\mathbb{Z}/3$ with $h = 1$ and for $\mathbb{Z}/4$ with $h = 1, 2$: $\text{CF} = 1, 0.9994, 0.98, 0.8, 0.4, 0$ at $\varepsilon = 0, 3 \cdot 10^{-4}, 0.01, 0.1, 0.3, 0.5$; flat cycles give 0 throughout.

The three layers read the same twist differently:

- AMB reads h itself, as a class in $\mathbb{Z}/n$.
- $\mathbb{R}^\Omega$ information cohomology reads $\gcd(n, h)$, i.e. the order of h , so it cannot tell $h = 1$ from $h = 3$ in $\mathbb{Z}/4$.
- Symmetric entropies read nothing (Proposition 33.2).

The outcome signal is also far more robust: CF survives to $\varepsilon = 2/L$, while the $O(\varepsilon)$ near-cocycle count fades by $\varepsilon \approx 10^{-2}$. Only the holonomy of a twist is observable, in either layer; which edge carries it is gauge. This matches the equal cost split of Proposition 42.1.

What this means for detection. The probability layer does feel an operation twist, but only through near-deterministic edges, the same stratum on which Section 21 places AMB's single-outcome obstruction. A soft statistic is the count of $O(\varepsilon)$ near-cocycles. Generic, full-support data are blind even with non-symmetric coefficients.

Corollary 33.5 (Fontaine-type reading; admissibility). *proved* In the setting of Proposition 33.3(2), write $D(h) = V_0^{(h)}$. This has the shape of Fontaine's functor $D_B(V) = (V \otimes B)^G$: the admissible local functionals V_0 play the role of B , the holonomy group $\langle h \rangle$ that of G , and Z^1 is the invariants. Since V_0 is free over $\mathbb{R}[\langle h \rangle]$,

$$\dim D(h) = \frac{\dim V_0}{\text{ord}(h)} \leq \dim V_0, \quad \text{with equality iff } h = 0.$$

Call a twist V_0 -admissible when equality holds. Then a twist is V_0 -admissible iff it is trivial. Proposition 33.2 is the degenerate case in which $\langle h \rangle$ acts trivially on the coefficients, so that every twist looks admissible.

Two exact sequences make this precise. They should not be conflated.

1. The loop complex.

$$0 \rightarrow V_0^{(h)} \rightarrow V_0 \xrightarrow{h-1} V_0 \rightarrow V_{0,\langle h \rangle} \rightarrow 0.$$

This is the cochain complex of the loop with coefficients in the local system V_0 , so $\ker(h-1) = H^0(\text{loop}; V_0) = Z^1$, which is the identification in Proposition 33.3. The cokernel, the coinvariants, is $H^1(\text{loop}; V_0)$. It is not an obstruction to admissibility. Indeed $\dim \ker = \dim \text{coker}$ (Euler characteristic 0), and for a finite group over $\mathbb{R}$ averaging makes the composite $V_0^{(h)} \hookrightarrow V_0 \twoheadrightarrow V_{0,\langle h \rangle}$ an isomorphism. This recovers the equal twisted Betti numbers of *twisted_betti* (Section 31).

2. The comparison map. The analogue of Fontaine's $\alpha_V: B \otimes D_B(V) \rightarrow B \otimes V$ is the inclusion $\alpha_h: V_0^{(h)} \hookrightarrow V_0$. As for α_V , it is always injective, and it is an isomorphism iff the twist is admissible. Its cokernel

$$0 \rightarrow V_0^{(h)} \xrightarrow{\alpha_h} V_0 \rightarrow \text{im}(h-1) \rightarrow 0, \quad \dim \text{im}(h-1) = \dim V_0 \left(1 - \frac{1}{\text{ord}(h)}\right),$$

is the non-admissible part, namely the sum of the non-trivial isotypic components of V_0 . This is the obstruction, and its dimension recovers the order of the holonomy.

Remark 33.6 (Analogies with p -adic Hodge theory: what transfers). *interpretation* Proposition 33.2 is sometimes read as the opposite of a comparison isomorphism. It is better read as the base case of Corollary 33.5, and the other suggested analogies need the following corrections.

- F_α is not a period ring. It is a one-parameter deformation family of modules, running from the support action at $\alpha = 0$ to Shannon at $\alpha = 1$. No module over which the $\mathbb{Z}$ -valued outcome sheaf and F_α become isomorphic can exist, by non-unification. Its flavour is that of a q -analogue: compare $S_2 = \frac{2}{n^2} \dim \text{Flag}$ (Remark 22.3).
- The representation ρ of Section 31 is a monodromy representation, not a Galois representation. It is a representation of the free group $\pi_1(\Gamma)$ with finite image, and it has no profinite topology, inertia or decomposition groups. The apt frame is the complex Riemann–Hilbert correspondence, in which a flat connection corresponds to a π_1 -representation. There the loop matrix $M = T_{AB}T_{BC}T_{CA}$ is the monodromy and the transports are the periods.

- *The arity index k is a grading, not a filtration.* The monodromy splits as a direct sum over k , which makes it a Hodge–Tate-type decomposition rather than a Hodge filtration. It is also the choice of an observable, not intrinsic to the complex.
- *There is no Frobenius from ∂_α .* We have $S_\alpha = H - \frac{\alpha-1}{2} \sum_x p_x (\ln p_x)^2 + O((\alpha-1)^2)$, so $\partial_\alpha S_\alpha|_{\alpha=1} = -\frac{1}{2} \mathbb{E}[(\ln p)^2] = -\frac{1}{2}(H^2 + \text{Var}(-\ln p))$. That is the second moment of surprisal, not the co-information. The identity $I(X;Y) = H(X) + H(Y) - H(XY)$ is instead the coboundary of H for the *trivial* action (Baudot–Bennequin). A structurally closer candidate is the escort monoid $E_\alpha(p) = p^\alpha / \sum p^\alpha$, which satisfies $E_\beta \circ E_\alpha = E_{\alpha\beta}$: a monoid of power maps, as Frobenius lifts $x \mapsto x^p$. *Open*: whether E_β intertwines the α -action on the complex, relating S_α to $S_{\alpha\beta}$.

- *A template with a monodromy defect.* Hansen and Li [32] prove an inequality between dimensions for proper smooth rigid spaces over a p -adic field with strictly semistable reduction: $\dim H^1(X, \mathcal{O}_X) \geq \dim H^0(X, \Omega_X^1)$. The defect equals the virtual torus rank of Pic_X^0 . Their proof uses the Hodge–Tate decomposition, the Kummer sequence and Raynaud uniformisation. They also prove $\dim \text{Alb}_X \leq \dim H^0(X, \Omega_X^1)$. The shape matches Corollary 33.5: there $\dim V_0 - \dim Z^1 = \dim \text{im}(h-1)$ is an inequality whose defect is governed by a monodromy.

The structural statement “in both cases the defect is the part on which the monodromy acts non-trivially” is *false* on the Hansen–Li side, and the difference is instructive. Write $\text{Pic}_X^0 = E/\Gamma$ (Raynaud), with E an extension of an abeloid B by a split torus T of dimension r , and Γ a lattice of rank $k \leq r$. The Tate module then carries a filtration with graded pieces $V_p(T)$, $V_p(B)$ and $\Gamma \otimes \mathbb{Q}_p$. Degeneration monodromy is *unipotent*: it acts through the extension, via $N: \Gamma \otimes \mathbb{Q}_p \rightarrow V_p(T)$, so its rank is at most k . It acts non-trivially on no graded piece. The defect $r - k$ counts the torus directions *not* reached from the lattice. The Tate curve $\mathbb{G}_m/q^\mathbb{Z}$ has $r = k = 1$, non-trivial monodromy and defect 0.

In Corollary 33.5, by contrast, h has finite order and so is *semisimple*, and the defect $\text{im}(h-1)$ is precisely the sum of the isotypic components on which h acts non-trivially. Semisimple versus unipotent is the real dividing line. The shared statement is only that each defect is a dimension shortfall controlled by monodromy data: the image of $h-1$ in our case, the torus part left uncovered by N in theirs. A unipotent analogue on our side would need a loop operator that is not semisimple, for example from non-invertible or non-diagonalisable transports. That is open.

We do not claim more than this shape. In particular, nothing in [32] involves a Frobenius or the escort map, and the Rényi dial is a family of divergences, not a comparison isomorphism.

34 Contextuality without holonomy: the weak edge

Sections 31–33 treat a twist, a relabelling whose holonomy is non-trivial, as the way a local change breaks gluing. That is not the only way, and the flat case of Proposition 22.2 already contains the other. It is easy to miss, so we state it separately. No relabelling, sign or twist is needed: *weakening a single correlation on an otherwise consistent loop is enough*.

Proposition 34.1 (Weak-edge contextuality). *proved Take a binary L -cycle with uniform marginals, all correlations $s_i \geq 0$ (flat: no holonomy), and deficits $\delta_i = 1 - s_i$.*

1. Imbalance criterion. *The model is contextual iff the weakest edge’s deficit exceeds the sum of all the others:*

$$\delta_{\max} > \sum_{i \neq \max} \delta_i, \quad \text{and then} \quad \text{CF} = \frac{1}{2}(\delta_{\max} - \sum_{i \neq \max} \delta_i).$$

This is the flat formula of Proposition 22.2 rewritten: $\sum s_i - 2 \min s_i - (L - 2) = \delta_{\max} - \sum_{i \neq \max} \delta_i$.

2. One weak edge. If $L - 1$ edges are perfect ($s = 1$) and one has correlation $e \in [-1, 1]$, then $\text{CF} = (1 - e)/2$ for every L .
3. Uniform weakening is harmless. If all $s_i = e \geq 0$, then $\text{CF} = 0$.
4. No loop, no contextuality. On a tree (acyclic cover) $\text{CF} = 0$ for any edge laws.

Proof. (2), lower bound: the formulas $x_0 = x_1, \dots, x_{L-2} = x_{L-1}$ and $x_{L-1} \neq x_0$ are jointly unsatisfiable, with probabilities $1, \dots, 1, \frac{1-e}{2}$. Proposition 22.1 gives $\text{CF} \geq (L - 1) + \frac{1-e}{2} - (L - 1)$.

(2), upper bound: the model equals

$$\frac{1+e}{2} (\text{all-equal global law}) + \frac{1-e}{2} (\text{Liar cycle}),$$

so its non-contextual fraction is at least $\frac{1+e}{2}$.

(3): with $s_i = e$, $\delta_{\max} = 1 - e \leq (L - 1)(1 - e)$.

(4): Vorob'ev [7]; locally consistent families on an acyclic cover always glue.

(1): Proposition 22.2. Its equality is computed for $L = 3, \dots, 6$, and the one-weak-edge case is proved in (2). $\square$

e (fourth edge of an $L = 4$ cycle)	CF	Δ_0 (support)	Δ_1 (KL)	strength
1	0	0	0	non-contextual
0.8	0.10	0	0.018	logical
0.5	0.25	0	0.046	logical
0	0.50	0	0.100	logical
-0.5	0.75	0	0.168	logical
-1	1	0.288	0.288	strong (twist, $\log \frac{4}{3}$)

Also measured:

- deficits $(0.1, 0.1, 0, 0.5)$ give $\text{CF} = 0.15$, while $(0.2, 0.2, 0.2, 0.5)$ gives 0;
- a perfect 5-cycle with one edge at 0.3 gives 0.35;
- the same edges on a path give 0.

Why this is not obvious, and what reads it.

- *It is a probability perturbation that changes the outcome verdict.* No value is relabelled, and each edge on its own is an ordinary, even independent, law. The obstruction comes from the *imbalance* between edges: the perfect edges force $x_0 = \dots = x_{L-1}$, and any disagreement mass on the weak edge cannot be glued. Lowering all correlations together removes it.
- *Strength.* The contextuality is logical, not strong: the all-0 and all-1 assignments survive. So the support-level dial Δ_0 and the $\mathbb{R}^\Omega$ twist machinery of Section 33 are blind to it. CF and the deficits $\Delta_{\alpha>0}$ grow continuously with the imbalance, and only the limit $e = -1$ is a twist.
- *Localisation.* AMB's obstruction γ flags exactly the sections $(0, 1), (1, 0)$ of the weak edge, and no others. This contrasts with a twist, whose cost splits equally over the loop because only the holonomy class is observable (Proposition 42.1). The repair hint is equally direct: raise the weak edge's correlation, or lower the others, until the imbalance criterion fails.
- *Layer 2 sees it.* Symmetric entropies read the moduli $|s_i|$ (Proposition 22.2), and on a flat cycle the moduli are the whole story. Mutual informations alone therefore decide weak-edge contextuality: the imbalance criterion is a statement about $I_i = \log 2 - h(\frac{1+s_i}{2})$. This is the regime where the probability layer is *not* blind.

35 Which coefficients can see contextuality

Proposition 35.1 (Two-sided acyclicity). *(1)–(2) proved; (3) computed* Take a finite measurement scenario and a no-signalling family $e = (e_C)$, i.e. one whose marginals agree on overlaps.

1. Signed and field-valued measures always glue. *For any field K of characteristic 0 ($\mathbb{Q}, \mathbb{R}, \mathbb{Q}_p, \mathbb{C}_p, \dots$), a K -valued no-signalling family has a K -valued global measure whose marginals are the e_C . Consequently no K -valued invariant built from gluing can detect contextuality. In particular this holds for non-Archimedean measures.*
2. Positive measures glue on every family iff the cover is acyclic. *This is Vorob’ev’s theorem [7]. On any cover with a cycle there are contextual families, and Propositions 21.3 and 34.1 give families with and without holonomy.*
3. Integrality, computed. *For the CHSH scenario the marginalisation matrix has all Smith invariant factors equal to 1. Every integer no-signalling family, the PR box included, therefore has an integer, and hence $\mathbb{Z}_p$ -valued, signed extension, and the rank is the same over every $\mathbb{F}_p$. There is no p -torsion.*

Proof. (1): the marginalisation map is a linear map with integer entries. The no-signalling families are exactly the image of the signed global measures: this is Abramsky–Brandenburger’s quasi-probability characterisation [1], a rank statement over $\mathbb{Q}$. Rank, and hence solvability, is preserved by extension of scalars to any field of characteristic 0. (2): [7]. (3): the Smith normal form (`tests/test_coefficients.py`). $\square$

Proposition 35.2 (A single source always glues). *proved; checked on MBIC* Suppose every regional comparison is a difference of one global assignment b evaluated on the same units, $d = \delta b$. Then every loop sum vanishes, because $\delta\delta b = 0$. Hence a global gluing failure among pairwise comparisons requires that the regions use different units (different events or sentences), or different scales.

On MBIC we compared annotator groups L, C and R on the same 1,600 sentences. The loop L–C–R–L closes to 10^{-16} , while the same loop built from disjoint thirds of the sentences is free to be non-zero. Outlets, by contrast, report different sentences, and there the outlet $\times$ topic loops fail globally (Freedman–Lane $p = 0.0005$; `examples/bias3`).

Reading. Contextuality is a phenomenon of *positivity* and of *supports*: of the semiring $\mathbb{R}_{\geq 0}$ and the Boolean semiring. It is not a phenomenon of linear gluing. Two consequences follow:

- the ring-dependence of AMB’s γ comes from the support subpresheaf, not from the measures;
- a theory of non-Archimedean *probability* measures cannot see contextuality at all.

Non-Archimedean structure can still enter through *parameters*, for example hierarchical codes in $\mathbb{Z}_p$ with real, positive probabilities formed from valuations, as in recent p -adic learning work. There the obstruction theory of this paper applies unchanged. Every obstruction exhibited in this paper lives on a cycle of the cover.

36 Three-level bias: house lean, selective loops and global gluing

Setting. Several sources describe the same events.

- *Level 1* is the set of descriptions.
- *Level 2* is a bias $b(\text{source}, \text{topic})$.

- *Level 3* is the relation between biases. On the bipartite source–topic graph we write

$$b(s, t) = u(s) + v(t) + h(s, t),$$

with u the house lean, v the topic effect, and h the loop residual. The residual h is the only topic-selective part. For two sources and two topics its loop sum is $\Phi = b(A, a) - b(A, b) - b(B, a) + b(B, b)$.

$\Phi = 0$ on every loop means bias is a coboundary. In probability coordinates the additive model is a log-linear (toric) model, and Φ is the log of its 2×2 binomial. Exact tests come from the model’s Markov basis. By Proposition 35.2, loops can fail only when regions use different units. Code and outputs are in `examples/bias3`.

BASIL [28]: 100 events, each covered by Fox, NYT and HuffPost (HPO). Themes (ELECT, INVEST, POLICY, SOCIAL), the focal party of each event (R, D, N), and the party of each span target are our hand labels, released for audit. Bias is measured in two ways: the annotated article-level relative stance, and span favourability (the log-odds that a directly aimed bias span toward a party is positive).

- *Stance* (annotated relative stance, $-2..2$; paired by event).
 - House lean: Fox–NYT $+0.81$, Fox–HPO $+0.87$, HPO–NYT -0.06 .
 - Fox’s lean rises over time. Over NYT it is $+0.40$, $+1.00$, $+1.17$ (2010–13, 2014–16, 2017–19); over HPO it is $+0.65$, $+0.70$, $+1.33$. The trend is about 0.09 per year in both pairs, with permutation $p = 0.11$ and 0.13.
 - No loop survives correction: 36 loop tests, smallest BH-adjusted $p = 0.45$. For example, Fox over NYT is $+0.97$ on Democratic and $+0.95$ on Republican stories.
 - Every gluing region fits one lean.
 - At stance resolution the bias is a coboundary: house lean plus drift. Power is limited, since $\text{sd}(d) \approx 1.7$ gives a minimum detectable Φ of about 1.1–1.6 stance points.
- *Span favourability* (1,274 directly aimed spans; source $\times$ target party $\times$ polarity; exact test of the no-three-way model on the one-move fiber, plus an event-cluster bootstrap):

pair	Φ	exact p	event-bootstrap 95% CI
Fox–NYT	$+0.63$	0.16	$[-0.32, +1.60]$
Fox–HPO	$+1.34$	0.0017	$[+0.48, +2.35]$
HPO–NYT	-0.71	0.066	$[-1.56, +0.08]$

- Fox–HPO survives Bonferroni over 24 tests.
- In 2017–19, HPO–NYT gives $p = 0.00025$ (Bonferroni 0.006). HPO had 0 of 94 Republican-directed spans positive, against 6 of 17 Democratic-directed.
- Excluding Trump as a target, Fox–HPO persists: $\Phi = +1.09$, $p = 0.015$, CI $[+0.10, +2.18]$. The two outlets are then selective in opposite directions. Fox’s favourability log-odds are -1.46 toward R and -2.11 toward D; HPO’s are -0.99 toward D and -1.44 toward R. Most of the 2017–19 HPO effect is Trump-specific: without Trump, Fox–HPO 2017–19 has $p = 0.25$, while HPO–NYT still has $p = 0.01$.
- Each party alone glues; only their union fails. This is a genuine global failure.

- *Planted global failure.* We split the events into three disjoint regions: Fox–NYT, NYT–HPO and HPO–Fox. Each region glues trivially. We then shift HPO’s scale by δ in one region only.
 - Detection at $|z| > 1.96$ is 0.052, 0.17, 0.48, 0.83, 0.98 for $\delta = 0, 0.5, 1, 1.5, 2$ over 2,000 splits.
 - Unplanted, real BASIL closes, $\Phi = +0.01 \pm 0.50$, and the test is calibrated at 5%. The mean Φ is +0.48, +0.93, +1.39, +1.84, so it tracks δ . The failure is invisible region by region.
 - On a ring of K outlets with BASIL’s noise, the holonomy h is spread evenly over the edges. Every proper arc glues, and only the whole ring can fail. Detection rates:

K , events per edge	$h = 0$	0.5	1	2	3
3, 33	0.05	0.18	0.55	0.98	1.00
6, 33	0.04	0.11	0.31	0.81	0.98
6, 100	0.06	0.25	0.70	1.00	1.00

Longer loops spread the same holonomy over more noisy edges, so they need more events per edge.

MBIC [33]: 1,700 sentences, 8 outlets, 14 topics, 17,775 crowd labels with annotator ideology.

- *Theorem check.* Annotator groups compared on the same 1,600 sentences close the loop L–C–R–L to $1.1 \cdot 10^{-16}$. Disjoint thirds of the sentences give +0.009.
- *Outlet $\times$ topic* (different sentences per outlet).
 - Propensity to be rated biased: Federalist 0.77, Alternet 0.74, HuffPost 0.64, Breitbart 0.61, MSNBC 0.56, Fox 0.52, USA Today 0.41, Reuters 0.36.
 - The additive model fails globally: Freedman–Lane $p = 0.0005$. Without white nationalism it is still 0.0005, and without sport as well it is 0.001.
 - 35% of the 91 two-topic regions fail on their own.
 - The strongest loops pass through Reuters $\times$ white nationalism. Reuters is rated 0.27–0.28 on three other topics but 0.71 there. For example, Breitbart/Reuters $\times$ coronavirus/white nationalism gives $\Phi = +0.47$, $z = 6.3$.
- *Annotator ideology $\times$ outlet type* (paired per sentence; hostile-media loop).
 - $\Phi = d(\text{right outlets}) - d(\text{left outlets}) = +0.056, +0.070$ and $+0.053$ for the ideology cut-offs $(\leq -4, > 3)$, $(\leq -1, > 0)$ and $(\leq -6, > 5)$, with exact permutation $p = 0.011, 0.0003$ and 0.030 . Left annotators see right outlets as relatively more biased, and right annotators see left outlets that way.
 - Right-leaning annotators rate centre outlets as much more biased than left-leaning annotators do: 0.50 vs 0.34. Loops +0.12 and +0.18, both $p = 5 \cdot 10^{-5}$.

Prediction (`predict.py`, `predict_sens.py`). We fit nested logistic models: lean (M1), +topic (M2), +loop (M3, source $\times$ topic) and +trend (M4, source $\times$ year). Scores are out-of-sample log loss, in nats per prediction.

task	split	M1	M2	M3	M4
BASIL stance (210)	rolling origin 2013–19	1.389	1.453	1.473	1.579
BASIL spans (911)	rolling origin 2013–19	0.462	0.465	0.463	0.491
MBIC labels (17,775)	10-fold by sentence	0.640	0.635	0.637	–

- *Level 2 predicts.* In MBIC the outlet lean gains +0.034 nats per prediction over the base rate (0.674), and the topic effect adds +0.004.
- *Level 3 detects but does not yet predict.* The loop term’s gain is about zero everywhere: +0.001 on BASIL spans and −0.001 on MBIC. This holds for $C \in [0.1, 10]$ and under both rolling-origin and event-level CV. The largest gain is Fox–HPO spans, +0.004 to +0.006, and every 95% CI includes 0.
- *Stance.* Forecasting later years, the topic and trend terms hurt (−0.06 and −0.11 nats): the lean is stable, while party-specific and drifting terms overfit a small, non-stationary sample.
- *Tree-pooled loops do not help either (predict_tree.py).* Pooling the loop one level up the outlet tree, to outlet type $\times$ topic, gives MBIC log loss 0.6364 against 0.6352 for the additive model (gain −0.0011, CI [−0.0022, −0.0001]). The free loop gives 0.6354. Outlets of the same type do not share their topic-specific deviations: the Reuters $\times$ white-nationalism effect belongs to Reuters, not to centre outlets.
- *Why.* A loop is one contrast among four cells. It can be significant when pooled over all the data ($p = 0.0017$) while being worth only about 0.005 nats per prediction, which is below these test sets’ resolution.

Two targeted forecasts (trump_predict.py).

- *HPO favourability toward Trump.* HPO had 0 positive out of 105 directly aimed Trump spans in all years (Fox 4/82, NYT 11/149). We forecast each year $Y \in \{2017, 2018, 2019\}$ from the years before it.
 - The HPO $\times$ Trump cell (Jeffreys prior) forecasts $p = 0.014, 0.009, 0.007$ and is calibrated: it gives $P(0 \text{ of } n) = 0.75, 0.83, 0.81$ for $n = 20, 20, 31$, with log loss 0.014, 0.009, 0.007.
 - The global and per-source rates forecast $p \approx 0.17\text{--}0.22$. Under them the observed zeros have probability 0.001–0.02, and the log loss is 0.19–0.25.
 - Here a selective, source $\times$ target term predicts, because it is large, stable and backed by 34–74 earlier spans.
- *Does a Fox article criticise Trump?* (19 of 100 do.)
 - Leave-one-event-out log loss: base 0.496; Trump in the event 0.271 (AUC 0.83); adding NYT/HPO criticism of Trump on the *same* event gives 0.210 (AUC 0.96); adding the focal party gives 0.186 (AUC 0.97).
 - Rolling origin over 2016–19 ($n = 40$): $0.748 \rightarrow 0.533$, AUC $0.56 \rightarrow 0.83$.
 - Among Trump events, Fox criticises in 18/28 when NYT does and in 1/5 when NYT does not.
 - The log-loss gain from the cross-source terms is +0.061 (CI [−0.022, 0.135]) leave-one-out and +0.215 ([−0.061, 0.464]) rolling, so it is suggestive at this size.
 - This is level-1 coupling: the other outlets’ coverage of the same event is observed at the same time, so this is nowcasting, not a forecast of the future.

Forecasting “the probability of bias the next time source A covers topic t ” is therefore carried by the house lean. The loop is a diagnostic of *where* the lean-only picture fails; it is not yet a better forecaster.

Arity-graded holonomy (`arity_holonomy`, `arity_graded.py`). The loops above are holonomy loops of an additive ($\mathbb{R}$ -valued) connection: crossing the edge $A \rightarrow B$ adds $d_u(A, B) = v(u, A) - v(u, B)$. The *arity grade* of a unit relative to a loop is the number of the loop’s sources that measured it. Grade- g holonomy uses only grade- g units on each edge.

Proposition 36.1 (Filled cells are flat). *proved; checked* A unit measured by all k sources of a loop fills a $(k - 1)$ -simplex. At full grade every edge uses the same units and $\delta\delta = 0$, so the grade- k holonomy is exactly 0. Non-zero holonomy lives only on loops that no filled cell bounds.

- *BASIL, full coverage.* Grade-3 $\Phi = 1.1 \cdot 10^{-16}$, and every per-event loop is 0.
- *BASIL, thinned coverage* (keep each article with probability 0.75; 1,000 thinnings). Grade 3 always closes ($\leq 2 \cdot 10^{-16}$), and the grade-2 test is calibrated (0.057).
- *A plant visible only at grade 2.* We shift HPO’s scale by δ only on events where NYT is missing. Grade-2 detection is 0.08, 0.22, 0.44, 0.71 for $\delta = 0.5, 1, 1.5, 2$, with mean $\Phi_2 \approx \delta$.
- *MBIC annotator groups.* 1,600 grade-3 sentences close ($\leq 1.1 \cdot 10^{-16}$). The 100 grade-2 sentences have no C–R-only edge, so the grade-2 loop is undefined.

Phase change (`phase_change.py`). We scan for a single change point over 2011–19, using $\max_\tau |\text{after} - \text{before}|/\text{se}$ with event years permuted as the null. The prespecified candidates are 2015 (start of the Trump campaign) and 2017 (Trump presidency).

- *Lean.* No change survives the scan: Fox–NYT $p = 0.16$, Fox–HPO $p = 0.22$. The 2017 Fox–HPO candidate shift is +0.66 at $p = 0.032$ alone, which is not significant after correction.
- *Power.* A 1.0 / 1.5 shift planted from 2016 on a time-shuffled baseline is detected 0.56 / 0.94 of the time, and located at 2016.
- *Holonomy.* The HPO–NYT span loop has a regime change at 2016 (scan $p = 0.010$; change at 2017 -4.18 , $T = 4.25$). It disappears without Trump as a target ($p = 0.109$), so the phase change is HPO’s treatment of Trump.
- The Fox–HPO scan picks 2011 ($p = 0.020$, persisting without Trump). That is 2010 against the rest, an edge effect of the early period rather than a Trump effect.

NewsWCL50 [34] (`newswcl50.py`). The data are 10 events (2018) covered by HuffPost (LL), NYT (L), USA Today (M), Fox (R) and Breitbart (RR), with 1,636 valenced word-choice/labelling property codes. Every event is covered by all five outlets, so loops among outlets close (Proposition 36.1); structure lives only on outlet $\times$ target loops.

- *Level 2.* Favourability toward Trump rises monotonically left to right: log-odds LL -2.17 , L -0.99 , M -0.68 , R -0.36 , RR -0.57 . The slope is +0.40 per step, event-bootstrap CI $[+0.22, +0.57]$.
- *Level 3.* Of 100 outlet-pair $\times$ target-pair loops, 6 survive BH, and all involve HuffPost $\times$ Iran. For example, LL–RR $\times$ Trump/IRN gives $\Phi = -4.39$, $p = 0.0004$, BH 0.014, CI $[-5.69, -0.46]$. HuffPost is the most negative outlet toward Trump and the caravan, but the least negative toward Iran (-0.51 against -1.4 to -3.3).
- This rests on one event: event 4, where the codes frame deal compliance.

- *Global.* The no-three-way model fails on $5 \times 5 \times 2$: $G^2 = 37.9$, parametric-bootstrap $p = 0.0015$, within-event outlet permutation $p = 0.006$.

Gluing lattice (§3.3). The regions that glue form a downward-closed family, and we find no monotonicity violations. On NewsWCL50 the minimal obstructed regions are exactly the pairs $\{\text{Iran}, X\}$ in the target cover and $\{\text{HuffPost}, Y\}$ in the outlet cover. The largest glueable regions are $\{\text{Trump}, \text{USA}, \text{caravan}, \text{Democrats}\}$ and $\{\text{NYT}, \text{USA Today}, \text{Fox}, \text{Breitbart}\}$. On MBIC 28 of 91 topic pairs are obstructed, and the largest glueable regions have 10 of the 14 topics.

Reading and caveats.

- House lean and selective bias separate cleanly. At BASIL’s stance resolution, lean is all there is. Span favourability and MBIC show real non-closing loops.
- A non-zero loop is a measurable selectivity, not intent.
- Topic, party and ideology cuts are choices, and we report several.
- BASIL is small, so only large loops are detectable. Spans are not independent, and the event-cluster bootstrap handles that. The relative stance is annotated relative to the other articles on the same event, which suits the paired design but compresses absolute differences.
- The MBIC reading of the Reuters $\times$ white-nationalism loop (that annotators react to topic vocabulary rather than outlet framing) is a hypothesis to test.

Checks.

- The exact toric test (hypergeometric enumeration of the fiber) agrees with MCMC using the degree-4 Markov-basis moves to within 0.009, which is Monte-Carlo error.
- In BASIL, annotations are joined to articles by filename, because four HPO annotation files carry a duplicated uuid. Events are keyed by triplet, because main-event strings differ between the articles of a triplet.

Next.

- Test predictive value of loops with more data per loop (prediction is done above: lean predicts, loops do not yet).
- Build an automatic bias measure, needed to go beyond BASIL’s 100 events.
- Draw a gluing map at span resolution over party $\times$ window.

37 Dynamics: phase changes as natural and non-natural transformations

Setting. Two corpora describe events before and after a triggering event t^* (a buyout, a public accusation of bias, an election). The phase change is the passage from the presheaf of local data F (corpus 1) to F' (corpus 2). A per-source recalibration η_s is *natural* if it commutes with the restriction maps, i.e. the same change is seen on every region the source covers.

Proposition 37.1 (Naturality of a phase change). *proved; module dynamics* Let $\Delta = b' - b$ be the change of a bias table on the bipartite source–topic graph. The following are equivalent:

1. $\Delta(s, t) = \eta_s + \eta_t$ for per-source and per-topic recalibrations; that is, the change is a natural transformation of the gluing data;
2. Δ is a coboundary;

3. $\Delta\Phi = 0$ on every loop.

The non-natural part of a phase change is the loop residual of Δ , and it is measured by $\Delta\Phi$.

Proof. (1) $\Leftrightarrow$ (2) is the definition of δ^0 on the bipartite graph. (2) $\Leftrightarrow$ (3) holds because the 4-cycles span the cycle space and a graph has no 2-cells. $\square$

A natural change moves house leans and is invisible to loops. A non-natural change alters holonomy. Outcome-level changes, such as an inversion of the outcome probabilities, live in Layer 1 and are invisible to both.

A three-layer generator with planted phase changes (examples/dynamics/three-layer_dynamics.py).

- *Layer 1.* Events carry a latent outcome, $P(o) = (0.6, 0.25, 0.15)$, and 4 items (claims) of structural arity $k \in \{2, \dots, 5\}$.
- *Layer 2.* Sources A, B, C each cover an event with probability 0.8. Each reports item labels through its own relabelling (noise 0.1) and a stance $b = u_s + v_t + \mathcal{N}(0, 1)$.
- *Planted phase changes at t^* , one at a time:*
 - A** outcome inversion, $P(o) \rightarrow (0.25, 0.6, 0.15)$;
 - B** a natural lean shift, $u_B += 1$;
 - C** a non-natural shift, +1 for B on topic 0 only;
 - D** a monodromy twist, C relabels odd-arity items;
 - E** a coverage-graded shift, +1 for C only where A is missing.

The detectors are the outcome distribution, the paired lean change, $\Delta\Phi$ (naturalness), arity-graded monodromy of the loop $A \rightarrow C$ (after) $\rightarrow A$ (before) per item-arity parity, and coverage-grade-2 holonomy. Detection rates over 200 runs with 400 events per period:

planted	outcomes	lean	$\Delta\Phi$	mono. odd k	mono. even k	cov. grade 2
none	0.06	0.07	0.06	0.00	0.00	0.04
A inversion	1.00	0.05	0.05	0.00	0.00	0.04
B natural	0.05	1.00	0.06	0.00	0.00	0.06
C non-natural	0.09	0.77	0.90	0.00	0.00	0.06
D relabel odd k	0.07	0.06	0.04	1.00	0.00	0.04
E coverage	0.07	0.06	0.06	0.00	0.00	0.82

- Each invariant catches its own failure and stays at the null rate on the others. The one expected exception is that a topic-selective shift also moves the average lean.
- A natural change is caught by the lean detector and correctly *not* by the naturalness test. The relabelling is seen only in the odd-arity grade. This separates arity-graded monodromy (the grade is a property of the item, the group is the permutations) from coverage-graded holonomy (the grade is a property of the cover, the group is $\mathbb{R}$).
- At 150 events per period the naturalness and coverage rates are 0.52 and 0.40, and the others are unchanged.
- *Projected outcomes at time t .* A change-point forecast of the outcome distribution of the last 50 events beats the pooled forecast after an inversion (log loss 0.967 vs 1.037) and does slightly worse without one (0.993 vs 0.974), so it does not over-fit.

BASIL: the 2016 phase change is non-natural (`examples/bias3/naturality.py`). Splitting the change at 2016:

- The natural part is null. The stance lean changes by +0.19, +0.43, −0.23 for Fox–NYT, Fox–HPO, HPO–NYT, with $|z| \leq 1.4$; the stance loop over party changes with $|z| \leq 0.5$.
- The non-natural part is large for HPO. The span-favourability loop changes by $\Delta\Phi = +2.90$ ($z = 3.2$) for Fox–HPO and -4.26 ($z = -5.7$) for HPO–NYT. Without Trump as a target these are +1.30 ($z = 1.2$) and -3.35 ($z = -2.9$).
- These z values are nominal, because 2016 was chosen by the scan. The scan-corrected p is 0.010 with Trump and 0.109 without.
- The phase change therefore altered HPO’s party-selective favourability, not its overall lean.

37.1 Slepian (prolate) tools for short series, and a corpus with big events

Remark 37.2 (What the prolate operator does and does not give). `interpretation` Discrete prolate spheroidal (Slepian) sequences [38, 39] are the best band-limited basis for a window of n points, and the multitaper spectrum built from them has about $2K$ degrees of freedom per $2NW/n$ of bandwidth. The Shannon number $2NW$ is the number of resolvable modes. Consequences for our data:

- The 9-year BASIL window has one point per year. It resolves about 2 modes: a trend and a remainder. For the 4 years after 2016, the frequency resolution already equals the whole band, so there is no spectrum to compare.
- A level shift is broadband. The best test for it is a step test against a smooth trend, not a spectral criterion. A spectral change tests something else: a change of persistence, of volatility, or a new cycle.
- The prolate index k orders eigenfunctions by concentration (roughly, by sign changes). It is not the arity grade. A spectrum can be computed per arity grade, but this is bookkeeping, not a structural link.
- No spectral-gap order parameter is used: ordered eigenvalues of one operator cannot cross, and no concentration operator for the gluing problem has been specified.

We therefore use Slepian sequences as *estimation* tools (module `spectral`): a step test against a Slepian trend (constant, linear term and the first $2NW-1$ sequences, AR(1)-corrected); multitaper band ratios before/after; and Thomson’s harmonic F -test for a new periodic component, calibrated by Monte Carlo over the frequency scan.

Which detector sees which change (`examples/dynamics/spectral_synthetic.py`). Weekly AR(1) series, 73 weeks before and 73 after; rejection rates over 400 runs at level 0.05 (bands at 0.05/3):

planted after τ	Welch mean	Slepian step	MT low	MT mid	MT high	line F
none	0.09	0.06	0.04	0.00	0.01	0.05
level shift (+0.6 sd)	0.90	0.36	0.03	0.01	0.00	0.04
persistence (ρ 0.2 $\rightarrow$ 0.8)	0.36	0.21	0.28	0.00	0.99	0.03
new 8-week cycle	0.10	0.04	0.04	0.03	0.01	0.20
smooth trend, no break	0.91	0.07	0.04	0.01	0.01	0.03

- The plain before/after mean test calls a smooth trend a phase change (0.91). The Slepian step test does not (0.07). The price is power: 0.36 instead of 0.90 for a true step, because in 146 points a step and a slow trend are partly confounded.
- A persistence change is seen in the high band, as a loss of week-to-week volatility (0.99).
- A new cycle of amplitude 0.6 sd is only found 20% of the time in 73 weeks. Detecting cycles needs longer or denser series.

AllSides: six big events, 2015–2020 ([examples/allsides/](#)). The data are the AllSides articles of [37]: 37,554 articles with outlet, date, topic and an AllSides side (left, center, right). We keep the 16,069 dated articles that mention Trump (June 2015 to July 2020). No model is trained. The measures are fixed lexicons applied to the sentences that mention Trump:

- *adversarial*: fact-check or condemnation vocabulary (falsely, baseless, lie, misleading, racist, ...);
- *directed*: the same vocabulary with Trump within three words before it (“Trump falsely said”);
- *tone*: VADER sentiment.

Tone is not favourability: right outlets score *lower* on it, because their Trump sentences quote attacks on opponents. The outcome of an article is whether it contains an adversarial (or a directed) sentence.

The events are pre-specified, not scanned. We compare 26 weeks before with 26 weeks after (COVID: to data end). Sources are the three sides, topic cells are coarse AllSides topic groups, and the bootstrap resamples AllSides story groups.

event	adv. gap L–R	$z(\Delta)$	directed gap	non-nat. p	MT high (p)
election 2016	+0.100 → +0.054	−1.7	+0.020 → +0.009	0.72	0.34 (0.004)
inauguration	+0.085 → +0.047	−1.5	+0.017 → +0.016	0.66	0.60 (0.16)
Charlottesville	+0.052 → +0.035	−0.6	+0.018 → +0.014	0.86	1.60 (0.20)
Mueller report	+0.098 → +0.063	−1.3	+0.026 → +0.027	0.013	1.67 (0.16)
impeachment inquiry	+0.064 → +0.031	−1.3	+0.027 → +0.017	0.41	1.99 (0.09)
COVID emergency	+0.040 → +0.027	−0.5	+0.018 → +0.031	0.39	3.02 (0.07)

- *Outcome layer*. Left outlets are more adversarial toward Trump than right outlets before every event (z from 2.2 to 4.9). The directed marker gives the same order, about $4\times$ (article rates 0.021, 0.012, 0.005 for L, C, R). No event inverts the outcome probabilities. The broad gap narrows after every event, but no single change is significant, and the windows overlap.
- *Naturality*. Only the Mueller report gives a non-natural change ($p = 0.013$; 0.08 after Bonferroni over six events). The loop change sits in the justice cell: center −0.18, right +0.13. Reading the sentences shows what it is. After the report, right outlets use the vocabulary *against the investigation* (“baseless collusion claims”, “debunked dossier”). The directed marker does not move (+0.026 → +0.027). The non-natural change is a change of *target* of the vocabulary, not a change of stance toward Trump. The naturality test sees it because the target moved in one topic only.
- *Time*. There are 267 weekly points. No event produces a Slepian step ($|z| \leq 1.33$). The only spectral change is after the 2016 election: high-frequency power of the gap falls to 0.34 of its campaign level ($p = 0.004$; 0.07 after Bonferroni over 18 band tests). This is a volatility change, not a level change. An unrestricted scan finds its largest step in late March 2018

($|z| = 3.6$, block-bootstrap scan $p = 0.050$). No new periodic component is found after any event (line- F adjusted $p \geq 0.45$).

- *The cover itself changes.* After the COVID declaration a new topic cell (coronavirus) appears. The naturality test compares the same cover before and after, so it cannot see a new open set. A change of site is a different kind of phase change and needs its own test.

In this corpus the house lean toward Trump is stable across big news events. What changes is volatility (after the election) and the target of contested-truth vocabulary in one topic (after the Mueller report). Both are changes that a mean-only before/after comparison would not identify.

38 Is the topology tearing? Descent, a spectral certificate, and outcome toggling

The homotopy-theoretic results that move statements between categories, such as the transfer of full faithfulness to algebras in [41, Prop. 15.2.3.1], have one mechanism. The global object is a homotopy *limit* of local mapping spaces, and a limit of equivalences is an equivalence. Our mapping spaces are discrete (tables and probability simplices), so the statement needed here is the 1-categorical one.

Lemma 38.1 (Descent). *proved Let $\alpha : \mathcal{F} \Rightarrow \mathcal{G}$ be a map of presheaves of sets on a cover $\{U_i\}$ (contexts, sources, topics). If α_U is a bijection for every U_i and every finite intersection $U_{i_0} \cap \dots \cap U_{i_k}$, then α is a bijection on glued sections $\lim \mathcal{F} \rightarrow \lim \mathcal{G}$.*

Proof. Glued sections are the limit (equaliser) of the Čech diagram $\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$. A natural isomorphism of diagrams induces an isomorphism of limits. $\square$

Contrapositive: *every global change is visible on some context or overlap.* This is what the naturality test, the loop tests and the gluing lattice localise. The colimit side is the nerve lemma: for a good cover the nerve is homotopy equivalent to the union, so β_1 and H^1 do not depend on which good cover is used. Its empirical form is invariance under refinement of the cover.

Spectral non-tearing.^{proved} Put the cell means b on the edges of the bipartite source–topic graph, with weights $w_e = n_e/\hat{\sigma}^2$. The Hodge split is $b = X\beta + h$, where $X\beta = u_s + v_t$ is the natural (exact) part and h the loop part. Let $L_0 = X^\top W X$ (the signless Laplacian, which is similar to the Laplacian because the graph is bipartite). Its kernel has dimension β_0 , and $\beta_1 = E - V + \beta_0$. If a perturbation satisfies $\|\Delta L_0\|_2 < \lambda_2(L_0)$ (the Fiedler value), then by Weyl’s inequality $\lambda_2(L_0 + \Delta L_0) > 0$. The cover stays connected, so β_1 and the loop coordinates are unchanged: the topology does not tear. The certificate compares λ_2 with the 95% bootstrap quantile of $\|\Delta L_0\|_2$ (margin > 1 is certified), and reports:

- the obstruction $Q = \sum_e w_e h_e^2 \sim \chi_{\beta_1}^2$;
- its stability: $\text{SNR} = \|h\|_W / \text{rms} \|h^* - h\|_W$, which is about 1 under the null, and the median cosine between h and its bootstrap copies h^* ;
- survival when the topic cover is refined by splitting each topic’s clusters into random halves (50 refinements);
- leave-one-source-out.

(`amb.vigneaux/certificate.py`, `examples/certificate/run_all.py`.)

data (sources $\times$ topics)	β_1	margin	connected	Q, p	SNR	refined sig.
BASIL spans (3×2)	2	1.89	1.00	13.0, 0.0015	2.27	1.00
BASIL, Trump excluded	2	1.64	1.00	6.0, 0.05	1.34	0.42
MBIC (8×14)	66	1.24	1.00	131, 3×10^{-6}	1.36	0.98
NewsWCL50 (5×5)	15	0.13	0.36	29.7, 0.013	1.32	1.00
NewsWCL50, no Democrats	12	0.22	0.57	28.5, 0.005	1.30	1.00
AllSides adversarial (3×9)	16	1.11	1.00	18.1, 0.32	1.18	0.02
AllSides directed (3×9)	16	1.11	1.00	16.6, 0.41	1.10	0.02

- *Certified and robust: BASIL and MBIC.* The cover cannot tear under resampling (margin > 1), and the obstruction survives refinement in 100% and 98% of refinements. It is also stable in direction (median cosine 0.97, 0.81).
 - Leave-one-source-out localises BASIL’s obstruction at HPO: without HPO $p = 0.10$; without Fox 0.04; without NYT < 0.001 .
 - MBIC’s obstruction survives removing any single outlet: it is spread, as the gluing lattice found.
- *Not robust: BASIL without Trump.* It survives only 42% of refinements. The Trump spans carry most of BASIL’s obstruction.
- *The cover tears: NewsWCL50.* Only 36% of event-bootstrap replicates stay connected. The reason is a one-event target: all Democrats codes come from a single event, so resampling events detaches that vertex. Dropping it helps but does not cure it (57%), because with ten events other targets are also thin. The obstruction survives refinement and localises at the far-left outlet (removing LL gives $p = 0.72$). Its topology, however, is *not* certified at the event level: more events are needed before its loops can be trusted as structure rather than sampling accident.
- *Glues, and stays glued: AllSides.* The side $\times$ topic structure has no loop obstruction, under both markers and under refinement. This is consistent with the stable house lean of Section 37.1.

Outcome toggling.^{proved} Outcomes are read from probabilities, and a phase change flips an outcome only if it moves the probabilities far enough, *in the right direction*.

Proposition 38.2 (Toggle directions). *Let $o = \text{sign } f(b)$ with $f(b) = \langle a, b \rangle_W - c$, and split the reading vector W -orthogonally as $a = a_{\text{ex}} + a_{\text{loop}}$ ($a_{\text{ex}} \in \text{im } X$). Among changes Δb restricted to a subspace V , the smallest that flips o has W -norm*

$$r_V = \frac{|f(b)|}{\|P_V a\|_W} \quad (= \infty \text{ if } a \perp_W V).$$

In particular, natural changes (re-calibrations $u_s + v_t$) cannot flip an outcome that reads only loops, and loop changes cannot flip an outcome that reads only the natural part. An AMB (possibilistic) outcome reads supports, so it flips only when some probability reaches or leaves 0. Its toggle radius in the simplex is the smallest probability that must vanish or appear.

Proof. $f(b + \Delta b) - f(b) = \langle a, \Delta b \rangle_W = \langle P_V a, \Delta b \rangle_W$ for $\Delta b \in V$. By Cauchy–Schwarz the smallest $\|\Delta b\|_W$ reaching $f = 0$ is $|f(b)|/\|P_V a\|_W$, attained along $P_V a$. The special cases are $V = \text{im } X$ and $V = (\text{im } X)^\perp_W$. $\square$

This states precisely the picture that the corpora carry information underneath, biases and loops (the syzygies) shape it, and an outcome is set up by enough change on the right edges. Which phase changes *can* toggle a given outcome is decided by the Hodge split of its reading vector. How close a toggle is is decided by r_V .

COVID and the centre (`examples/allsides/toggle.covid.py`). The outcome is the alignment of centre outlets in coverage of Trump: $f = 2P_C - P_L - P_R$, negative when the centre is closer to the right. Pooled over topics, f reads only the natural part (share 1.00), so a uniform re-calibration of the centre could flip it.

- *Directed fact-checks*. f moves from -0.019 (se 0.009) in the 26 weeks before the emergency to -0.004 (se 0.012) in the 19 weeks after. The centre closed about 80% of the distance to the midpoint, but did not toggle.
- The change is not unusual: 48% of before/after pairs at other dates in 2016–2020 moved as much. Before COVID the toggle radius was 0.009, or 1.2 placebo sd.
- The broad adversarial marker barely moves: $-0.056 \rightarrow -0.039$, matched by 80% of placebos.

These data describe media alignment, not votes. They end in July 2020 and contain no electoral outcome. A single election observed in hindsight cannot validate a forecast. The claim “COVID set the stage for the election loss” therefore needs an outcome series in the same framework, such as approval or polling, and a toggle radius committed to in advance. The present data show a move of the centre toward the left on fact-checking Trump that is real in direction but within normal variation.

Gluability to outcome, on a stream (`examples/allsides/stream.outcome.py`, Figure 3). We treat the corpus as a stream of 26-week windows, stepped every 2 weeks from 2016 to 2020, with corpus 1 and corpus 2 the windows before and after an event. Each window records four things:

- the *outcome*, $O = [f \geq 0]$, where $f = 2P_C - P_L - P_R$ on directed fact-checks of Trump: “yes” means the centre sits with the left;
- its confidence $P(\text{yes}) = \Phi(f/\text{se})$;
- the window’s *gluability*: a parametric-bootstrap test of the loop-free (no-three-way) model on side $\times$ topic $\times$ outcome;
- the *counterfactual*: the fewest centre articles whose relabelling, or whose addition, would flip O .

For each event, Δf is split into a *rate* part (topic-specific rates change, topic mix fixed) and a *mix* part (the topics covered change). The possibilistic (AMB) outcome “a directed fact-check is possible for the centre on topic t ” is tracked with its exact toggle radius (Proposition 40.1).

event	O	$P(\text{yes})$	glue p	Δf (rate, mix)	flip cost	AMB
election	yes→yes	0.81 → 0.57	0.04 → 0.33	-0.017 ($-0.026, +0.010$)	3 of 294	4 topics
inauguration	yes→ no	0.66 → 0.28	0.46 → 0.14	-0.016 ($-0.020, +0.003$)	2 of 330	3 topics
COVID	no→ yes	0.03 → 0.72	0.21 → 0.48	$+0.036$ ($+0.044, -0.009$)	7 of 560	none

(Flip cost: centre articles relabelled, or added before COVID, in the window before the event, out of all centre articles; AMB: topics whose possibilistic outcome toggled.)

- *How yes became no, or no became yes*. At the inauguration the outcome flipped to “no”, and at COVID it flipped to “yes”. Both changes are driven by *rates* within topics, not by which topics were covered: the rate parts are -0.020 and $+0.044$, the mix parts only $+0.003$ and -0.009 . At the 2016 election the outcome stayed “yes” but weakened ($P(\text{yes})$ 0.81 → 0.57).

- *What more could have been done.* Before COVID the outcome was one of the few decided ones ($P(\text{yes}) = 0.03$). Seven more directed fact-checks by centre outlets, out of 560 centre articles over 26 weeks, would have flipped it. After the inauguration, 2 relabelled or 2 added centre articles would have restored “yes”. Across the stream the cost to flip a “no” never exceeds 7 articles (Figure 3, bottom). The outcome is decided ($|f| > 1.96 \text{ se}$) in only 3% of windows, and its window-to-window standard deviation is 0.023. *This binary outcome sits on its own boundary almost all the time*, so its flips are cheap and mostly within noise. The COVID flip ($z = 1.54$) is the largest move but not significant.
- *Gluability and outcome.* The loop-free model holds in 92% of windows. There the pooled outcome reads only the natural part of the bias table (Proposition 38.2), so a uniform shift of the centre’s rate flips it, and the flip costs above are exact. The windows that fail to glue (before the 2016 election, early and mid 2019) are those where topic-specific loops exist. There the same pooled f can also be moved by changing the topic mix.
- *The AMB layer moves independently.* Around the election and the inauguration the set of topics where the centre fact-checks Trump at all changed in 3–4 topics, while the probability-layer outcome barely moved. Around COVID the support did not change while the outcome flipped. The nearest AMB toggles after an event need only 2–5 articles to vanish under the glued model, against about 1–3 for the cell alone. The possibilistic outcome is fragile at these counts.

38.1 Gluing-lattice dynamics: what jumps and what moves continuously

A natural picture ties three things together: the gluing lattice, the possibilistic class γ , and the contextual fraction CF. In that picture an “obstructed” region has $\text{CF} > 0$ and $\gamma \neq 0$, a “glueable” region has $\text{CF} = 0$ and $\gamma = 0$, and crossing between them is a sharp phase transition. Part of this is right.

- *The two layers behave differently.* Support-level (AMB) invariants jump when a support changes, while probabilities move continuously inside a face. This matches Propositions 38.2 and 40.1.
- *Gluing, extension and IPF.* When the marginals of a region have a joint, the extension set is a non-empty convex polytope, and IPF from the uniform law converges to its maximum-entropy member, the I-projection [57]. When no joint exists, IPF does not settle.
- *A face function and continuous entropies.* S_0 , the log of the support size, is a face function and registers support changes. S_α for $\alpha > 0$ varies continuously inside a face.

The three objects nevertheless come apart:

1. $\gamma = 0$ is not gluing. AMB cohomology is sound but incomplete: $\gamma \neq 0$ certifies contextuality, but $\gamma = 0$ does not certify a global section.
2. $\text{CF} > 0$ is not $\gamma \neq 0$. A full-support contextual model has $\text{CF} > 0$ and $\gamma = 0$, because its possibilistic collapse is non-contextual. The two-box picture leaves out this whole region.
3. *Obstruction does not pin probabilities.* Marginals locked at $1/2$ is the special result for two-outcome non-flat cycles (Proposition 21.3), not a property of obstructed regions. In MBIC and NewsWCL50 the obstructed regions have freely varying probabilities.
4. *CF has no jump.* CF is the value of a linear program, continuous and convex in the model, and it leaves 0 continuously at the boundary of the non-contextual polytope. The discontinuities

are only in support-level quantities.

5. *The data gluing lattice is statistical.* Its verdicts come from a hypothesis test, so they depend on N (Section 38). It computes neither CF nor γ .

quantity	what it measures	at a support change	inside a face
γ (AMB)	possibilistic obstruction; sound, not complete	jumps	constant
S_0	log support size	jumps	constant
CF	probabilistic contextuality (LP)	continuous	continuous, piecewise linear along mixtures
$S_\alpha, \alpha > 0$	entropies	continuous	continuous
IPF	converges iff a joint exists (CF = 0)	–	–
gluing (data)	lattice statistical fit of the loop-free model on sub-covers	depends on N	depends on N

The path experiment (`examples/dynamics/gluing_lattice_dynamics.py`, Figure 4). On the CHSH scenario (a 4-cycle of contexts) take $e(\lambda) = \lambda \text{PR} + (1 - \lambda) \text{white noise}$. The empirical support uses a tolerance $\tau = 0.02$: a cell with $p < \tau$ counts as impossible. Along λ :

- *CF.* CF is 0 up to $\lambda = 1/2$, then exactly linear, $\text{CF} = 2\lambda - 1$. There is no jump.
- *γ and S_0 .* Both change once, at the support collapse $\lambda = 1 - 4\tau = 0.92$. There S_0 falls from $\log 4$ to $\log 2$ and $\gamma \neq 0$ switches on (strong contextuality).
- *The stretch the two-box picture misses.* On $\lambda \in (0.5, 0.92)$, $\text{CF} > 0$ but $\gamma = 0$: the model is probabilistically contextual and possibilistically clean.
- *IPF.* IPF converges exactly where $\text{CF} = 0$ (checked at every grid point). Beyond $\lambda = 1/2$ its residual stays bounded away from 0 and grows with CF.
- *The statistical verdict.* With N samples per context, a one-sided CHSH test flags the full cover as obstructed. At $\lambda = 0.52$ ($\text{CF} = 0.04$) its power is 0.11, 0.15, 0.45 and 0.93 for $N = 50, 200, 1000, 5000$. It tracks CF only when N is large.
- *The lattice.* Every proper sub-cover (3 of the 4 contexts) is a tree and always glues (Theorem 12.4), so the full cycle is the only possible minimal obstruction.

39 Probing inside arity: hull trees and the p -adic radius profile

The arity grading records how many sources cover a unit, but not how those sources relate on it. Proposition 36.1 explains why holonomy cannot help. Inside a filled cell every loop of differences on the same unit sums to zero. What survives is the *configuration* of the k values modulo translation, a point of $\mathbb{R}^k / \mathbb{R}\mathbf{1}$, and its combinatorial shadow is a tree.

Proposition 39.1 (Hull invariants of a filled cell). *proved; checked* Let a unit be covered by k sources with values $x_1, \dots, x_k$.

1. Every loop sum $\sum (x_{i_r} - x_{i_{r+1}})$ over sources on the unit is 0. The class $[x] \in \mathbb{R}^k / \mathbb{R}\mathbf{1}$ is not constrained.

2. The hull tree (single-linkage dendrogram of $|x_i - x_j|$) and its first merge, the coalition, are invariant under the unit effect $x \mapsto x + v_e \mathbf{1}$. After subtracting source effects they are invariant under every natural change $u_s + v_e$. Coalition frequencies in excess of an additive null are therefore structure inside arity that neither lean nor holonomy detects.
3. For integer values in $\mathbb{Z}_p$ the hull tree is the convex hull of the k points in the Berkovich line. Its vertices are the discs where the points split. The number of discs of radius p^{-m} containing them is a step function of m with integer jumps, and the class sizes give a composition of k that is refined as m grows.
4. For sum claims $x_i + x_j = r_{ij}$ over $\mathbb{Z}_p$, let $e(m) = \log_p$ of the order of the obstruction class of r in $\text{coker}(A) \otimes \mathbb{Z}/p^m$. This is canonical: it does not depend on the Smith basis. Then $e(m) = \max_i(\min(m, v_p(d_i)) - v_p(c_i))^+$ is piecewise linear with integer slopes in $\{0, 1\}$ and finitely many breakpoints. It vanishes exactly for $m \leq m^*$ (the local hulls glue at radius p^{-m^*} , Proposition 41.2), and it flattens at the torsion ceilings $v_p(d_i)$.

Proof. (1) telescopes. (2) Translation preserves every $|x_i - x_j|$; subtracting $\hat{u}_s$ removes source effects. (3) Two integers lie in a common disc of radius p^{-m} iff they agree mod p^m , and these classes refine as m grows. (4) In Smith form $\text{coker}(A) \otimes \mathbb{Z}/p^m \cong \bigoplus_i \mathbb{Z}/p^{\min(m, v_p(d_i))}$. The i -th coordinate c_i of r has order $p^{(\min(m, v_p(d_i)) - v_p(c_i))^+}$, and the order of the class is the maximum. $\square$

Item (4) is the elementary, arithmetic counterpart of the continuity theorems for radius functions on Berkovich curves: continuous, piecewise log-linear with integer slopes, and controlled by a finite graph [40]. Only this elementary part transfers. The deformation theory there (canonical Σ -module structures, quasi-unipotent monodromy) needs a p -adic differential equation, which our data do not have. We also use the p -adic metric only where the data are arithmetic (counts and sum claims). On real-valued bias scores the 2-adic distance is semantically arbitrary, so the real dendrogram is used instead.

Results (`amb.vigneaux/hull.py`, `examples/arity_probe/run.py`).

- *MBIC, annotators inside a sentence.* We use the 1,640 sentences labelled by left, centre and right annotators (ideology ≤ -4 , middle, ≥ 3), with the value being each group's bias rate. The null permutes the group labels within each sentence, which keeps ties, group sizes and rater heterogeneity (ties: 225 observed, 228 expected).
 - Left and centre annotators agree first far more often than chance: 578 against 461 expected, $p < 0.002$. The right group is the odd one out: left–right 441 against 484 ($p = 0.008$), centre–right 396 against 467 ($p < 0.002$).
 - The isolation depends on the outlet. On sentences from left outlets the right annotators split from the *left* (left–right 157 against 194, $p < 0.002$; centre–right 180 against 194, $p = 0.20$). On sentences from right outlets they split from the *centre* (centre–right 150 against 187, $p < 0.002$; left–right 191 against 197, $p = 0.59$).
 - The ideology leans are about ± 0.01 , so none of this is lean. It is structure inside a 3-ary cell, invisible to holonomy by Proposition 36.1. It refines the hostile-media loop of Section 36.
- *AllSides, sides inside a story.* In the 736 story groups covered by left, centre and right, using the tone of Trump sentences, coalition counts equal the additive null (258/251, 236/240, 242/246). They do not shift at the 2016 election or at COVID ($p \geq 0.20$). Tone carries no structure inside arity here.

- *NewsWCL50*. Only 6 event $\times$ target cells have at least 3 codes from all five outlets, which is too few to test.
- *BASIL*. Stance is coded as a relative ranking within each triplet, for example $(1, 0, -1)$, so equal spacing is built in: 68 of 100 triplets are ties against about 30 under the additive null. The probe needs a continuous value for each source.
- *Arity grading, p -adically* (synthetic; `examples/arity_probe/padic_arity.py`). Units of arity k measure integer source values with errors divisible by p^{a_k} ($a_2 = 1$, $a_3 = 2$, $a_4 = 4$), and each unit contributes sum claims for the pairs it covers.
 - The claims of any *single* unit glue at full precision (24 of 24 units): a filled cell is flat, the p -adic form of Proposition 36.1.
 - Keeping the units of arity $\geq K$, the claims glue to radius 3^{-1} , 3^{-2} and 3^{-4} for $K = 2, 3, 4$. The order profiles are $e = (0, 0, 1, 2, \dots)$, $(0, 0, 0, 1, \dots)$ and $(0, 0, 0, 0, 1, \dots)$.
 - The arity filtration therefore gives *nested gluing radii*, each set by the precision of its grade, with integer slopes above each radius.
 - Real bias scores are not arithmetic, so this check is synthetic. On data the real dendrogram of Proposition 39.1(2) plays this role.
- *p -adic profile* ($p = 3$, two independent even loops on $K_{2,3}$). Errors of 3^4 on one edge give $e = (0, 0, 0, 0, 0, 1, 2)$, so $m^* = 4$. Adding an error of 3^2 on the other loop gives $e = (0, 0, 0, 1, 2, 3, 4)$, so $m^* = 2$. An error of $18 = 2 \cdot 3^2$ gives the same profile as 9, because only the valuation matters. The breakpoint is the gluing radius and the slope is an integer.

39.1 Replication, the lean-invariant variogram, and the bridge to toggling

Coalition counts depend on the leans. The quantity that does not is the *variogram* of the sources on shared units,

$$\gamma_{ij} = \frac{1}{2} \text{Var}_{\text{units}}(x_i - x_j), \quad \gamma_{ij}^* = \gamma_{ij} - \frac{1}{2} \mathbb{E}[v_i + v_j],$$

where v_s is the sampling variance of source s 's value on a unit (for a mean of $n \geq 2$ binary raters, $\hat{p}(1 - \hat{p})/(n - 1)$). The correction stops groups with few raters from being pushed apart by noise alone.

Proposition 39.2 (Variogram tree). *proved; checked γ_{ij} is invariant under every natural change $x_{s,e} \mapsto x_{s,e} + u_s + v_e$. Its single-linkage tree is a tree-level invariant of the family of filled cells. The first merge of that tree, $\arg \min \gamma^*$, is the population coalition.*

Proof. $(x_i + u_i + v_e) - (x_j + u_j + v_e) = x_i - x_j + (u_i - u_j)$, and a constant shift does not change a variance. $\square$

A proposed alternative, “the coalition tree is non-trivial iff the values are not all equal, its shape is given by the p -adic order, and it is invariant under natural recalibrations”, needs three corrections:

- the first clause is true but empty;
- the p -adic order applies only to arithmetic data;
- the tree of the raw values is *not* invariant under source recalibration u_s . Only the variogram tree, or the tree after removing source effects, is.

Proposition 39.3 (Majority reads the tree). *proved; checked* For three binary sources with marginals p_i and $\gamma_{ij} = \frac{1}{2}P(x_i \neq x_j)$,

$$P(\text{majority} = 1) = p_1 + p_2 + p_3 - (\gamma_{12} + \gamma_{13} + \gamma_{23}) - 2P(x_1 = x_2 = x_3 = 1).$$

With fixed marginals, and hence the same bias cochain, lean and loops, the majority outcome can lie on either side of $\frac{1}{2}$. For example, with all $p_i = 0.45$:

- independent votes give 0.425;
- all three always agreeing gives 0.45;
- exactly two agreeing on “biased” (probability 0.675, each pair equally often, otherwise all “unbiased”) gives 0.675.

Proof. $1[\text{maj}] = x_1x_2 + x_1x_3 + x_2x_3 - 2x_1x_2x_3$, and $E[x_ix_j] = \frac{1}{2}(p_i + p_j) - \gamma_{ij}$. The examples are checked in `tests/test_hull.py`. $\square$

This is the bridge to Proposition 38.2. That proposition covers outcomes that are linear in the mean cochain b . A majority outcome, or any AMB outcome that reads the joint support, also reads the variogram tree. A phase change that leaves every mean unchanged, with no natural part and no loop part, can therefore still toggle it. Such a change sits at second order, at the level of the coalition tree, and holonomy cannot see it.

Replication (`examples/arity_probe/replicate.py`).

- *MBIC, noise-corrected variogram, 3 groups.*
 - $\gamma_{LC}^* = -0.002 \pm 0.003$, $\gamma_{LR}^* = 0.005 \pm 0.003$, $\gamma_{CR}^* = 0.014 \pm 0.004$. The first merge is left–centre (bootstrap support 0.97). The only clearly non-zero disagreement is centre–right, at 3.3 se.
 - By outlet type: centre–right is largest on right-outlet sentences (0.021 ± 0.007) and weaker on left-outlet sentences (0.008 ± 0.006).
 - The coalition counts are the same for articles up to 2019 and from 2020, though the ratings were all collected in 2020.
- *MBIC, 5 finer groups (≤ -7 , $-6..-2$, $-1..1$, $2..6$, ≥ 7).* No structure: every $|\gamma^*| \leq 0.011$ with se 0.004–0.007, and the first merge has bootstrap support 0.37. With fewer raters per group the noise correction dominates. The 3-group result is therefore not refuted, but it does not refine into a finer ideology gradient.
- *BABE expert annotators.* Eight annotators rate all 1,700 sentences (SG1), or about five per sentence over 3,674 sentences (SG2). At the annotator level the variogram does not follow ideology: the Mantel correlation of γ_{ij} with $|\text{ideology}_i - \text{ideology}_j|$ is -0.26 ($p = 0.95$) and -0.02 ($p = 0.50$). Annotator 1 (right, +5) is closest to annotator 10 (left, –15). Group-level comparisons in BABE are confounded by individuals: SG1 has one right annotator, and in SG2 groups of at least two raters rarely overlap.

Assessment. The coalition structure inside MBIC’s 3-ary cells is real and stable over article years: the right crowd is the odd one out, mainly against the centre and mainly on right-outlet sentences. It does not refine to finer ideology groups, and trained experts show no ideology-driven agreement geometry. It is a crowd-annotation effect, not yet a general law.

40 Moduli: toric strata, tree space, and the singular strata of grammars

Moduli language earns its place only where the quotient is not a vector space. The bias tables modulo natural changes form $C^1/\text{im } \delta = H^1 \cong \mathbb{R}^{\beta_1}$, with the loop sums as coordinates. So do the $\mathbb{R}$ -valued character varieties $\text{Hom}(\pi_1, \mathbb{R})$. Calling these moduli spaces is correct, but it adds no theorem. Three places do add content: the *boundary* of the toric model (AMB outcomes), the *space of trees* (hull and coalition trees), and the *singular strata* of grammar space (hierarchy recovery). Derived and stacky structure is not needed for any result below. It becomes relevant only for long chains of hierarchy levels, and it is deferred.

40.1 AMB outcomes are boundary strata of the toric model

A hierarchical log-linear model with design matrix A (the columns a_j record which margins cell j contributes to) is a toric variety. Its closure is a disjoint union of strata, one for each *facial set* S : a set of cells that is the zero set of a supporting hyperplane of $\text{cone}(A)$, i.e. $h \cdot a_j = 0$ on S and $h \cdot a_j > 0$ off S . The stratum of S is the model restricted to the cells in S [42, 43, 44, 45]. Under the moment map, strata correspond to faces of the marginal polytope [46, 47]. The support of a point, meaning which cells are possible, is its possibilistic (AMB) outcome [1], and Section 21 already stratifies the engine by support. What the toric model adds is which supports are *reachable* as limits of the model, and how far away they are. So the outcome layer says *which stratum* a point is in, and the probability layer says *where* it is inside the stratum.

Proposition 40.1 (AMB toggle radius). *proved; checked* Let p lie in the interior of a toric model and let S be a facial set.

1. The reverse-KL closest point of the S -stratum is $p|_S/p(S)$, at distance $-\log p(S)$. Consequently the smallest change of the probability layer, within the model, that makes cell c impossible is

$$r(c) = \min\{-\log(1 - p(F)) : c \in F, F^c \text{ facial}\}.$$

2. Every Markov move $m = m^+ - m^-$ gives $\sum_{j \in m^+} a_j = \sum_{j \in m^-} a_j$. Hence a co-facial set F that meets one side of a move meets the other side. In particular, in the no-three-way (glued) model with $I, J \geq 2$, a single cell is never co-facial: a possibilistic outcome cannot toggle in one context alone.

Proof. (1) $p|_S/p(S)$ lies in the closure, since facial restrictions of the model are in its closure [42]. For any q supported on S , $\text{KL}(q||p) = \text{KL}(q||p|_S/p(S)) - \log p(S) \geq -\log p(S)$ by Gibbs' inequality. (2) Apply $h \geq 0$ to the move identity. If $h \cdot a_c > 0$ for some $c \in m^+$, then some $j \in m^-$ has $h \cdot a_j > 0$. The basic move of any $2 \times 2 \times 2$ sub-box then shows that a single-cell F is impossible. $\square$

We compute $r(c)$ exactly by a mixed-integer program: minimise $\sum_j p_j z_j$ subject to $h \cdot a_j \geq 0$, $h \cdot a_j \leq M z_j$, $h \cdot a_c \geq 1$. An LP relaxation gives an achievable upper bound. The exact values were checked against brute-force enumeration of all facial sets on $2 \times 3 \times 2$. The minimal co-facial sets there include the context, the slice, and loop-shaped sets such as $\{(0, 0, 0), (1, 1, 1)\}$ on $2 \times 2 \times 2$ (`amb_vigneaux/strata.py`).

Results (`examples/strata/amb_toggle.py`; evidence in sentences or articles, $N \cdot p(F)$).

- *MBIC* (outlet $\times$ topic $\times$ majority label; 176 live cells, with empty outlet–topic contexts as structural zeros). Under the glued model, making a label impossible in one context costs a median of $2.0\times$ the evidence of the saturated model, and up to $17\times$.

- The cheapest toggle removes the *whole context* for 163 cells, a source–label slice for 9, and a topic–label slice for 4. The possibilistic outcome of a single outlet–topic context can only change by losing that context or a whole slice.
- The nearest toggles are thin contexts with 3–5 sentences, such as HuffPost×environment and Reuters×student debt.
- *AllSides* (side × topic group × adversarial, 2015–2020; 48 cells). The median ratio is 1.6×. The cheapest toggle is a whole context for 25 cells, a topic slice for 20, and a source slice for 3. For example, “adversarial is impossible on healthcare” can only happen for all three sides at once (7 articles).
- Around COVID, the centre’s “adversarial is possible” outcome moves *closer* to a toggle on justice (3.7% → 1.8% of the window’s articles), and *further* from one on media (1.1% → 3.7%) and politics (2.5% → 4.7%). These are two opposite moves of the same outcome layer, invisible to the pooled alignment of Section 38.

40.2 Tree space: the coalition test in BHV space

Hull, variogram and hierarchy trees are points of the space of phylogenetic trees [48]. That space is the tropical moduli space $\overline{M}_{0,n}^{\text{trop}}$ and the Berkovich skeleton of $\overline{M}_{0,n}$ [49, 50]. The p -adic and real hull trees of Section 39 are therefore the same object read in two settings, and the real points of $\overline{M}_{0,n}$ are tiled by associahedra [51], which links trees to the A_∞ structure of Section 41.

For three sources the space T_3 is a spider: three half-lines, one per coalition, glued at the star tree. A cell gives the point (coalition, internal edge $h_2 - h_1$). The metric is $|s - t|$ on the same leg and $s + t$ across legs. The Fréchet mean is *sticky* [52]: with leg moments $m_a = E[s \mathbf{1}_a]$, it lies on leg a at $m_a - \sum_{b \neq a} m_b$ when that is positive, and at the origin otherwise. Changes are tested by the energy distance [53] with a permutation null (`amb.vigneaux/treespace.py`, `examples/strata/treespace_run.py`).

- *MBIC* (1,640 sentences). The mean sticks at the star. The left–centre leg carries excess mass (moment 0.118 against 0.082 under within-sentence label permutation, $p < 0.002$), and the centre–right leg carries a deficit (0.067 against 0.080). So there is a strong left–centre *tendency*, but not a population coalition: its mass is less than the other two legs together. Left against right outlets: energy $p = 0.066$. Articles up to 2019 against 2020: $p = 0.23$.
- *AllSides* (736 stories, tone). The mean sticks at the star. Before/after the 2016 election $p = 0.17$; before/after COVID $p = 0.058$, with the left–centre moment rising from 0.058 to 0.095 on 85 post-COVID stories. This is suggestive, not significant.

In tree space, then, the coalition structure of Section 39 is a displacement of mass toward one leg. It never unsticks the mean, and no event moves it significantly at 0.05.

40.3 The singular strata of grammar space

The parameters of a hierarchical grammar, modulo relabelling of the symbols at each level, form a space that is singular along the strata where two symbols at one level coincide. Singular learning theory [54, 55] says that such strata, not the parameter count, govern what data can resolve. We measured the likelihood distance from the true grammar to each merge stratum. For each level l and each pair of level- l symbols, we average the pair’s rows (weighted by usage), collapse the child axes above, and compute the held-out difference in $\log p$, with a paired se over 4,000 sequences (`benchmarks/o6/merge_strata.py`).

Proposition 40.2 (The root level is not identifiable). *proved; checked* The likelihood depends on the root level only through the mixture $\sum_s \pi_s P_L[s]$. Merging any two root symbols with usage weights leaves $\log p(x)$ exactly unchanged. Root-symbol structure lies on a positive-dimensional fibre of constant likelihood, and no learner can recover it from data.

Proof. $p(x) = \sum_s \pi_s \sum_{a,b} P_L[s, a, b] \beta_a \beta_b$, where β are the inside values of the children. Replacing s, s' by one symbol with prior $\pi_s + \pi_{s'}$ and rule row $(\pi_s P_L[s] + \pi_{s'} P_L[s']) / (\pi_s + \pi_{s'})$ leaves the sum unchanged. (Checked exactly in `tests/test_root_identifiability.py`.) $\square$

Δ : median (min)	level 1	2	3	4	5	6
$V = 8, L = 6$	5.07 (1.37)	2.28 (0.93)	0.99 (0.43)	0.37 (0.07)	0.13 (0.035)	0 (all)
$V = 16, L = 4$	0.69 (0.17)	0.27 (0.13)	0.14 (0.0016)	0 (all)		

- Every pair of used symbols below the root is resolvable ($\Delta > 2$ se). The merge distance falls by about $2.5\text{--}3\times$ per level toward the top, and it is exactly 0 at the root.
- This resolves the O6 finding that “likelihood does not certify structure”. The worst top-level shortfalls of the baseline, 0.55 against 0.88 at level 4 of $V = 16, L = 4$ and 0.39 against 0.74 at level 6 of $L = 6$, are at the *root*. Root recovery is not a well-posed target, and the “Bayes ceiling” there uses the true grammar and is unattainable. The O6 scorer now marks the root level as non-identifiable, and structure is scored on levels $1 \dots L - 1$.
- At $L = 6$ the likelihood gap EM leaves (0.005) is below every level-5 merge distance (≥ 0.035). The level-5 shortfall is therefore slow EM along weakly curved directions, not a likelihood blind spot. At $V = 16, L = 4$ one level-3 pair costs only 0.0016 nats, below the 0.02 gap: there the likelihood genuinely cannot separate that pair at this sample size.
- Split-merge (Section 11) moves the model onto a merge stratum and leaves it along another branch. It crosses between basins through the singular locus, a move that EM’s small uphill steps do not make.
- Below the root, generic identifiability of latent tree models up to relabelling [56] agrees with the positive merge distances measured here.

40.4 Local window EM: learning one level at a time

The strata above suggest a gradient-free strategy that avoids global coupling: fit the hierarchy bottom up, one level at a time, from random starts, looking only at a local neighbourhood (`benchmarks/o6/local_em.py`).

- *A lone node is not identifiable.* A node’s child pair is a single categorical draw, and in a mixture of single draws, merging two symbols with usage weights is exactly free (the argument of Proposition 40.2; checked in `tests/test_root_identifiability.py`). The smallest identifying neighbourhood is a *two-level window*, parent $\rightarrow$ (left, right) $\rightarrow$ child pairs, in which the sibling is the second view [56]. Level l is kept, the parent table is refitted in the next window, soft inside messages are passed up, and the root is one symbol (Proposition 40.2).
- *Symmetry must be broken at the start, not annealed.* Tempered E-steps from $T_0 = 2$ drove every symbol to the same table, the symmetric fixed point exactly, which EM cannot leave (level-1 NMI 0.05). Random Dirichlet starts at $T = 1$ break the symmetry.

- *Restarts land on merge strata.* At the upper levels the fitted symbols duplicate one another (held-out merge cost ≈ 0). A targeted local split-merge helps: merge the pair that the diagnostic finds redundant, split the most loaded symbol, and keep the change only if the held-out window likelihood improves. At $L = 6$ this raised level-4 NMI from 0.16 to 0.40 and level-5 from 0.03 to 0.23.

NMI on the identifiable levels (test split, $N = 20,000$):

$V = 8, L = 6$	L1	L2	L3	L4	L5	gap
baseline (full split-merge)	0.909	0.970	0.957	0.794	0.655	0.005
local window EM only	0.901	0.894	0.663	0.398	0.229	5.71
local + 60 global EM sweeps	0.948	0.923	0.809	0.593	0.413	1.75
hybrid (local + short split-merge)	0.963	0.970	0.957	0.768	0.668	0.049
Bayes ceiling	0.993	0.970	0.956	0.866	0.872	0
$V = 16, L = 4$	L1	L2	L3	gap		
baseline	0.983	0.986	0.810	0.020		
local window EM only	0.978	0.934	0.748	0.79		
local + 60 global EM sweeps	0.987	0.981	0.815	0.28		
hybrid	0.977	0.982	0.832	0.021		

- *As an initialiser, local EM helps.* After 10 warm-up sweeps the hybrid is at a validation gap of 2.47 nats, where the clustered initialisation needs 60 sweeps to reach 5.85. With half the global split-merge budget (2 rounds instead of 4), the hybrid matches or beats the baseline's structure on most identifiable levels. The exceptions are L4 at $L = 6$ and L1 at $V = 16$. At $V = 16$, local EM followed by 60 plain EM sweeps, with no global split-merge at all, already equals the baseline on levels 1–3.
- *On its own it does not reach the top.* Going only deeper discards the top-down (outside) evidence, and errors compound upward: local-only NMI falls from 0.90 at L1 to 0.23 at L5. This agrees with the merge distances of Section 40.3, which shrink about $2.5\text{--}3\times$ per level.
- *Cost.* The window E-step is $O(K^3)$ per node. The local fits cost more than the clustered initialisation, so the total time exceeds the baseline's. A sparse parent table is the obvious remedy.

41 Massey products, and the p -adic and hierarchy loop tests

Massey products (massey). The averaged Shannon action is not multiplicative, which is why Vigneaux's cup product needs $F \otimes F$. The *pointwise* action $(X.f)(P)(\omega) = f(P|_{X=X(\omega)})(\omega)$ is multiplicative, so local pointwise cochains form a differential graded algebra. The coboundary is Hochschild's,

$$(\delta f)[X_1 | \dots | X_{n+1}] = X_1.f[X_2 | \dots] + \sum_i (-1)^i f[\dots | X_i X_{i+1} | \dots] + (-1)^{n+1} f[X_1 | \dots | X_n],$$

and the cup product is $(f \smile g)[X_1 | \dots | X_{n+m}] = f[X_1 | \dots | X_n] \cdot (X_1 \cdots X_n).g[X_{n+1} | \dots]$.

Proposition 41.1 (Surprisal is formal). *proved; checked numerically* Let $s[X](P)(\omega) = -\log P(X = X(\omega))$ and $u_n = s^n/n!$, both local 1-cochains. Then

$$\delta u_n = - \sum_{i+j=n, i,j \geq 1} u_i \smile u_j.$$

Hence $s \smile s = -\delta u_2$ is a non-zero but exact cochain, and every Massey power $\langle s, \dots, s \rangle$ is defined and contains 0. The A_∞ structure generated by the surprisal is formal. The generating function of the defining system is $\sum_n u_n = e^s - 1 = 1/P - 1$.

Proof. With $a = \log P(x)$ and $b = \log P(y|x)$ we have $\delta(\log^n P)[X|Y] = b^n - (a+b)^n + a^n = -\sum_{0 < i < n} \binom{n}{i} a^i b^{n-i}$, while $(u_i \smile u_j)[X|Y] = (-a)^i (-b)^j / (i! j!)$. Numerically, on three binary variables, the identity holds to $3 \cdot 10^{-14}$ for $n \leq 4$, the Leibniz rule to $3 \cdot 10^{-15}$, and $s \smile s$ has size 12.5 (`tests/test_massey.py`). $\square$

On full-support laws the only class is s (Proposition 33.3), so there are no non-trivial Massey products there. For $\alpha \neq 1$ the pointwise Tsallis action $P(x)^{\alpha-1} f(P|x)$ is not multiplicative, so this DGA does not exist. The one non-trivial higher structure the paper does have is the arity of parse trees: faces of the associahedron correspond to planar trees, and a node of arity k spans a cell of dimension $k - 2$. These are combinatorial, not homotopical.

The p -adic loop test (`padic_loops`). Take claims $x_i + x_j = r_{ij}$ in $\mathbb{Z}_p$.

Proposition 41.2 (Sign twist and gluing radius). *proved; checked*

1. The edge acts by $x \mapsto -x + r$, so the obstruction lives in H^1 with sign-twisted coefficients. An even cycle needs its alternating sum to vanish. An odd cycle is always solvable when 2 is a unit (p odd), and carries the parity obstruction when $p = 2$.
2. Let m^* be the largest m for which the system is solvable mod p^m , computed exactly by the Smith normal form over $\mathbb{Z}$. There is x with $|x_i + x_j - r_{ij}|_p \leq \rho$ for all claims iff $\rho \geq p^{-m^*}$. So the local Berkovich hulls glue exactly at radius p^{-m^*} , and on one even cycle $m^* = v_p(\text{alternating sum})$.

Checks:

- a 3-adic 4-cycle with alternating sum 1, 3, 9, 27, 54 gives depths 0, 1, 2, 3, 3;
- triangles always glue for $p = 3$, and for $p = 2$ fail exactly when $\sum r$ is odd;
- an error planted at digit m of one claim on K_8 gives depth m , for $m = 0, \dots, 5$.

No gradients are needed: the system is linear over $\mathbb{Z}_p$ and is solved digit by digit.

The hierarchy loop test. A claim “ i and j meet at depth exactly d ” is realisable by a tree iff no claim (i, j, d) has i and j already joined by claims all deeper than d . Equivalently, on every cycle the shallowest depth is attained at least twice (ultrametric). We check this by union-find in decreasing depth, with a witness path for each violation.

Proposition 41.3 (Node-wise distortions are invisible). *proved; checked* If every claim’s depth is changed by a function of its true meeting node only, no violation appears. The two shallowest claims on a cycle meet at the same node and move together. Detectable failures need item-level disagreement between sources. This is the hierarchy analogue of Proposition 35.2.

Synthetic test with graded arity (`examples/hierarchy_loops.py`). The tree has depth 5, and each node draws an arity in $\{2, 3, 4, 5\}$. Two sources report exact meeting depths on random leaf pairs. Source B misplaces 4 of 80 leaves, and only under odd-arity nodes.

- Shifting every odd-arity node by one level gives 0 violations.
- Noiseless, there are no false alarms. Every violation’s witness touches a misplaced leaf and lies in the odd-arity grade.

- Detection of top-level misplacements rises with data: 0.10, 0.33, 0.82 at 100, 200, 400 claims per source. Deep misplacements are rarely seen (≤ 0.07), because only pairs meeting below that level can reveal them.
- Full results (100 runs per cell; the null 95th percentile matches the number of claims):

misplacement level	100 claims/source		200		400	
	1	3	1	3	1	3
detection, noise 0	0.10	0.00	0.33	0.01	0.82	0.03
detection, noise 0.05	0.06	0.09	0.01	0.04	0.08	0.05
detection, noise 0.10	0.05	0.04	0.10	0.04	0.13	0.06

Null violation means are 0 without noise, 23 / 174 / 558 at noise 0.05, and 38 / 215 / 585 at noise 0.10, for 100 / 200 / 400 claims. With noise, witness localisation is 0.21–0.32 and the odd-grade share is 0.31–0.55, against an odd-node base rate of 0.51.

- With 5–10% depth noise the violation count is dominated by noise: detection is ≤ 0.13 , and witness localisation falls to about 0.25. A noise-robust version (for example an EM noise model, as in the contradiction meter) is open.

42 Detection: a map of the ideas and theorems

One idea. A contradiction is a failure to glue locally consistent data. The meter keeps three failures apart:

- *self-contradiction*: a unit’s support is empty;
- *overlap disagreement*: $d^0 \neq 0$, the deficit;
- *contextuality*: $d^0 = 0$ but $H^1 \neq 0$, which is “locally consistent, globally inconsistent” [27].

Proposition 42.1 (Localisation). *[S]proved Let g be an operation system with values in an abelian group A .*

1. *On a cycle graph with holonomy $h \neq 0$, for every edge e there is a flat system that differs from g on e alone, namely $g - h \mathbf{1}_e$ with the orientation of e along the cycle. So the edge data do not determine which edge is wrong. On a general graph, the minimal corrections to a flat system are likewise not unique.*
2. *Declaring a spanning tree T trusted removes the ambiguity. There is exactly one flat system that agrees with g on T : its value on each cotree edge is the integral of g along the tree path. The discrepancy sits on the cotree edges of the non-flat fundamental cycles.*

Proof. 1. Changing one edge of the cycle by $-h$ changes the holonomy by $-h$.

2. A flat system is δc for a potential c , and $\delta c = g$ on T determines c up to a constant.

□

Design rules.

1. Build the cover from provenance, not from token distance.
2. Record overlap disagreements; do not average them.
3. Report cycles. Blame an edge only against a declared trusted set (Proposition 42.1).

4. Read phase with loop quantities (holonomy, parity product, deficit), and magnitude with per-context information.
5. Give every detector a planted positive and a null.

Part IV

Status

43 Status of every claim

- **Proved:**

- T0 and its corollary; holonomy obstruction and connecting class;
- Propositions 8–10 of [O]; non-unification; vanishing (trivial action); Corollary 28;
- the p -adic proposition;
- identifiability (Prop. 17.1) and its T0 corollary; the Chernoff bound;
- Theorem 18.1 and Layer-2 blindness; character eigenvalues;
- Theorem 19.4; the lower bound of Theorem 19.5;
- orientation on $\mathbb{Z}/3$ and $\mathbb{Z}/5$; convexity of KL balls and exact pair radii;
- $\text{Aff}(\mathbb{Z}/p)$ invariants; the codimension formula;
- R and E ; the zero set of the deficit;
- support stratification (Proposition 21.1); localisation (Proposition 42.1), its corollary, and two-outcome supports (Proposition 21.3);
- the logical Bell bound (Proposition 22.1); the lower bound in Proposition 22.2; the identities in Remark 22.3; dimension as a cocycle on affine supports over $\mathbb{F}_p$ (Proposition 22.4);
- O2 for the true grammar: Propositions 27.1, 27.2, 27.5, Theorem 27.3, and the root collapse;
- the band proposition (Proposition 33.1) and twist-invariance for symmetric local solutions (Proposition 33.2); holonomy on the graph stratum with $\mathbb{R}^\Omega$ coefficients (Proposition 33.3), non-exact under Vigneaux locality; weak-edge contextuality without holonomy (Proposition 34.1); the admissibility criterion (Corollary 33.5); two-sided acyclicity (Proposition 35.1); single-source consistency (Proposition 35.2); formality of the surprisal (Proposition 41.1); the p -adic gluing radius and sign twist (Proposition 41.2); invisibility of node-wise distortions (Proposition 41.3); filled cells are flat (Proposition 36.1); naturality of phase changes (Proposition 37.1); descent (Lemma 38.1); toggle directions (Proposition 38.2); hull invariants of a filled cell and the p -adic order profile (Proposition 39.1); the variogram tree (Proposition 39.2); majority reads the tree (Proposition 39.3); the AMB toggle radius and Markov-move closure of co-facial sets (Proposition 40.1); non-identifiability of the root level (Proposition 40.2); decoding as Bayesian inversion (Proposition 3.2); functoriality of mutual-information persistence (Proposition 3.4).

- **Computed:**

- V0–V4a identities; H^0 , H^1 , H^2 of small semilattices;
- the mod-3 grid over five rings; possibilistic fragility; Hardy false negative;
- Baudot–Bennequin identities; the Tsallis δ_α cocycle;
- Counterexamples 19.7 and 27.4; the O2 rank table and root collapse;
- Z^1 of the operation-twisted Vigneaux complex on $\mathbb{Z}/3$ and $\mathbb{Z}/2$ cycles, and the $\mathbb{R}^\Omega$ table with the $O(\varepsilon)$ near-cocycle count (Section 33).

• **Measured:**

- transformer negative; split–merge recovery; instrument calibrations;
- Wald calibration and N_{80} ; equality in Theorem 19.5;
- wall absorption; $\text{Aff}(\mathbb{Z}/5)$ recovery; graded-extension uniqueness;
- the gated-test table; deficit behaviour; ridge; threshold windows;
- grammar-guided transformers on Corpus B (Section 28);
- equality in Proposition 22.2 for $n = 4$; the dimension deficit from estimated supports; S_2 versus Shannon estimation (no gain); the contradiction-meter benchmark (Section 23); recursive-training collapse (Section 29); unknown branching with chunks and split–merge (Section 30); monodromy and the local certificate phase (Section 31, Proposition 31.3); the covariant perturbation split (Theorem 31.4); the Rényi deficit dial (Proposition 32.1); the O6 scaling ladder (Section 11); three-level bias on BASIL, MBIC and NewsWCL50, arity-graded holonomy and phase change (Section 36).

• **Open:**

- O1 identifiability threshold $\theta(V, L, \ell, m)$; O2 on fitted tables (statistical equality of upward rows); O4 power of β_1 ; O6 at $V = 1024$, $L = 16$ (benchmark and baseline in Section 11; our baseline breaks at $V \approx 64$ –128 and $L \approx 6$ –8); O7 recurrent-depth transformers;
- Problems 21, 22 (23 blocked; 24 settled);
- $H^{\geq 2}$ for δ and δ_α ;
- the packing proof for Theorem 19.5; an invariant separating noise from holonomy under non-covariant noise;
- integer-coefficient persistence for decorated barcodes;
- the conjecture that operations visible to Layer 2 are lossy and admit no *cohomological* invariant. It is refuted in part: the loop spectrum is a non-cohomological invariant;
- extending the gated test to triple contexts;
- equality in Proposition 22.2 for general n ; the $\mathbb{Z}/n$ modulus–phase analogue; non-triviality of the dimension class; the deficit under non-covariant noise; the rank of V_0 for richer ansätze; the soft near-cocycle count on estimated (noisy) data; an escort-map Frobenius on the α -complex (Remark 33.6); a twist-sensitive H^1 off the stratum (full support is blind in our ansätze).

• **Withdrawn or corrected:**

- the embedding probe [H];

- “ $I_2 = \delta H$ ”;
- the two-point entropy formula for noisy shifts;
- “information bounds dictate CF”;
- “the spectrum reads h ”: it is the labelled eigenvalue;
- “the optimal non-contextual part is generally not unique”: it depends on the scenario;
- “deficit independent of CF”: it was then established by the sweep;
- a claim in the nerve module that non-identity bars die only at non-flat events: it fails for $\mathbb{Z}/2$ coefficients and non-abelian groups;
- the “BBV” expansion as a Bell-nonlocality framework: no such framework could be identified;
- the “sheaf–geometric bridge” $\chi(\gamma_{\text{AMB}}) = \text{Hol}(\alpha, \partial M)$ and contextuality at single-outcome supports (Remark in Section 21);
- our prediction that linear labelled cocycles carry a twisted Čech part at $\alpha = 1$: they vanish (Section 33); and our first exactness remark for $\mathbb{R}^\Omega$ coefficients, which holds only without locality in degree 0.

44 Method notes

- Every detector needs a planted positive and a null. The engine’s T0 bug and the thresholded- γ false positives were caught this way.
- Use Wasserstein, not bottleneck, distance for near-degenerate diagrams [H].
- Plug-in MI bias is $(K - 1)(L - 1)/2N$; bootstrap nulls cancel it [H].
- Read the proposition, not the summary.

45 Reproduction

Code availability. All code, tests, examples and this paper’s source are at <https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/AMB-Vigneaux/>.

- [H]: the command appendix of the hierarchy paper (`build_corpus_syn.py`, `splitmerge.py`, `estimator4.py`, `estimator6.py`).
- [O]: `holonomy.py`, `amb_gamma.py`, `parity_glue.py`, `h2_rational.py`, `h2_integral.py`.
- O6: `benchmarks/o6` (`o6_generate.py`, `baseline_splitmerge.py`, `o6_score.py`; results and logs in `results/`, summary in `BASELINE.md`).
- [S]: `python -m pytest` (219 tests), `examples/holonomy_estimated.py`, `examples/channels.py`, `examples/o2_ranks.py`, `examples/o2_verify.py`, `examples/train_tf.py` with `grid.sh` (JAX), and modules `outcome`, `holonomy`, `channels`, `nerve`, `functors`, `na_contexts`, `magphase`, `meter`, `meter_general`, `monodromy`, `deficits`, `relclaims`, `twisted_info`, `dynamics`, `spectral`, `certificate`, `toggle`, `hull`, `strata`, `treespace`, `giry`.

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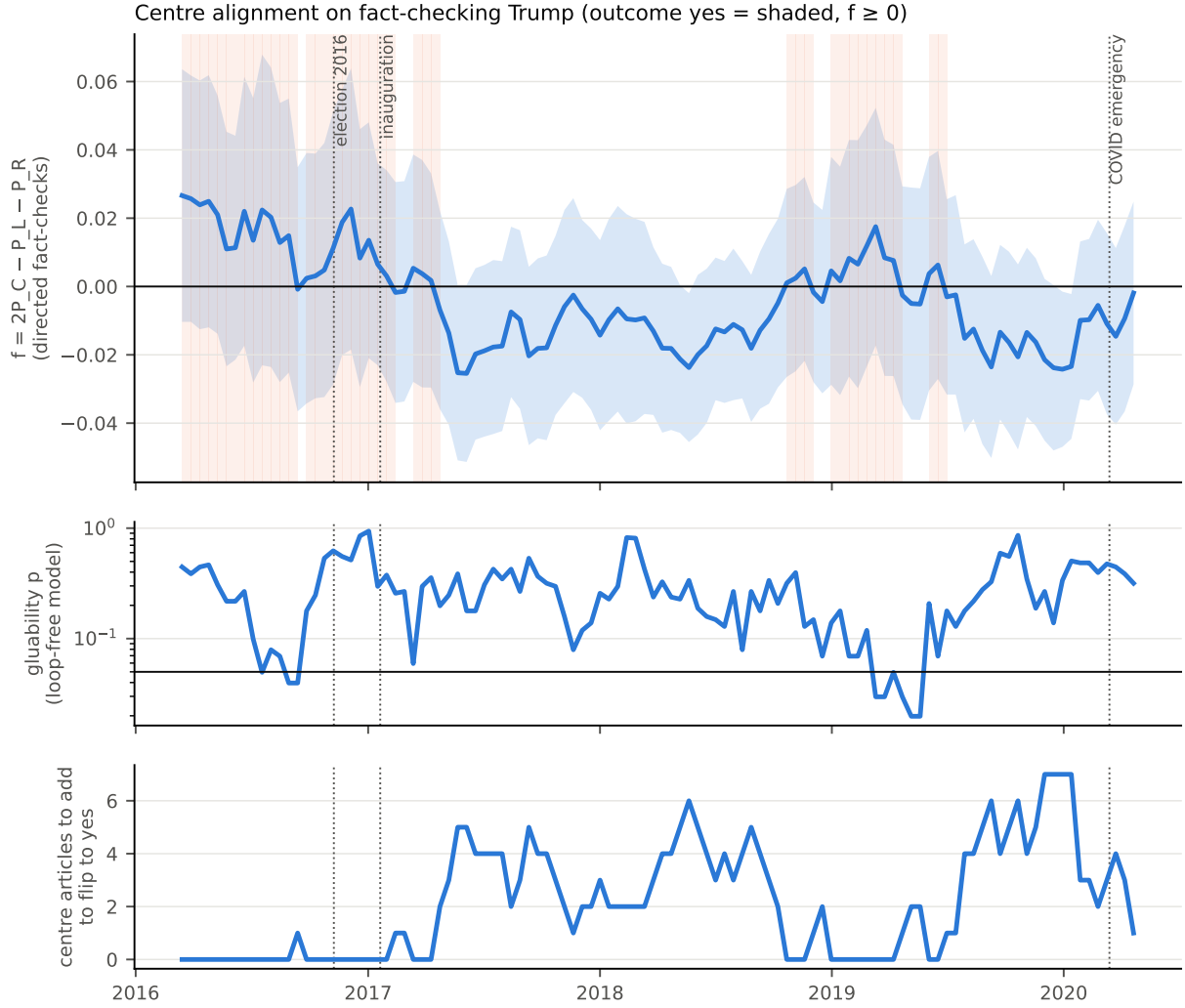


Figure 3: Streaming gluability-to-outcome map. Top: f with 95% band, shaded where the outcome is “yes”. Middle: gluability p of the loop-free model, with the 0.05 line. Bottom: centre articles that would have to be added to flip “no” to “yes”. Windows are 26 weeks long, plotted at their midpoint, with events as dotted lines.

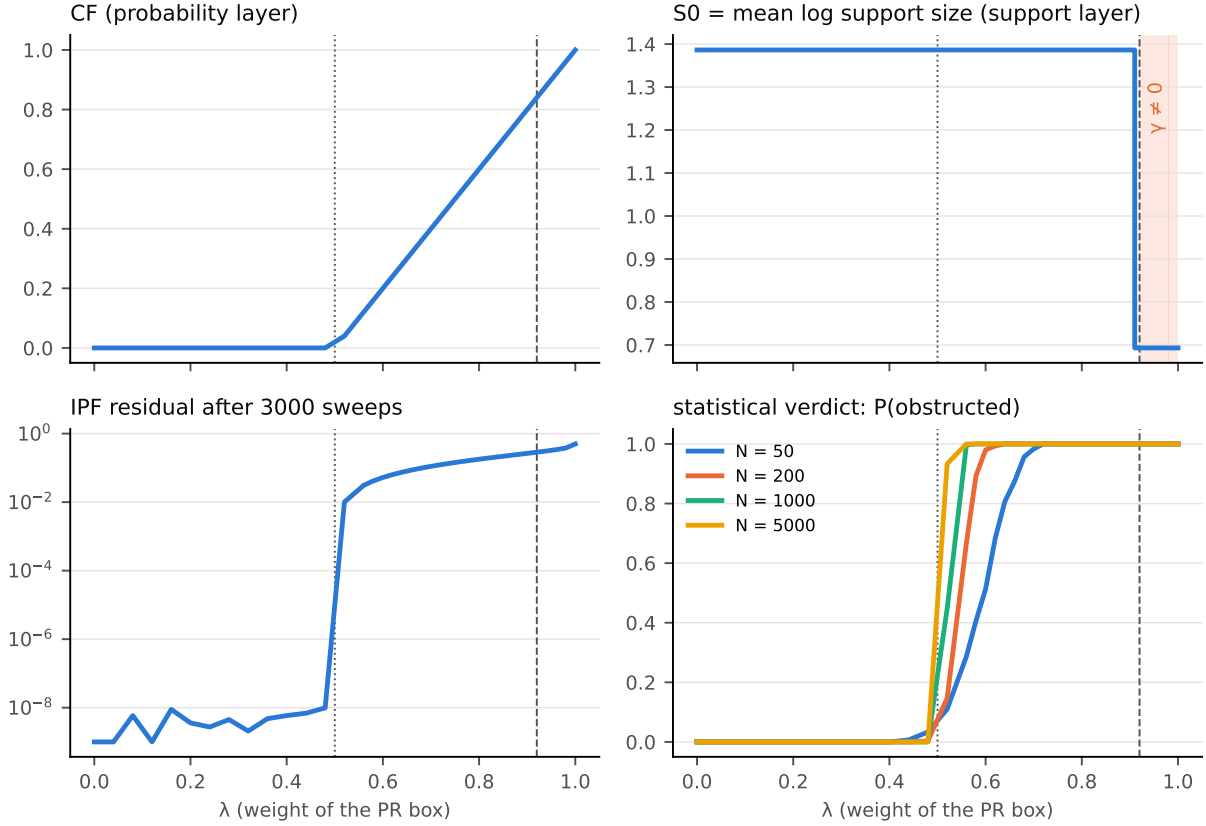


Figure 4: Along $e(\lambda) = \lambda \text{PR} + (1 - \lambda) \text{noise}$: CF is continuous and piecewise linear; S_0 and γ jump only at the support collapse (dashed, $\lambda = 0.92$, support tolerance 0.02); IPF converges exactly where $\text{CF} = 0$ (dotted, $\lambda = 1/2$); the statistical verdict sharpens toward the CF boundary as N grows.

role	result	where
<i>What to detect</i>		
obstruction	AMB γ , ring-dependent; sufficient but not necessary (Hardy false negative)	§16
degree	contextual fraction (LP; the dual is a Bell inequality); CF = 1 iff strong	§16, §21
where γ can be nonzero	support stratification; interior versus vertices	Prop. 21.1
binary pairwise claims	all-or-nothing; odd $\mathbb{Z}/2$ cycle = Liar = PR box	Prop. 21.3
operations to outcomes	holonomy $[g]$ and the connecting class	Thm. 18.1
cycles are necessary	acyclic and pairwise consistent implies global (BFMY; Vorob'ev)	Thm. 12.4, §23
<i>What cannot be detected</i>		
information lens	Layer-2 blindness to $[g]$	Prop. 18.2
what it does read	moduli $ s_i $; CF from moduli plus one bit	Prop. 22.2
falsehood	coherent fabrications have CF = 0	§23
which edge	localisation needs a trusted tree	Prop. 42.1
long loops	girth bound $\ell_{\min}$	Thm. 19.5
<i>How to compute</i>		
cheap CF bound	logical Bell bound $\sum p_i - (n - 1)$	Prop. 22.1
noisy cycles	closed form $1 - L \varepsilon/2$	Thm. 19.4
parity claims	Gaussian elimination over $\mathbb{F}_2$ with a certificate	§23
affine operations	dimension deficit $\text{rank}(M - I)$; section count	§22
dimension as a class	cocycle on affine supports over $\mathbb{F}_p$	Prop. 22.4
shortest contradictions	BFS in the parity double cover	§23
<i>From samples</i>		
holonomy	Wald test with delta-method standard error; N_{80}	§17
overlaps	deficit with a pinned bootstrap null; threshold sweep with a null band	§26
meter	pooled flip rate by EM; Bonferroni over cycles	§23
real data	permutation null (BASIL)	§23

Table 20: The detection toolkit, with pointers to where each result is stated.